

Modern Fractal Geometry
Higher-Dimensional Inverse Theorems and Projections
Appendix Volume

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Preface and proof contract

The line inverse theorems are not repeated here. Volume II.5 proves the Euclidean inverse-entropy theorem and its convolution-power structure theorem; Volume II.11 proves the line inverse L^q theorem; Volume III.6 proves the rank-one Bourgain–He projection mechanism. The present volume takes their *proved statements*, with their hypotheses, as earlier internal results. We do prove the one-dimensional Bourgain regular-tree, grouped-porosity, packet-amplification, and corrected $A + xA$ chain, because it supplies the scalar comparison for the higher-dimensional theory. The new higher-dimensional tasks are Shmerkin’s inverse L^q theorem and He’s arbitrary-rank discretized projection theorem. The Falconer–Jin invariant-fiber input is a cited authorized omission.

This is a reader manuscript, not an independent referee certificate. The companion file `NO_FULL_PROOF_RESULTS.md` classifies all results not proved in full and distinguishes a written local proof from an externally certified one. No exercises or solutions are included.

The He chapter contains full *local* implication proofs, and its scalar segment supplier in Volume VI.12 uses the Volume VI.10 Bourgain sum–product chain. The latter now prints its own complete dyadic-tree, packet-amplification, metric-transfer and corrected all-parameter $A + xA$ proof route, independent of this later appendix. The published Bourgain Theorem 2 has a false lower-only cardinality hypothesis; the corrected upper-cardinality version and its grid counterexample are stated explicitly below. No proof status is promoted by a source citation alone.

Chapter A.1

The entropy branch without repetition

The problem that changes in dimension $d > 1$ is not entropy bookkeeping. It is the possibility that two measures have almost no convolution entropy growth because one is nearly uniform along a nonzero proper plane and the other is nearly supported near a translate of that plane. Thus “small growth implies concentration near points,” the one-dimensional shorthand, is false in general.

The definitions of dyadic components, (V, ε, m) -saturation and (V, ζ) -concentration are those of Volume II.5, Definitions D-II.5.01–D-II.5.03. We refer to its *proved* Euclidean inverse theorem G-II.5.18 and repeated-convolution theorem G-II.5.17. In particular the planes in G-II.5.18 depend on the scale, not on the sampled component. The dependence of its lower window bound on d, ε , but not on the concentration tolerance ζ , is part of the theorem and is important in the application.

Here is the exact boundary between that analysis and the self-similar application. Let $\Phi = (x \mapsto r_i U_i x + a_i)_{i \in \Lambda}$ be a finite contracting similarity system on $\mathbb{R}^d$, let $p_i > 0$ sum to one, and write μ for its invariant probability. The closure of the group generated by the U_i is denoted G_Φ . A linear subspace V is *invariant* when $gV = V$ for every $g \in G_\Phi$. We write $\pi_{V^\perp}$ for the orthogonal projection to $V^\perp$ and μ_{V+x} for conditional probabilities in the disintegration of μ over this projection. The detailed construction and its measurable qualification are in Volume II.7, Definitions D-II.7.08–D-II.7.11.

Theorem A.1.1 (Falconer–Jin invariant-fiber input; authorized cited theorem). *Suppose first that the U_i preserve orientation. For each orthogonal projection π of rank $1 \leq k < d$, some $g \in G_\Phi$ has the following properties: $(\pi g)_* \mu$ is exact-dimensional; almost every conditional probability of μ over πg is exact-dimensional with a common dimension β ; and*

$$\dim \mu = \dim((\pi g)_* \mu) + \beta. \quad (\text{A.1.1})$$

Here exact-dimensional means that $\lim_{r \downarrow 0} \log v(B(x, r)) / \log r$ exists and is constant for v -almost every x , and $\dim v$ denotes this constant.

Source and proof policy. This is the deterministic specialization of Falconer–Jin, Theorem 3.1(ii)–(iii) and Corollary 3.1 [3], arXiv:1212.1345v4, printed pp. 10–11. At the user’s direction its research proof is omitted and classified as an authorized omission (Category 5),

not as a locally proved theorem. Volume II.7 proves the orientation lift and invariant-plane reduction; those deductions remain conditional on this cited input.

The rest of the higher-dimensional Hochman chain is already written as genuine deductions in Volume II.7: persistent saturation produces full-dimensional invariant fibers (G-II.7.16); its local action argument yields the entropy alternative (G-II.7.19); and the measure/set trichotomies, irreducible and algebraic consequences, and the homothetic small-dimension projection bound are G-II.7.20, .21, .23, .27 and .29. These deductions must not be recopied as new proofs. Their global status is conditional on the authorized Falconer–Jin input, even though the local implications in Volume II.7 are proved. The projection bound G-II.7.29 has the indispensable hypothesis $s_\Phi \leq 1$. We do not use the unrestricted formula printed as Theorem 1.14 in the frozen Hochman preprint: Volume II.7 records counterexamples to it and to a sign-only repair.

Chapter A.2

Inverse L^q theory in $\mathbb{R}^d$

The line theorem of Volume II.11 has only two geometric modes at a given scale: no branching of one measure or nearly full branching of the other. In $\mathbb{R}^d$ an intermediate-dimensional plane can carry the obstruction. The following finite-tree lemma, which is a useful part of the higher-dimensional proof, illustrates why one needs uniform branching before the geometric planes can be compared across scales.

A.2.1 Finite dyadic regularization

Definition A.2.1. For $j \geq 0$, $\mathcal{D}_j$ is the partition of $\mathbb{R}^d$ into half-open cubes of side 2^{-j} ; its cubes inside $[0, 1)^d$ form the rooted tree used below. The unique member of $\mathcal{D}_j$ containing x is denoted $\mathcal{D}_j(x)$. A 2^{-m} -set is a subset of $2^{-m}\mathbb{Z}^d \cap [0, 1)^d$. If $m = SL$, a 2^{-m} -set A is (L, S) -uniform when integers $R_0, \dots, R_{S-1}$ exist such that every $I \in \mathcal{D}_{sL}$ meeting A has exactly R_s children in $\mathcal{D}_{(s+1)L}$ meeting A . We write $\mathcal{D}_j(A) = \{I \in \mathcal{D}_j : I \cap A \neq \emptyset\}$. If $I \in \mathcal{D}_{sL}$, the set $A_I^{(L)}$ is the 2^{-L} -set of lower-left corners of the children of I meeting A , after the affine map taking I to $[0, 1)^d$.

This definition fixes the dyadic-grid convention at boundaries. In particular $\#\mathcal{D}_m(A) = \#A$.

Lemma A.2.2 (Uniformization with finitely many local statistics). *Let $L, S, V \geq 1$ and let, for each $0 \leq s < S$, F_s be a function on 2^{-L} -sets taking at most V values. Every nonempty 2^{-SL} -set A_0 contains an (L, S) -uniform subset A such that $F_s(A_I^{(L)})$ is independent of $I \in \mathcal{D}_{sL}(A)$ for every s , and*

$$\#A \geq [2(dL + 1)V]^{-S} \#A_0. \quad (\text{A.2.1})$$

Proof. We prune the finite tree from its leaves toward its root. At the start, every level- SL cube that meets A_0 contains one point. Inductively suppose the set retained after the levels greater than s have been treated has the same number P of points in every occupied level- $(s + 1)L$ cube. This is true at $s = S - 1$ with $P = 1$.

For each occupied parent $I \in \mathcal{D}_{sL}$ let b_I be its number of occupied children; $1 \leq b_I \leq 2^{dL}$. Split the parents into the $dL + 1$ classes $2^k \leq b_I < 2^{k+1}$, $0 \leq k \leq dL$. Their total point masses sum to the current cardinality, so one class carries at least a proportion $(dL + 1)^{-1}$.

In each parent of that class keep exactly 2^k of its occupied children, retaining their entire already-pruned descendant trees. Since each child carries P points and $2^k > b_I/2$, this loses at most another factor 2 of the mass in the selected class. All retained parents now have the same child count 2^k , and each retained parent has $2^k P$ points.

Compute F_s on the resulting normalized child set of each retained parent. One of its at most V values accounts for at least V^{-1} of the remaining point mass. Delete all parents with other values. This costs at most a factor V . It does not change the descendant set inside any surviving parent or child. Hence all earlier choices at levels $> s$ remain valid, while every surviving level- sL cube has exactly 2^k occupied children and the same F_s value. The invariant for the next induction step holds with P replaced by $2^k P$. After S stages the total loss is at most $[2(dL+1)V]^S$. The final root-to-leaf subtree has all asserted properties. $\square$

Corollary A.2.3. *For the set A in Theorem A.2.2, $\#A = \prod_{s=0}^{S-1} R_s$ and therefore*

$$\log_2 \#A = \sum_{s=0}^{S-1} \log_2 R_s. \quad (\text{A.2.2})$$

Proof. The occupied cubes at level sL number $\prod_{j < s} R_j$ by induction, beginning with the single root cube. At level SL each occupied cube contains exactly one point of A . $\square$

The same rooted-tree viewpoint permits prescribed pruning at selected scales. This is the step that later removes the small number of scales at which an entropy argument does not certify the desired plane dimension.

Lemma A.2.4 (Prescribed branching reduction). *Let A_0 be an (L, S) -uniform 2^{-SL} -set with branching numbers R_s , and let $T \subset \{0, \dots, S-1\}$. For each $s \in T$ choose an integer $1 \leq R'_s \leq R_s$, and for every $I \in \mathcal{D}_{sL}(A_0)$ choose exactly R'_s of the R_s occupied children of I . Then there is an (L, S) -uniform $A \subset A_0$ whose occupied children at a retained level- sL parent are exactly the chosen ones when $s \in T$, and are all the original children when $s \notin T$. In particular,*

$$\#A = \#A_0 \prod_{s \in T} \frac{R'_s}{R_s}. \quad (\text{A.2.3})$$

Proof. Retain a point $x \in A_0$ exactly when, for every $s \in T$, the level- $(s+1)L$ cube containing x is one of the chosen children of its level- sL parent. A surviving parent at a selected level has exactly its R'_s chosen children still occupied: each chosen child has a full descendant subtree except for later selected-scale pruning, and every prescribed later branching number is at least one. At an unselected level all R_s original children survive for the same reason. Hence A is uniform with branching numbers R'_s for $s \in T$ and R_s otherwise. Apply Theorem A.2.3 to A and A_0 to obtain (A.2.3). No child set inside a surviving parent is changed by pruning an earlier level. $\square$

The inverse theorem's final central-cell convention can also be obtained by a direct translation argument. For a cube I , write $\frac{1}{3}I$ for its concentric subcube of one-third the side length. The block length is assumed large, as it is in the theorem; no statement is made for every positive block length.

Lemma A.2.5 (Central-cell translation at separated scales). *Let $L \geq 5$, $S \geq 1$, and let $A \subset [1/3, 2/3]^d$ be finite. There are $A' \subset A$ and $y \in [-1/3, 1/3]^d$ such that*

$$\#A' \geq 4^{-dS} \#A, \quad x + y \in \frac{1}{3} \mathcal{D}_{sL}(x + y) \quad (x \in A', 0 \leq s < S). \quad (\text{A.2.4})$$

Proof. First work in one dimension. Inside the root interval $[0, 1)$ retain its central third $J_0 = [1/3, 2/3]$. If J is the central third of a dyadic level- sL interval I , then J contains at least $2^L/3 - 2 \geq 2^L/4$ entire dyadic level- $(s+1)L$ intervals when $L \geq 5$. Retain the central third of each of those intervals. Their total length is at least one-quarter the length of J . Induction produces a set $T \subset [1/3, 2/3]$ of length at least $3^{-1}4^{-(S-1)}$ whose points lie in the central third of every ancestral level- sL interval, $0 \leq s < S$. Taking the Cartesian power, T^d has measure at least $3^{-d}4^{-d(S-1)}$ and has the stated central-cell property in every coordinate.

For each $x \in A$, the translated cube $x + [-1/3, 1/3]^d$ contains $[1/3, 2/3]^d$, and hence T^d . Thus the set of $y \in [-1/3, 1/3]^d$ for which $x + y \in T^d$ has measure at least $3^{-d}4^{-d(S-1)}$. Integrate the number of such x over the translation cube, whose volume is $(2/3)^d$. Some y therefore has at least

$$\frac{3^{-d}4^{-d(S-1)}}{(2/3)^d} \#A = 2^{-d}4^{-d(S-1)} \#A \geq 4^{-dS} \#A$$

successful points. Take those points as A' . Their translations lie in T^d , so they meet every central-cell condition in (A.2.4). $\square$

For a finitely supported nonnegative measure λ on $\mathbb{R}^d$ and $1 < q < \infty$, set $\|\lambda\|_q^q = \sum_x \lambda(\{x\})^q$. If μ and ν are finitely supported probabilities, their convolution is the law of the sum of independent random variables with laws μ and ν .

Proposition A.2.6 (Convolution does not increase the discrete L^q norm). *For finitely supported probabilities μ, ν on $\mathbb{R}^d$ and $1 < q < \infty$,*

$$\|\mu * \nu\|_q \leq \|\mu\|_q. \quad (\text{A.2.5})$$

If μ is nonzero, equality holds exactly when ν is a point mass.

Proof. For each x , Jensen's inequality for the strictly convex map $t \mapsto t^q$ gives

$$\left(\sum_y \nu(\{y\}) \mu(\{x - y\}) \right)^q \leq \sum_y \nu(\{y\}) \mu(\{x - y\})^q.$$

Sum over x and interchange the finite sums. Translation invariance of counting measure makes the right side $\|\mu\|_q^q$. If equality holds and $y_1, y_2 \in \text{spt } \nu$, strict convexity forces $\mu(\{x - y_1\}) = \mu(\{x - y_2\})$ for every x . Thus μ is periodic with period $y_1 - y_2$. A nonzero finitely supported measure cannot have a nonzero period. Therefore $y_1 = y_2$ and ν is a point mass. The converse is immediate. $\square$

An inequality close to equality on the exponential scale does *not* imply that ν is nearly a point mass: μ may be nearly uniform in some directions. Before stating the higher-dimensional theorem, it is useful to see the one-dimensional tree mechanism behind the earlier line result. It also shows exactly what porosity contributes. The proof below is transplanted from Volume II.11, not taken as a black box; the stronger multilevel amplification claimed in Volume VI.10 is distinguished afterward.

A.2.2 Bourgain's regular and porous trees in one dimension

In this section $\delta = 2^{-m}$, $m = m_1 T$, and a δ -set is a nonempty subset of $\delta\mathbb{Z} \cap [0, 1)$. A level- j dyadic interval is $[k2^{-j}, (k+1)2^{-j})$; it is *occupied* if it meets the set under discussion. For intervals I, J we use the distance between their closures. A *block* runs from level sT to level $(s+1)T$, $0 \leq s < m_1$. The distinction between an occupied intermediate cell and an occupied terminal child matters: a priori one intermediate cell can hold many terminal children. For bounded $E \subset \mathbb{R}$, $N(E, r)$ is the least number of closed radius- r intervals covering E ; changing this radius by a fixed factor changes N by at most a fixed factor in one dimension.

Lemma A.2.7 (Bourgain regular-tree extraction). *There is an absolute $c > 0$ such that, for sufficiently large T , every δ -set A contains A_1 with the following properties. There is a set of active blocks $S \subset \{0, \dots, m_1 - 1\}$ and, for each $s \in S$, integers n_s, R_s with*

$$sT < n_s < (s+1)T - 4, \quad 2 \leq R_s \leq 2^{n_s - sT},$$

where R_s is a power of 2. In every occupied level- sT parent, independently of the parent:

1. if $s \notin S$, exactly one level- $(s+1)T$ child is occupied;
2. if $s \in S$, exactly R_s level- n_s cells are occupied, and each contains exactly one occupied level- $(s+1)T$ child;
3. if $s \in S$, two of those level- n_s cells have distance in $[2 \cdot 2^{-n_s}, 10 \cdot 2^{-n_s}]$.

Moreover

$$\#A_1 = \prod_{s \in S} R_s \quad \text{and} \quad \#A_1 \geq (cT^{-2})^{m_1} \#A. \quad (\text{A.2.6})$$

Proof. We first establish the geometric selection rule inside one level- p parent. Let $q > p + 8$ and suppose at least 1000 level- q children are occupied. The distance D between the leftmost and rightmost occupied level- q cells obeys

$$998 \cdot 2^{-q} \leq D \leq 2^{-p}.$$

Choose r so that $5 \leq D2^r < 10$. The two endpoint cells at level r then have distance between $3 \cdot 2^{-r}$ and $10 \cdot 2^{-r}$; the bounds on D give $p < r < q - 4$. Choose the largest r for which some occupied level- r pair has distance in $[2 \cdot 2^{-r}, 10 \cdot 2^{-r}]$. Every occupied level- r cell contains fewer than 1000 occupied level- q cells. Indeed, a cell with 1000 such descendants would, by the same selection rule inside that cell, produce a qualifying level $r' > r$. The inequality $q > r + 8$ needed for that repetition follows from the presence of 1000 distinct level- q children.

Now work upward through the T -blocks. Assume the blocks below $(s+1)T$ have been regularized, so each occupied level- $(s+1)T$ child carries the same number P of retained terminal points. In a level- sT parent I , let $M(I)$ be its number of occupied terminal children. If $M(I) < 1000$, keep one such child and call I inactive. Otherwise apply the preceding rule with $p = sT$, $q = (s+1)T$ and let $n(I)$ be the maximal qualifying level. If $M_n(I)$ cells at level $n(I)$ are occupied, then $M(I) < 1000M_n(I)$. Choose a power of two $R(I)$ with

$M_n(I)/2 < R(I) \leq M_n(I)$; retain $R(I)$ cells, choosing them to include a qualifying pair, and retain one terminal child in each selected cell. This keeps more than $M(I)/2000$ of the point mass of I . If the chosen count is 1, declare the parent inactive instead.

An active parent has type $(n(I), \log_2 R(I))$, and there is one inactive type. There are $O(T^2)$ types: the intermediate level has at most T possibilities, and $0 \leq \log_2 R(I) \leq T$. Among all current level- sT parents, retain a type carrying the largest point mass and delete the other parents. Because each surviving terminal child carried the same mass P , this global pigeonhole, together with the local $1/2000$ loss, retains at least a proportion cT^{-2} at block s . Deletion is by whole previously regularized descendant trees, so it cannot disturb choices at deeper blocks. The surviving type is identical in every occupied level- sT parent.

After all m_1 blocks, the retained mass is at least $(cT^{-2})^{m_1} \#A$. Branching is uniform from the single root parent: an inactive block multiplies the number of occupied terminal cells by 1, and an active one by R_s . Every level- m cell contains one point of the δ -set. This proves both identities in (A.2.6). Finally a level- sT parent has at most $2^{n_s - sT}$ intermediate cells, giving the stated upper bound on R_s . $\square$

The next step is a small-doubling pressure calculation. We use the finite-set Petridis–Ruzsa inequality proved in Volume II.11: if $\#(A + A) \leq K\#A$, then $\#(hA) \leq K^h\#A$ for every positive integer h . It is an internal additive-combinatorial result, not a research theorem imported here. The point of the proof is a scale-respecting injection into the 2^T -fold sumset; simply counting branches of A_1 would not control their locations.

Proposition A.2.8 (Small doubling forces near-maximal regular branching). *There is an absolute C_0 such that, for sufficiently large T , if a δ -set A satisfies*

$$\#(A + A) \leq 2^{2^{-2T-1}m} \#A, \quad (\text{A.2.7})$$

then it contains a regular subtree A_2 with active-block data as in Theorem A.2.7, except that some formerly active blocks may have been made inactive, and

$$R_s > 2^{(1-T^{-1/2})(n_s - sT)} \quad (s \in S_2), \quad \#A_2 \geq 2^{-C_0 m(\log_2 T)/\sqrt{T}} \#A. \quad (\text{A.2.8})$$

Proof. Apply Theorem A.2.7; put $d_s = n_s - sT$ and $C_s = 2^{d_s}$. We prove

$$\prod_{s \in S} C_s \leq \#(2^T A_1). \quad (\text{A.2.9})$$

In each active parent fix two occupied intermediate cells as in the lemma and order them left/right. For every binary word $c = (c_s)_{s \in S}$, descend through these choices to obtain a point $x_c \in A_1$. Given integers $1 \leq k_s \leq C_s$, define

$$\Phi(k) = \sum_{u=1}^{2^T} x_{c^{(u)}}, \quad c_s^{(u)} = \mathbf{1}_{\{u \leq k_s\}}.$$

The sum has exactly 2^T terms because $C_s \leq 2^T$. If $k \neq \ell$, let s be their first differing active block, and interchange the vectors if necessary so that $k_s < \ell_s$. For the indices u whose threshold choices first disagree at s , the two points lie in the distinguished cells of the same

level- sT parent; all their displacements have one sign and magnitude at least $2 \cdot 2^{-n_s}$. There are $\ell_s - k_s \geq 1$ such indices.

For a later block $t > s$, at most $|k_t - \ell_t| \leq C_t$ indices can have their first disagreement at t . The two points for each such index lie in distinguished cells of the same parent and differ by at most $12 \cdot 2^{-n_t}$. Their entire contribution, in either sign, is therefore bounded by

$$12 \sum_{t>s} C_t 2^{-n_t} = 12 \sum_{t>s} 2^{-tT} < 15 \cdot 2^{-(s+1)T}.$$

Grouping by the *first* disagreement is essential; bounding all later movement separately for each of the 2^T summands would be false. Since $n_s < (s+1)T - 4$, the first-disagreement contribution is larger than $32 \cdot 2^{-(s+1)T}$, so it dominates the tail and in particular $|\Phi(k) - \Phi(\ell)| > \delta$. Thus Φ is injective and its image lies in $2^T A_1$, proving (A.2.9).

Set $K = 2^{2^{-2T-1}m}$. The cited Petridis–Ruzsa inequality and Equation (A.2.7) give

$$\#(2^T A_1) \leq \#(2^T A) \leq K^{2^T} \#A = 2^{2^{-T-1}m} \#A.$$

By Equation (A.2.6), for an absolute C_1 ,

$$\#A_1 = \prod_{s \in S} R_s \geq 2^{-C_1 m (\log_2 T)/T} \#A.$$

Consequently

$$\prod_{s \in S} \frac{C_s}{R_s} \leq 2^{[2^{-T-1} + C_1 (\log_2 T)/T]m}. \quad (\text{A.2.10})$$

Call s bad when $R_s \leq C_s^{1-T^{-1/2}}$. For bad s , $C_s/R_s \geq C_s^{T^{-1/2}}$; hence (A.2.10) bounds the product of the bad C_s , and therefore of the bad R_s , by

$$2^{[\sqrt{T} \cdot 2^{-T-1} + C_1 (\log_2 T)/\sqrt{T}]m}.$$

In every occupied parent at every bad block retain one terminal child and discard the other branches. This costs exactly the product of the bad R_s and preserves all surviving deeper subtrees. The remaining active blocks satisfy the strict inequality in (A.2.8); the preceding bound and the retention bound for A_1 give the asserted estimate for A_2 after enlarging C_0 . $\square$

A gap of many dyadic generations between the intermediate branching cells and their terminal descendants makes the terminal pieces small relative to their mutual separation. The following precise statement extracts that geometry. It deliberately does not assert Bourgain's stronger grouped-level product estimate.

Proposition A.2.9 (Small-doubling porous-scale subtree). *Fix $0 < \sigma < 1$ and set $\beta = (1 - \sigma)/4$. For sufficiently large $T = T(\sigma)$, suppose Equation (A.2.7) holds and $\#A \leq 2^{\sigma m}$. There are a subtree $A_3 \subset A$, block data S_3, n_s, L_s , and a set $P \subset \{0, \dots, m_1 - 1\}$ with*

$$\#P \geq \frac{1}{2}(1 - \sigma)m_1, \quad \#A_3 \geq 2^{-m_1 - C_0 m (\log_2 T)/\sqrt{T}} \#A, \quad (\text{A.2.11})$$

such that A_3 has uniform block branching L_s on S_3 and one terminal child on inactive blocks. Every $s \in P$ is either inactive or satisfies $s \in S_3$ and

$$2 \leq L_s \leq R_s, \quad L_s \geq R_s/2, \quad \mu_s = 2^{-n_s}, \quad \text{dist}(J, J') \geq \mu_s, \quad \text{diam } J \leq 2^{-\beta T} \mu_s \quad (\text{A.2.12})$$

for distinct occupied terminal intervals J, J' inside any one level- sT parent. Here R_s, n_s are the data of A_2 in [Theorem A.2.8](#); the distance assertion concerns the retained terminal intervals, not arbitrary points at different higher-level parents.

Proof. Take A_2 from [Theorem A.2.8](#) and put $g_s = (s+1)T - n_s$ on active blocks. Call an active block nonporous when $g_s < \beta T$ and put every other block in P , including inactive blocks. On a nonporous active block,

$$n_s - sT = T - g_s > (1 - \beta)T, \quad \log_2 R_s > (1 - T^{-1/2})(1 - \beta)T.$$

Uniform branching gives $\#A_2 = \prod_{s \in S_2} R_s \leq \#A \leq 2^{\sigma m}$. Thus the number u of nonporous active blocks satisfies

$$u < \frac{\sigma}{(1 - T^{-1/2})(1 - \beta)} m_1.$$

Our choice $\beta = (1 - \sigma)/4$ makes the coefficient less than $(1 + \sigma)/2$ once T is sufficiently large. Therefore $\#P = m_1 - u \geq (1 - \sigma)m_1/2$.

Now thin only the active blocks in P and set $S_3 = S_2$. In a given parent, if $R_s = 2$, keep its distinguished separated pair and put $L_s = 2$. If $R_s \geq 4$, one parity class of the integer indices of its occupied level- n_s cells contains at least $R_s/2$ cells; retain exactly $L_s = R_s/2$ of them, together with their unique terminal children. The same count L_s works in every parent because R_s was uniform, although the chosen parity may vary. Distinct selected intermediate cells have at least one entire level- n_s cell between them, so their terminal intervals have distance at least $\mu_s = 2^{-n_s}$. For $R_s = 2$ the distinguished pair has even larger distance. Each terminal interval has diameter $2^{-(s+1)T} = 2^{-g_s} \mu_s \leq 2^{-\beta T} \mu_s$. This proves [\(A.2.12\)](#). Pruning every selected block by at most a factor 2 costs at most 2^{m_1} overall, giving [\(A.2.11\)](#); uniformity at other blocks is unaffected. $\square$

Theorem A.2.10 (Bourgain's grouped small-doubling porous subtree). *Fix $0 < \sigma < 1$, $\kappa > 0$, and an integer $D > 1000$. There are $\varepsilon_0 > 0$, $\beta > 0$, and a sufficiently large block size $T = T(\sigma, \kappa, D)$ with the following property for all sufficiently large m_1 . Put $m = Tm_1$, $\delta = 2^{-m}$, and $M = \lfloor 2^{T^{2/3}} \rfloor$. Suppose that a δ -set A satisfies*

$$\#A \leq \delta^{-\sigma}, \quad \#(A + A) \leq 2^{2^{-2T-1}m} \#A, \quad \#(A \cap I) \leq |I|^\kappa \#A \quad (\delta < |I| < \delta^{\varepsilon_0}) \quad (\text{A.2.13})$$

for every interval I in the displayed range. Then A contains a nonempty retained root $C_\emptyset$ partitioned into a uniform finite tree of some positive depth t_2

$$C_{\ell_1, \dots, \ell_{i-1}} = \bigsqcup_{\ell_i=1}^{L_i} C_{\ell_1, \dots, \ell_i}, \quad 1 \leq i \leq t_2,$$

where $L_i \geq 2$ is independent of the parent. There is a number $\gamma > (1 - \sigma)/5$ and, for each i , numbers $\mu_i > 0$ and ϑ_i such that all distinct level- i pieces have mutual distance at least μ_i ,

every level- i piece has diameter at most $\vartheta_i \mu_i$, and

$$\delta^{1/2} < \vartheta_i < 2^{-\beta T}, \quad \prod_{i=1}^{t_2} \min\{L_i, \vartheta_i^{-1/D}\} \geq M^{\gamma m_1}, \quad \gamma > \frac{1-\sigma}{5}. \quad (\text{A.2.14})$$

The retained root satisfies

$$\#C_\emptyset \geq M^{-\gamma m_1/80} \#A. \quad (\text{A.2.15})$$

Proof. The proof has two tree reductions. First apply [Theorem A.2.8](#) and write S_2 for the active blocks of A_2 . Set

$$\beta_0 = \frac{1-\sigma}{8}, \quad b = \lfloor \beta_0 T \rfloor, \quad P = \{0, \dots, m_1 - 1\} \setminus \{s \in S_2 : (s+1)T - n_s < \beta_0 T\}.$$

Indeed, if u active blocks have $(s+1)T - n_s < \beta_0 T$, then $\#A_2 = \prod_{s \in S_2} R_s \leq 2^{\sigma m}$ and [Equation \(A.2.8\)](#) give $u < \sigma m_1 / [(1-T^{-1/2})(1-\beta_0)]$. For T large this is less than $(1+\sigma)m_1/2$, so

$$t_1 := \#P > \frac{1-\sigma}{2} m_1 \quad (\text{A.2.16})$$

when T is large. Write $P = \{s_1 < \dots < s_{t_1}\}$ and put $s_0 = -1$. Give every inactive block the formal branching number $R_s = 1$.

We build an intermediate uniform tree B indexed by these t_1 blocks. For $1 \leq j \leq t_1$ put $q_j = (s_j + 1)T$ and

$$a_j = \begin{cases} n_{s_j}, & s_j \in S_2, \\ s_j T, & s_j \notin S_2. \end{cases} \quad Q_j = \prod_{s_{j-1} < s \leq s_j} R_s, \quad K_j = \max\{1, Q_j/2\}.$$

All Q_j are powers of 2, so K_j is an integer. The defining inequality for P gives $a_j \leq q_j - b$. Inside each already selected level- $(j-1)$ piece, uniform branching of A_2 gives exactly Q_j occupied level- a_j cells. Each contains exactly one occupied level- q_j child. This last assertion is [Theorem A.2.7\(2\)](#) when s_j is active and (1) when it is inactive; the intervening blocks merely supply the product Q_j .

The integer indices of the Q_j level- a_j cells split into two parity classes. Retain K_j cells from a larger class, and with each cell retain its whole unique level- q_j terminal subtree. If $Q_j = 1$, retain that cell. Do this inside every level- $(j-1)$ piece, then continue at the next j . The number K_j is independent of the parent, and every child carries the same terminal point count because A_2 had uniform branching. Hence the resulting tree, whose root is $B_\emptyset$, satisfies

$$\#B_\emptyset = \#A_2 \prod_{j=1}^{t_1} \frac{K_j}{Q_j} \geq 2^{-t_1} \#A_2 \geq 2^{-m_1} \#A_2. \quad (\text{A.2.17})$$

Put $\lambda_j = 2^{-(q_j-b)}$. Two selected level- a_j cells in the same parent have an entire such cell between them, so their terminal subtrees are separated by at least $2^{-a_j} \geq \lambda_j$. Pieces with different level- $(j-1)$ parents were already separated by at least $\lambda_{j-1} > \lambda_j$. Each level- j piece lies in one level- q_j cell. Thus, globally,

$$\text{dist}(B_{\mathbf{k}}, B_{\mathbf{k}'}) \geq \lambda_j \quad (\mathbf{k} \neq \mathbf{k}', |\mathbf{k}| = j), \quad \text{diam } B_{\mathbf{k}} \leq 2^{-q_j} = 2^{-b} \lambda_j. \quad (\text{A.2.18})$$

Moreover all level- j pieces have cardinality $\#B_\phi / \prod_{h \leq j} K_h$. This is the first tree reduction: inactive porous blocks contribute geometric separation but may have $K_j = 1$.

We now collapse runs of the B -tree. Set $t = t_1$. As long as possible, choose the *least length* $r \in \{1, \dots, t\}$ for which

$$Q(t, r) := K_{t-r+1} \cdots K_t > M^r. \quad (\text{A.2.19})$$

The use of the actual length r removes an indexing ambiguity in the printed source: a product from $t - r$ through t contains $r + 1$ factors. Minimality in (A.2.19) implies

$$K_{t-h+1} \cdots K_t \leq M^h \quad (1 \leq h < r). \quad (\text{A.2.20})$$

Let $a = t - r + 1$.

If $r \leq 1000$, collapse levels $a, \dots, t$ to one level. Each level- $(a-1)$ parent has $L = Q(t, r) > M^r$ children represented by its level- t descendants. By Equation (A.2.18), they are $\mu = \lambda_t$ separated and have diameter at most $2^{-b}\mu$. There is no pruning loss at this collapse.

If $r > 1000$, set $h = \lceil r/100 \rceil < r$. Keep all level- $(t-h)$ descendants beneath each level- $(a-1)$ parent, but beneath each such descendant keep just one of its level- t descendant subtrees. In this collapse,

$$L = K_a \cdots K_{t-h} = \frac{Q(t, r)}{K_{t-h+1} \cdots K_t} > M^{r-h} \geq M^{0.98r}, \quad (\text{A.2.21})$$

by Equation (A.2.20); the last inequality uses $r > 1000$. The selected children are separated by $\mu = \lambda_{t-h}$. Their diameter is at most $2^{-b}\lambda_t$, so their relative diameter is at most $2^{-b}\lambda_t/\lambda_{t-h}$. Since $s_t - s_{t-h} \geq h$,

$$\frac{\lambda_t}{\lambda_{t-h}} = 2^{-(s_t - s_{t-h})T} \leq 2^{-hT}. \quad (\text{A.2.22})$$

The frozen tail costs a factor at most $K_{t-h+1} \cdots K_t \leq M^h \leq M^{r/90}$.

For either case define the advertised relative-diameter parameter by

$$\vartheta = 2 \max\{\delta^{1/2}, 2^{-b}\lambda_t/\mu\}.$$

It is strictly larger than $\delta^{1/2}$ and bounds the diameter divided by μ . For T sufficiently large in terms of σ and D , it is smaller than $2^{-\beta T}$ with $\beta = \beta_0/2$. In the small case, $\vartheta^{-1/D} \geq M^r$ follows from $r \leq 1000$ and $b/D \gg 1000T^{2/3}$. In the large case, Equation (A.2.22) and $h \geq r/100$ give

$$\vartheta^{-1/D} \geq 2^{-1/D} \min\{2^{m/(2D)}, 2^{(b+hT)/D}\} \geq M^r$$

once $T^{1/3}$ is large compared with $100D$. Therefore each collapsed level, of length r , contributes

$$\min\{L, \vartheta^{-1/D}\} \geq M^{0.98r}. \quad (\text{A.2.23})$$

The construction is performed identically in number, not necessarily in the positions chosen, in every occupied parent. It consequently gives uniform child counts. Previously fixed deeper descendant subtrees are retained whole except for the explicitly frozen tails, so bottom-up collapsing preserves all separation and diameter assertions.

Replace t by $a-1 = t-r$ and repeat. Suppose the algorithm stops with $t \geq 1$ uncollapsed B -levels. Since no length satisfies Equation (A.2.19), in particular

$$\prod_{j=1}^t K_j \leq M^t \leq M^{m_1} \leq \delta^{-T^{-1/3}}.$$

Every level- t piece of the *original* B -tree has the same cardinality, so Equation (A.2.17) and Equation (A.2.8) give

$$\#B_{\mathbf{k}} = \frac{\#B_{\emptyset}}{\prod_{j \leq t} K_j} \geq 2^{-m_1 - C_0 m(\log_2 T)/\sqrt{T}} \delta^{T^{-1/3}} \#A \geq \delta^{2T^{-1/3}} \#A \quad (\text{A.2.24})$$

for sufficiently large T . The reference to the original tree is important: later frozen-tail pruning need not preserve this cardinality, but it does not enter the stopping test. The piece $B_{\mathbf{k}}$ lies in one level- $s_t T$ interval I . If $s_t/m_1 > \varepsilon_0$, then $|I| < \delta^{\varepsilon_0}$, and the non-concentration hypothesis in Equation (A.2.13) combined with Equation (A.2.24) yields $s_t/m_1 \leq 2T^{-1/3}/\kappa$. Otherwise $s_t/m_1 \leq \varepsilon_0$. Since $t \leq s_t + 1$, choosing

$$\varepsilon_0 < \frac{1-\sigma}{16}, \quad \frac{2T^{-1/3}}{\kappa} < \frac{1-\sigma}{16}$$

and then m_1 large gives $t < (1-\sigma)m_1/4$. The same inequality is automatic if the algorithm stops at $t = 0$.

Let $R = t_1 - t$ be the total number of collapsed B -levels; it is exactly the sum of their lengths r_i . By Equation (A.2.16),

$$R > \frac{1-\sigma}{4} m_1.$$

Set $\gamma = 0.98R/m_1$; then $\gamma > (1-\sigma)/5$. Multiplying Equation (A.2.23) over the collapsed levels proves the product estimate in Equation (A.2.14).

It remains to account for mass rather than only geometry. The frozen tails cost at most $M^{R/90}$ in total. Thus Equation (A.2.17) and Equation (A.2.8) yield

$$\#C_{\emptyset} \geq 2^{-m_1 - C_0 m(\log_2 T)/\sqrt{T}} M^{-R/90} \#A.$$

Because $R > (1-\sigma)m_1/4$, $\log_2 M \asymp T^{2/3}$, and

$$\frac{0.98}{80} - \frac{1}{90} > 0,$$

the factor preceding $M^{-R/90}$ is at least $M^{-(0.98/80 - 1/90)R}$ once T is sufficiently large. This is exactly Equation (A.2.15). If $t > 0$ uncollapsed levels remain above the highest collapsed block, absorb them into the first collapsed level: its child count is multiplied by $\prod_{j \leq t} K_j$, while its separation and diameter bounds remain those of that collapsed level because distinct level- t parents were farther apart. This can only increase its factor in Equation (A.2.14). The retained pieces, ordered from the highest collapsed block to the lowest, now form the asserted disjoint uniform tree from the single root $C_{\emptyset}$. $\square$

Remark A.2.11 (Source correction and scope). *The proof follows the geometric architecture of Bourgain's published Sections 2–3, pp. 198–207 [2]. We have used the fully proved finite-set version of Volume II.11 for the first two steps, fixed the block index range to $0 \leq s < m_1$, and made the collapsed-block length in the stopping rule equal to its number of factors. The source's displayed product from $t-r$ to t has $r+1$ factors, whereas its subsequent identity counts only r consumed levels; the present formulation avoids that off-by-one. It is a dyadic-lattice, cardinal-doubling version of the source construction. No claim is made here that every continuous δ -separated formulation follows without the ordinary rounding and covering-number comparisons.*

The grouped tree is useful only if its geometric gaps survive forming two independent sums. The next proposition gives the amplification step with its packet accounting explicit. Its proof applies to a geometric tree and does not assume that the tree was obtained from an integer lattice.

Proposition A.2.12 (Bourgain's packet amplification). *Let $C \subset \mathbb{R}$ be a finite δ -separated set with a uniform rooted tree of depth $t \geq 1$. Every parent at level $i-1$ has L_i children; distinct level- i pieces are μ_i -separated and each has diameter at most $\vartheta_i \mu_i$, where $0 < \vartheta_i < 10^{-3}$ and $\delta \leq \vartheta_i \mu_i$. Put*

$$M_i = \min\{L_i, \vartheta_i^{-1/D}\}.$$

Suppose $D > 10/\tau$, $M_i \geq M_$ for a sufficiently large $M_* = M_*(\tau, C_F)$, and $\nu \in (0, 1]$ obeys $\nu \vartheta_i \geq \delta$ for every i . Let λ be a probability on $[0, \nu]$ satisfying*

$$\lambda(I) \leq C_F(|I|/\nu)^\tau \quad (\delta \leq |I| \leq \nu).$$

For $C^{(\delta)} = \bigcup_{c \in C} [c - \delta/100, c + \delta/100]$, provided each ϑ_i is sufficiently small in terms of τ, C_F, D , one has

$$\int |\nu C^{(\delta)} + x C^{(\delta)}| d\lambda(x) \geq c \nu \delta \#C \prod_{i=1}^t M_i^{1/3}, \quad (\text{A.2.25})$$

where $c > 0$ is absolute and $|\cdot|$ is Lebesgue measure.

Proof. We use one interval observation, valid for two families $J_1, \dots, J_R$ and $J'_1, \dots, J'_{R'}$ of internally pairwise disjoint intervals:

$$\sum_{r,s} \mathbf{1}_{J_r + J'_s}(z) \leq C \max\{R, R'\} \quad (z \in \mathbb{R}). \quad (\text{A.2.26})$$

Indeed, for pairs with $|J_r| \leq |J'_s|$ and fixed r , at most two disjoint intervals J'_s of length at least $|J_r|$ can meet the interval $z - J_r$; the other pairs are symmetric. Integration of (A.2.26) bounds the area of a union of sets inside these interval sums from below by the sum of their areas divided by $C \max\{R, R'\}$.

The estimate at scale δ gives $\lambda(\{0\}) \leq C_F(\delta/\nu)^\tau \leq C_F \vartheta_i^\tau$. By the smallness assumption this is at most $1/2$. We may condition on $(0, \nu]$, increasing C_F by at most a factor two, prove the claimed lower bound for that conditional law, and lose only another factor two at the end. Hence below $x > 0$ almost surely.

Fix a pair of level- $(i-1)$ parent pieces. Order their L_i child pieces from left to right. Set $m_i = \lfloor M_i \rfloor$. Partition the children in each parent into consecutive packets of between m_i and $2m_i$ children, merging a final shorter packet with its predecessor. The convex hulls of distinct packets are disjoint, as are the convex hulls of distinct children inside one packet. There are at most L_i/m_i packets per parent. Applying (A.2.26) to the packet hulls of the two parents, after scaling them by ν and x , gives

$$|\nu C_{\mathbf{a}}^{(\delta)} + x C_{\mathbf{b}}^{(\delta)}| \geq c \frac{m_i}{L_i} \sum_{P,Q} |\nu D_P + x D'_Q|, \quad (\text{A.2.27})$$

where D_P, D'_Q are the unions of thickened children in their packets. For every packet pair the same interval inequality on its at most $2m_i$ children gives

$$|\nu D_P + x D'_Q| \geq \frac{c}{m_i} \sum_{a \in P, b \in Q} |\nu C_{\mathbf{a}\mathbf{a}}^{(\delta)} + x C_{\mathbf{b}\mathbf{b}}^{(\delta)}|. \quad (\text{A.2.28})$$

There is a better estimate for most x . Choose one point z_a in each child of the first parent and z'_b in each child of the second. The corresponding thickened child sum lies inside an interval of radius $C\nu\vartheta_i\mu_i$ centered at $\nu z_a + x z'_b$; the assumptions $x \leq \nu$ and $\delta \leq \vartheta_i\mu_i$ include the thickening error. If two distinct pairs $(a, b), (a', b')$ in a fixed packet pair have overlapping sum intervals and $b = b'$, the μ_i separation of the first children contradicts $\vartheta_i < 10^{-3}$. Otherwise $|z'_b - z'_{b'}| \geq \mu_i$, and overlap confines x to an interval of length at most $C\nu\vartheta_i$ about $-\nu(z_a - z_{a'})/(z'_b - z'_{b'})$. There are at most Cm_i^4 choices. The Frostman hypothesis, including the case $C\nu\vartheta_i > \nu$ by a trivial bound, therefore shows

$$\lambda\{x : \text{one of these overlaps occurs}\} \leq C_{\tau, C_F} m_i^4 \vartheta_i^{\tau} \leq C_{\tau, C_F} \vartheta_i^{\tau-4/D}. \quad (\text{A.2.29})$$

Here the interval length is at least δ up to an absolute dilation because $\nu\vartheta_i \geq \delta$; at larger or smaller constant multiples the same estimate follows by covering or enlarging the interval. On the complement of the bad set the child sum intervals are disjoint, so the right side of (A.2.28) has coefficient 1 instead of c/m_i .

Combining with (A.2.27), for every ordered child pair we obtain a measurable weight $w_{\mathbf{a}\mathbf{a}, \mathbf{b}\mathbf{b}}(x)$ such that the parent sum area is at least the sum of child sum areas times their weights. The weight is cm_i/L_i on the good event and c/L_i on the bad event. Since $D > 10/\tau$, the probability in (A.2.29) is at most $C_{\tau, C_F} \vartheta_i^{\tau/2}$. With the required lower bound on M_i and upper bound on ϑ_i , the constant and this bad probability are absorbed by a third of $\log M_i$, giving, uniformly over all parent and child addresses,

$$\int \log w_{\mathbf{a}\mathbf{a}, \mathbf{b}\mathbf{b}}(x) d\lambda(x) \geq \log(M_i^{1/3}/L_i). \quad (\text{A.2.30})$$

Iterate the area inequality through the uniform tree. For every pair of terminal addresses, Jensen's inequality and (A.2.30) give an expected product of weights at least $\prod_i M_i^{1/3}/L_i$. Every terminal piece has the same number $\#C/\prod_i L_i$ of δ -separated points. Taking one point of $x C_{\mathbf{b}}^{(\delta)}$ and translating $\nu C_{\mathbf{a}}^{(\delta)}$ shows that every terminal sum has area at least $c\nu\delta\#C/\prod_i L_i$. There are $(\prod_i L_i)^2$ terminal address pairs. Summing proves (A.2.25). $\square$

For the grouped tree above, $M_i \geq M^{0.98r_i}$ and $\prod_i M_i \geq M^{\gamma m_1}$. If $\nu \geq \delta^{\varepsilon_0}$ and $\varepsilon_0 < 1/2$, then $\nu \vartheta_i \geq \delta$, and the definition of ϑ_i in its proof gives $\delta \leq \vartheta_i \mu_i$. Consequently [Theorems A.2.10](#) and [A.2.12](#) yield an expected area at least $c\nu\delta M^{(1/3-1/80)\gamma m_1} \#A$. This quantitative margin is why the mass loss in the tree theorem cannot be omitted from the proof.

Here is the precise passage from the lattice model to the metric covering-number problem. It avoids assuming that rounding is harmless at the nonconcentration endpoint.

Corollary A.2.13 (Metric small-doubling amplification). *Fix $0 < \sigma < 1$, $\kappa > 0$, $\tau > 0$, and $C_F \geq 1$. There are $\varepsilon_{\text{nc}}, \varepsilon_d, \varepsilon_g > 0$ such that, for all sufficiently large multiples m of a fixed block size, $\delta = 2^{-m}$, the following holds. Suppose $A \subset [1, 2]$ is δ -separated,*

$$\#A \leq \delta^{-\sigma}, \quad N(A + A, \delta) \leq \delta^{-\varepsilon_d} \#A, \quad \#(A \cap I) \leq |I|^\kappa \#A \quad (\delta < |I| < \delta^{\varepsilon_{\text{nc}}}).$$

Let $\delta^{\varepsilon_{\text{nc}}} \leq \nu \leq 1$ and let λ be a probability on $[0, \nu]$ with $\lambda(I) \leq C_F(|I|/\nu)^\tau$ for $\delta \leq |I| \leq \nu$. Then some $x \in \text{spt } \lambda$ satisfies

$$N(\nu A + xA, \delta) \geq \delta^{-\varepsilon_g} \#A. \quad (\text{A.2.31})$$

Proof. Choose $D > \max\{1000, 10/\tau\}$ and apply [Theorem A.2.10](#) with the exponent $\kappa' = \kappa/2$. Its parameters T, β and its allowed nonconcentration exponent are fixed first. We may then choose $\varepsilon_{\text{nc}} > 0$ smaller than half that exponent, smaller than $1/4$, and smaller than $\frac{1}{100}(1 - \sigma) \log_2 M/T$. Choose $\varepsilon_d < 2^{-2T-2}$.

Partition $[1, 2]$ into half-open δ -cells and retain one point of A per occupied cell, discarding the possible point 2. Rounding the selected points down to the cell's left endpoint and subtracting 1 produces a δ -set $A_0 \subset [0, 1)$ with $\#A_0 \geq c\#A$, since a δ -cell contains at most two points of the separated set. Each element of A_0 has a designated original representative within δ . An interval of length $\rho \in (\delta, \delta^{\varepsilon_0})$, where ε_0 is the tree theorem's exponent, pulls back to an interval of length at most $\rho + 2\delta < \delta^{\varepsilon_{\text{nc}}}$ for small δ . Hence

$$\#(A_0 \cap I) \leq C\rho^\kappa \#A_0 \leq \rho^{\kappa'} \#A_0.$$

The last inequality is uniform in the displayed range because $\rho \leq \delta^{\varepsilon_0}$ and the fixed constant is absorbed by $\rho^{-\kappa/2}$.

Each sum in $2 + (A_0 + A_0)$ differs by less than 2δ from a point of $A + A$. Since translation does not change cardinality and a radius- δ ball meets only an absolute number of δ -lattice points,

$$\#(A_0 + A_0) \leq CN(A + A, \delta) \leq C'\delta^{-\varepsilon_d} \#A_0 \leq 2^{2-2T-1} m \#A_0$$

for sufficiently large m . Thus the grouped tree theorem applies to A_0 . For its tree, $\prod_i M_i \geq M^{\gamma m_1}$, $\gamma > (1 - \sigma)/5$, and $\#C_\emptyset \geq M^{-\gamma m_1/80} \#A_0$. The proof of that theorem defines ϑ_i so that $\delta \leq \vartheta_i \mu_i$; also $\nu \vartheta_i > \delta$ because $\vartheta_i > \delta^{1/2}$ and $\varepsilon_{\text{nc}} < 1/4$. Taking T larger if needed, all smallness requirements of [Theorem A.2.12](#) hold, so

$$\int |\nu C_\emptyset^{(\delta)} + x C_\emptyset^{(\delta)}| d\lambda(x) \geq c\nu\delta M^{(1/3-1/80)\gamma m_1} \#A_0.$$

For any fixed $x \in [0, \nu]$, first translate the rounded sum by $\nu + x$ and then replace each rounded point by its designated original representative. The displacement is at most 2δ ,

including the fixed thickening radius. Translation preserves area, so the translated set inside the area integral lies in the 3δ -neighborhood of $vA + xA$, whose measure is at most $C\delta N(vA + xA, \delta)$. It follows that

$$\int N(vA + xA, \delta) d\lambda(x) \geq c v M^{(1/3-1/80)\gamma m_1} \#A.$$

Our choice of ε_{nc} and the strict lower bound for γ make the right side at least $\delta^{-\varepsilon_g} \#A$ for some $\varepsilon_g > 0$ and all sufficiently small δ . Because the integrand is nonnegative, at least one $x \in \text{spt } \lambda$ satisfies (A.2.31). $\square$

Remark A.2.14 (The cardinality hypothesis in Bourgain's Theorem 2). *The published statement on p. 194 prints $\#A > \delta^{-\sigma}$ without an upper bound. The literal assertion is impossible: the full δ -grid in $[1, 2]$ obeys the stated interval nonconcentration for $0 < \kappa < 1$, but every $A + xA$ lies in a bounded interval and has $O(\delta^{-1}) = O(\#A)$ covering number at scale δ . Our corrected metric statement has the necessary upper bound $\#A \leq \delta^{-\sigma}$. The same correction is recorded at the earlier Volume VI.10 theorem; it is not silently attributed to the published statement.*

To finish the one-dimensional argument, two short covering-number comparisons turn the metric amplification into a coefficient from the original parameter measure.

Lemma A.2.15 (Mixed sums and four coefficients). *Let $A \subset \mathbb{R}$ be finite and δ -separated, $M = \#A$, and $0 < x \leq 1$. Then*

$$N(A + A, \delta) \leq Cx^{-3} \left(\frac{N(A + xA, \delta)}{M} \right)^6 M. \quad (\text{A.2.32})$$

Moreover, if $N(A + A, \delta) \leq LM$ and $N(A + y_j A, \delta) \leq KM$ for $j = 1, \dots, 4$, where $y_j \in [0, 1]$, then

$$N(y_1 A + y_2 A - y_3 A - y_4 A, \delta) \leq C(LK)^4 M. \quad (\text{A.2.33})$$

Proof. Round to the lattice $h\mathbb{Z}$, $h = \delta/10$. Let U be the rounded image of A and V that of xA . Then $|U| = M$, $|V| \geq cxM$, and $|U + V| \leq CN(A + xA, \delta)$. Ruzsa's triangle inequality and sum-difference inequality give

$$|U - U| \leq \frac{|U + V|^2}{|V|}, \quad |U + U| \leq \frac{|U - U|^3}{|U|^2}.$$

Since $N(A + A, \delta) \leq |U + U|$, the first claim follows.

For the second claim let V_j be the rounded image of $y_j A$. Then $|U + V_j| \leq CKM$ and $|U + U| \leq CLM$. The same Ruzsa triangle inequality, now with intermediate set $-U$, yields

$$|U - V_j| \leq \frac{|U + U| |U + V_j|}{|U|} \leq CLKM.$$

Put $B = V_1 \cup V_2 \cup (-V_3) \cup (-V_4)$. The union bound and the preceding estimates give $|U + B| \leq CLKM$. Apply the single-set Petridis–Ruzsa inequality proved in Volume II.11 to (U, B) : $|4B| \leq (CLK)^4 M$. Since $V_1 + V_2 - V_3 - V_4 \subset 4B$, this bounds the required lattice sum. Rounding each summand changes it by at most $h/2$, so these lattice sums cover the displayed metric sum at radius δ . This proves (A.2.33). $\square$

Lemma A.2.16 (Frostman difference measure at a first crossing). *Let μ be a probability on $[0, 1]$ satisfying $\mu(I) \leq |I|^\kappa$ for all intervals with $\delta < |I| < \delta^{\varepsilon_0}$. Put $\mu_1 = \mu * \tilde{\mu}$, where $\tilde{\mu}(E) = \mu(-E)$, and $\tau = \kappa/2$. For sufficiently small δ there are $v \in [\delta^{\varepsilon_0}/4, 1] \cap \text{spt } \mu_1$ and a probability λ supported on $[0, v] \cap \text{spt } \mu_1$ with*

$$\lambda(I) \leq C_\kappa(|I|/v)^\tau \quad (\delta \leq |I| \leq v). \quad (\text{A.2.34})$$

Proof. For any interval I of length ℓ , conditioning on the second variable gives $\mu_1(I) \leq \sup_b \mu([b, b + \ell])$. There is also an absolute C such that

$$\mu_1(I) \leq C\mu_1([- \ell, \ell]). \quad (\text{A.2.35})$$

Indeed, partition the line into ℓ -cells J_j and put $a_j = \mu(J_j)$. If $u - v \in I$, the indices of the cells containing u, v differ by at most five possible integers. For each offset k , $\sum_j a_j a_{j+k} \leq \sum_j a_j^2$ by Cauchy–Schwarz, and pairs within one cell give $\mu_1([- \ell, \ell]) \geq \sum_j a_j^2$.

Set $r_0 = \delta^{\varepsilon_0}/4$ and $F(r) = \mu_1([0, r])$. For small δ , $F(r_0) \leq r_0^\kappa < \frac{1}{4}r_0^\tau$, whereas $F(1) \geq 1/2$. Let

$$v = \inf\{r \in [r_0, 1] : F(r) \geq \frac{1}{4}r^\tau\}.$$

Right continuity gives $F(v) \geq v^\tau/4$. An open support gap around v would make $F(r)/r^\tau$ strictly decreasing there; thus $v \in \text{spt } \mu_1$. Define $\lambda = \mu_1|_{[0, v]}/F(v)$.

For $\delta \leq \ell < r_0$, the direct estimate and the hypothesis at scale 2ℓ if necessary give $\mu_1(I) \leq C\ell^\kappa \leq C\ell^\tau$. For $r_0 \leq \ell < v/4$, (A.2.35), symmetry, and minimality of v give $\mu_1(I) \leq 2CF(\ell) < C\ell^\tau$. For $\ell \geq v/4$ the probability bound $\lambda(I) \leq 1 \leq 4^\tau(\ell/v)^\tau$ suffices. Divide the first two estimates by $F(v) \geq v^\tau/4$ to obtain (A.2.34). $\square$

Theorem A.2.17 (Corrected one-dimensional Bourgain $A + xA$ theorem). *Fix $0 < \sigma < 1$ and $\kappa > 0$. There are $\varepsilon_0, \varepsilon_1 > 0$ such that for all sufficiently small $\delta = 2^{-m}$, every δ -separated $A \subset [1, 2]$ and probability μ on $[0, 1]$ satisfying*

$$\#A \leq \delta^{-\sigma}, \quad \#(A \cap I) \leq |I|^\kappa \#A, \quad \mu(I) \leq |I|^\kappa \quad (\delta < |I| < \delta^{\varepsilon_0})$$

have a coefficient $y \in \text{spt } \mu$ for which

$$N(A + yA, \delta) \geq \delta^{-\varepsilon_1} \#A. \quad (\text{A.2.36})$$

The same conclusion holds for all sufficiently small non-dyadic δ after changing the positive constants.

Proof. First take m to be a multiple of the block size in Theorem A.2.13. Apply that corollary with $\tau = \kappa/2$ and $C_F = C_\kappa$ from Theorem A.2.16. Denote its gain by $\varepsilon_g > 0$. Its small-doubling threshold may be decreased, so choose $0 < \varepsilon_d < \varepsilon_g/100$. Choose the present ε_0 smaller than the corollary's nonconcentration exponent and $\varepsilon_d/100$. Finally choose $0 < \varepsilon_1 < \min\{\varepsilon_d/20, \varepsilon_g/100\}$.

If $N(A + A, \delta) > \delta^{-\varepsilon_d} \#A$, the measure hypothesis at the interval $[0, \delta^{\varepsilon_0}/2]$ gives a point $y \in \text{spt } \mu$ with $y \geq \delta^{\varepsilon_0}/2$. If its mixed sum had $N(A + yA, \delta) \leq \delta^{-\varepsilon_1} \#A$, then (A.2.32) would imply $N(A + A, \delta) \leq C\delta^{-3\varepsilon_0 - 6\varepsilon_1} \#A$, contrary to the present case for small δ .

Otherwise the metric small-doubling hypothesis holds. Apply [Theorem A.2.16](#). Since ε_0 is smaller than the metric corollary's exponent, $\nu \geq \delta^{\varepsilon_0}/4$ exceeds its required lower scale for small δ ; the same hypothesis on A covers its nonconcentration range. Hence some $x \in \text{spt } \lambda \subset \text{spt } \mu_1$ satisfies

$$N(\nu A + xA, \delta) \geq \delta^{-\varepsilon_g} \#A.$$

The compact support identity $\text{spt}(\mu * \tilde{\mu}) = \text{spt } \mu - \text{spt } \mu$ provides $y_1, y_2, y_3, y_4 \in \text{spt } \mu$ such that $\nu = y_1 - y_3$ and $x = y_2 - y_4$. Consequently $\nu A + xA \subset y_1 A + y_2 A - y_3 A - y_4 A$. If [\(A.2.36\)](#) failed for every $y \in \text{spt } \mu$, [\(A.2.33\)](#) with $L = \delta^{-\varepsilon_d}$ and $K = \delta^{-\varepsilon_1}$ would bound this four-term set by $C\delta^{-4(\varepsilon_d + \varepsilon_1)} \#A$. Our parameter choices make its exponent strictly smaller than ε_g , a contradiction.

For arbitrary sufficiently small δ , first fix $\sigma < \sigma' < 1$ and $0 < \kappa' < \kappa$, and decrease the theorem's ε_0 if needed so that the dyadic conclusion just proved also applies to (σ', κ') . Choose $\delta' = 2^{-m'} \in [\delta, 2^T \delta)$ with m' a multiple of T . Choose a maximal δ' -separated subset $A' \subset A$. Because A was δ -separated, $\#A' \geq c_T \#A$. For small δ , $\#A' \leq 2^{T\sigma} \delta'^{-\sigma} \leq \delta'^{-\sigma'}$. The original interval bounds, on a slightly wider upper range chosen when ε_0 was fixed, give the required κ' -bounds for A' and μ : the factor c_T^{-1} in the A' bound is absorbed by $\rho^{\kappa' - \kappa}$ uniformly for $\delta' < \rho < \delta'^{\varepsilon'_0}$. Apply the just-proved dyadic theorem with $(\sigma', \kappa', \delta')$. Its coefficient y lies in $\text{spt } \mu$, and monotonicity of covering numbers gives

$$N(A + yA, \delta) \geq N(A' + yA', \delta') \geq c_T \delta'^{-\varepsilon'_1} \#A \geq \delta^{-\varepsilon_1} \#A$$

for any fixed $0 < \varepsilon_1 < \varepsilon'_1$ and sufficiently small δ . □

A.2.3 The higher-dimensional inverse theorems

We now give the exact higher-dimensional theorem whose proof must explain near-equality in convolution beyond the point-like and one-dimensional tree modes.

The intermediate object is a set theorem. For $E \subset \mathbb{R}^d$ and $r > 0$, write $E(r) = \{x : \text{dist}(x, E) < r\}$; for a finite set A , kA is the set of sums of k elements of A . An affine k -plane is a translate of a k -dimensional linear subspace.

Theorem A.2.18 (Higher-dimensional small-sumset structure). *Fix $d \geq 1$ and $\delta_0 > 0$. For every sufficiently large integer $L = L(d, \delta_0)$ there is $\sigma_0 = \sigma_0(d, \delta_0, L) > 0$ such that, if $0 < \sigma < \sigma_0$, S is sufficiently large in terms of d, δ_0, L, σ , $m = SL$, and a nonempty 2^{-m} -set $A \subset [0, 1)^d$ satisfies $\#(A + A) \leq 2^{\sigma m} \#A$, then a subset $A' \subset A$ and integers $k_s \in \{0, \dots, d\}$ exist with*

$$\#A' \geq 2^{-\delta_0 m} \#A. \tag{A.2.37}$$

The set A' is (L, S) -uniform, with branching numbers R_s satisfying $\log_2 R_s \geq (k_s - \delta_0)L$. For every s and every $I \in \mathcal{D}_{SL}(A')$, some affine k_s -plane P_I has

$$A' \cap I \subset P_I(2^{-(s+1)L}). \tag{A.2.38}$$

Moreover $\log_2 R_s \leq k_s L + c_d$, where c_d depends only on d .

The statement is Shmerkin's Theorem 1.5 [4]. The proof below does not invoke the literally false fixed-window formulation of the intermediate source theorem. It uses the differently quantified, fully proved Volume II.5 result G-II.5.17 with its window chosen *after* the exceptional probability. A short offset is then selected by averaging, and a finite-tree pruning returns to the original block grid. The loss of that reblocking is included in the cardinality and entropy estimates.

Proof. We may assume $0 < \delta_0 < 1/4$. All entropies below have base two. For a compactly supported probability ρ , write $H_j(\rho) = H(\rho, \mathcal{D}_j)$ and $h_L(\rho) = L^{-1}H_L(\rho)$; dyadic components $\rho^{x,i}$ are normalized as in Volume II.5. The chain rule gives, for $0 \leq r < L$ and $i_s = r + sL$,

$$H_{r+(S-1)L}(\rho) - H_r(\rho) = L \sum_{s=0}^{S-2} \int h_L(\rho^{x,i_s}) d\rho(x). \quad (\text{A.2.39})$$

This identity holds for measures outside $[0, 1)^d$ as well, since dyadic conditional entropy is translation- and rescaling-compatible within each dyadic cube.

Choose a small offset proportion $0 < \alpha < 1/10$ and an exceptional probability $e > 0$ in that order. Their exact smallness conditions are given below. Let L be sufficiently large that $L \geq M(d, e)$ in G-II.5.17, put $\zeta = 2^{-2L}$, and let $K = K(d, e, L, \zeta)$ be the finite power bound from that result. Define

$$u_L = \frac{\log_2(2(dL + 1))}{L}, \quad h = \lceil \log_2(2\sqrt{d}) \rceil.$$

Choose $\sigma_0 > 0$ so small that $K\sigma_0$ satisfies the loss inequality below, then take $0 < \sigma < \sigma_0$ and S sufficiently large. The order is

$$\delta_0 \longrightarrow (\alpha, e) \longrightarrow L \longrightarrow \zeta \longrightarrow K \longrightarrow \sigma_0 \longrightarrow S;$$

in particular the lower bound for L does not depend on ζ .

Apply Theorem A.2.2 with $V = 1$ to A . This gives an (L, S) -uniform $A_0 \subset A$ with

$$\#A_0 \geq 2^{-u_L m} \#A. \quad (\text{A.2.40})$$

Let μ be uniform on A_0 . By the finite-set Petridis–Ruzsa inequality of Volume II.11, for every $1 \leq k \leq K$,

$$\#(kA_0) \leq \#(kA) \leq 2^{k\sigma m} \#A \leq 2^{(k\sigma + u_L)m} \#A_0. \quad (\text{A.2.41})$$

Apply G-II.5.17 to μ with exceptional and saturation tolerance e , window L , and concentration tolerance ζ . It supplies $1 \leq k \leq K$, $\theta = \mu^{*k}$, and level-only subspaces $V_i \leq \mathbb{R}^d$, $0 \leq i \leq m$, for which each of the saturation and concentration failure probabilities, averaged over all these levels, is at most e . Put

$$R = \lfloor \alpha L \rfloor \geq 1.$$

The pairs (r, s) with $0 \leq r < R$ and $0 \leq s < S - 1$ give distinct levels $r + sL \in [0, m]$. Averaging the two failure probabilities over these pairs therefore selects one $r < R$ such that, with

$$\begin{aligned} p_s^\theta &= \theta\{x : \theta^{x,i_s} \text{ is not } (V_{i_s}, e, L)\text{-saturated}\}, \\ p_s^\mu &= \mu\{x : \mu^{x,i_s} \text{ is not } (V_{i_s}, \zeta)\text{-concentrated}\}, \end{aligned}$$

one has

$$\sum_{s=0}^{S-2} (p_s^\theta + p_s^\mu) \leq \frac{2e(m+1)}{R} \leq qS, \quad q := \frac{8e}{\alpha}, \quad (\text{A.2.42})$$

once L, S are large enough. This is the offset selection: the exceptional levels cannot occupy every short residue class.

For a saturated component the definition in Volume II.5 yields $h_L(\theta^{x, i_s}) \geq \dim V_{i_s} - e$; for any component the entropy is nonnegative. Set $k_s = \dim V_{i_s}$ for $s < S-1$ and $k_{S-1} = 0$. Since $H_m(\theta) \geq H_{r+(S-1)L}(\theta)$, [Equations \(A.2.39\)](#) and [\(A.2.42\)](#) imply

$$H_m(\theta) \geq L \sum_{s=0}^{S-1} k_s - (e + dq)m. \quad (\text{A.2.43})$$

The measure θ is supported on the 2^{-m} -lattice set kA_0 , so $H_m(\theta) \leq \log_2 \#(kA_0)$. Combining [Equations \(A.2.41\)](#) and [\(A.2.43\)](#) gives

$$\log_2 \#A_0 \geq L \sum_{s=0}^{S-1} k_s - \Xi m, \quad \Xi := K\sigma + u_L + e + dq. \quad (\text{A.2.44})$$

We next make the concentration information hereditary before any new uniformization. For each $s < S-1$ and each level- i_s cube J for which μ^J is (V_{i_s}, ζ) -concentrated, fix an affine translate in the normalized unit cube whose ζ -tube has μ^J -mass at least $1 - \zeta$, and denote its pullback to the original coordinates by P_J . Call $x \in A_0$ bad at level s if its component fails this concentration condition or x lies outside the corresponding original-coordinate tube $P_J^{(\zeta 2^{-i_s})}$. Conditioning on J gives

$$\mu\{x : x \text{ is bad at level } s\} \leq p_s^\mu + \zeta.$$

Choose $\beta = 4(q + \zeta) < 1/2$. Markov's inequality and [Equation \(A.2.42\)](#) show that at least half of A_0 consists of points bad at no more than βS levels among $s < S-1$. There are at most $2^{H_2(\beta)S}$ possible such bad-level patterns, where $H_2(t) = -t \log_2 t - (1-t) \log_2 (1-t)$. Indeed the binomial theorem with $x = \beta/(1-\beta)$ gives $\sum_{j \leq \beta S} \binom{S}{j} \leq x^{-\beta S} (1+x)^S = 2^{H_2(\beta)S}$. Thus a set $A_1 \subset A_0$ and a common pattern $B \subset \{0, \dots, S-2\}$ exist with

$$\#A_1 \geq \frac{1}{2} 2^{-H_2(\beta)S} \#A_0, \quad \#B \leq \beta S, \quad (\text{A.2.45})$$

such that every $x \in A_1$ lies in its selected plane tube at every $s \notin B$. Add $S-1$ to B , increasing its cardinality by one.

For each $s \notin B$, and separately inside each occupied level- sL parent of the current set, retain the most populous level- $(sL+r)$ descendant. There are at most 2^{dr} choices, so this pass retains at least 2^{-dr} of the current points. For each $s \in B$, instead retain the most populous level- $(s+1)L$ child in each level- sL parent, losing at most 2^{-dL} . Perform these finitely many passes in any fixed order; each later restriction preserves the earlier geometric properties. Finally, for each $1 \leq j < S$, inside every occupied level- jL cube retain a most populous level- $(jL+h)$ subcube. This *centering pass* loses at most 2^{-dh} per boundary

and ensures that every occupied level- jL cube of the resulting set contains its points in one subcube of relative side 2^{-h} . Uniformize the resulting set once more by [Theorem A.2.2](#) with $V = 1$, obtaining an (L, S) -uniform set A_2 . All operations are restrictions of A_1 , hence the selected pointwise tube conditions remain true. The preceding retention estimates and $r/L < \alpha$ give

$$\log_2 \#A_2 \geq \log_2 \#A_0 - (\Lambda - u_L)m, \quad (\text{A.2.46})$$

where, for $m = SL$ and S large,

$$\Lambda = 2u_L + \frac{H_2(\beta)}{L} + d\alpha + d\beta + \frac{dh}{L} + \frac{d+1}{S}. \quad (\text{A.2.47})$$

In particular [Equation \(A.2.40\)](#) gives $\#A_2 \geq 2^{-\Lambda m} \#A$.

Let R_s be the branching numbers of A_2 . At a scale $s \notin B$, all points in any level- sL parent lie in one selected level- $(sL + r)$ cube J and in the original-coordinate tube $P_J^{(\zeta 2^{-i_s})}$. Since $\zeta 2^{-i_s} < 2^{-(s+1)L}$, this is the desired plane containment with dimension k_s . Moreover an affine k_s -plane inside a cube of side 2^{-sL} meets at most $C_d 2^{k_s L}$ level- $(s+1)L$ cubes after thickening by $2^{-(s+1)L}$: cover the plane's intersection with a fixed enlargement of the parent by $O_d(2^{k_s L})$ balls of radius $2^{-(s+1)L}$, each of which meets $O_d(1)$ dyadic children. At $s \in B$, only one child per parent survived, so $R_s = 1$ and k_s is reset to zero. The centering pass puts every surviving child's points in one subcube of side $2^{-h} 2^{-(s+1)L}$ when $s < S-1$; at $s = S-1$ the child contains one lattice point. The center of that subcube, or the point itself at the last scale, is a zero-dimensional affine plane whose $2^{-(s+1)L}$ -tube contains the points, because $2^{-h} \sqrt{d}/2 \leq 1$. Consequently, with $c_d = \log_2 C_d$, at every scale,

$$\log_2 R_s \leq k_s L + c_d. \quad (\text{A.2.48})$$

Equations [\(A.2.44\)](#) and [\(A.2.46\)](#), together with $\log_2 \#A_2 = \sum_s \log_2 R_s$, imply

$$\sum_{s=0}^{S-1} (k_s L + c_d - \log_2 R_s) \leq (\Xi + \Lambda + c_d/L)m. \quad (\text{A.2.49})$$

Each summand is nonnegative by [Equation \(A.2.48\)](#). Let E be the scales with $\log_2 R_s < (k_s - \delta_0/2)L$. Then

$$\frac{\#E}{S} \leq \frac{2(\Xi + \Lambda + c_d/L)}{\delta_0}. \quad (\text{A.2.50})$$

Apply [Theorem A.2.4](#) to collapse the branching at every $s \in E$ to one child, obtaining $A' \subset A_2$. The centering property is hereditary, so these newly collapsed scales also have the exact zero-plane tube property. At every uncollapsed scale the branching remains R_s and satisfies $\log_2 R_s \geq (k_s - \delta_0/2)L$; at every collapsed scale put $k_s = 0$, for which $R_s = 1$.

To close the parameter ledger, first choose α small in terms of d, δ_0 , then e so small compared with α that $q = 8e/\alpha$ and $\beta = 4(q + 2^{-2L})$ are small. Increase L to make $u_L, dh/L, c_d/L$ small and to ensure $L \geq M(d, e)$; finally choose σ_0 after the resulting K and increase S to absorb the terms $1/S$. These choices can and do satisfy the explicit inequalities

$$\Lambda < \frac{\delta_0^2}{100d}, \quad \Xi < \frac{\delta_0^2}{100d}, \quad \frac{c_d}{L} < \frac{\delta_0^2}{100d}. \quad (\text{A.2.51})$$

By Equation (A.2.50), $\#E/S < 3\delta_0/(50d)$. Since $R_s \leq 2^{dL}$,

$$\#A' \geq 2^{-dL\#E}\#A_2 \geq 2^{-[\Lambda+d\#E/S]m}\#A \geq 2^{-\delta_0 m}\#A.$$

The uniformity, branching lower bounds and exact plane tubes just proved are all the conclusions of the theorem. $\square$

The remaining transfer begins with two finite-set reductions. They are recorded here because a near equality of L^q norms does not by itself give a useful sumset inequality: the weights must first be made nearly constant, and the resulting asymmetric energy must be converted to a small-doubling model. In this section $N(E, r)$ denotes the least number of closed Euclidean balls of radius r covering $E \subset \mathbb{R}^d$.

Lemma A.2.19 (Weighted L^q data yield asymmetric energy). *Fix $d \geq 1$, $1 < q < \infty$, and $\tau > 0$. There is $\varepsilon_* = \varepsilon_*(d, q, \tau) > 0$ such that, for $0 < \varepsilon < \varepsilon_*$ and all sufficiently large m , if μ, ν are probabilities on $2^{-m}\mathbb{Z}^d \cap [0, 1)^d$ and $\|\mu * \nu\|_q \geq 2^{-\varepsilon m}\|\mu\|_q$, there are nonempty sets $A_1 \subset \text{spt } \mu$, $B_1 \subset \text{spt } \nu$ on each of which the respective atomic masses differ by a factor at most two, and*

$$\|\mu|_{A_1}\|_q \geq 2^{-\tau m}\|\mu\|_q, \quad \nu(B_1) \geq 2^{-\tau m}, \quad E(A_1, B_1) \geq 2^{-\tau m}|A_1||B_1|^2, \quad (\text{A.2.52})$$

where $E(A, B) = \|1_A * 1_B\|_2^2$.

Proof. Write $q' = q/(q-1)$ and $D = dm$, so both supports have at most 2^D points. Partition the positive atoms into the dyadic classes

$$A_j = \{x : 2^{-j-1}\|\mu\|_q^{q'} < \mu(x) \leq 2^{-j}\|\mu\|_q^{q'}\}, \quad B_j = \{y : 2^{-j-1}2^{-D} < \nu(y) \leq 2^{-j}2^{-D}\}.$$

Hölder's inequality gives $\|\mu\|_q^{q'} \geq 2^{-D}$; consequently a nonempty class in either family has $j \geq -D - 1$. Let $J = \lceil 4\varepsilon q'm \rceil$ and put $E_\mu = \bigcup_{j \geq J} A_j$ and $E_\nu = \bigcup_{j \geq J} B_j$. Directly,

$$\|\mu|_{E_\mu}\|_q^q \leq (2^{-J}\|\mu\|_q^{q'})^{q-1} \mu(E_\mu) \leq 2^{-J(q-1)}\|\mu\|_q^q, \quad \nu(E_\nu) \leq 2^D 2^{-J} 2^{-D} = 2^{-J}.$$

Young's inequality bounds each convolution term involving one of these tails by $2^{-4\varepsilon m}\|\mu\|_q$. Since the original convolution norm is at least $2^{-\varepsilon m}\|\mu\|_q$, for large m the sum of terms with $-D - 1 \leq j, j' < J$ has norm at least half the original norm. There are at most $(D + J + 2)^2$ such pairs. The triangle inequality therefore gives one pair, denoted A_1, B_1 , for which

$$\|\mu|_{A_1} * \nu|_{B_1}\|_q \geq [2(D + J + 2)^2]^{-1} 2^{-\varepsilon m}\|\mu\|_q \geq 2^{-2\varepsilon m}\|\mu\|_q. \quad (\text{A.2.53})$$

Young's inequality bounds the left side by $\|\mu|_{A_1}\|_q \nu(B_1)$. Since both factors are at most $\|\mu\|_q$ and 1, respectively, (A.2.53) forces $\|\mu|_{A_1}\|_q / \|\mu\|_q$ and $\nu(B_1)$ each to be at least $2^{-2\varepsilon m}$.

If a, b are the largest atomic masses on A_1, B_1 , then their smallest masses exceed $a/2, b/2$ respectively. Pointwise $\mu|_{A_1} * \nu|_{B_1} \leq ab(1_{A_1} * 1_{B_1})$, while $\|\mu\|_q \geq (a/2)|A_1|^{1/q}$ and $b|B_1| \leq 2\nu(B_1) \leq 2$; applying the one-pair bound and absorbing the fixed factor into $2^{\varepsilon m}$ yields

$$\|1_{A_1} * 1_{B_1}\|_q \geq 2^{-3\varepsilon m}|A_1|^{1/q}|B_1|. \quad (\text{A.2.54})$$

For completeness, put $f = 1_{A_1} * 1_{B_1}$. If $1 < q < 2$, Hölder gives $\sum f^q \leq (\sum f)^{2-q} (\sum f^2)^{q-1}$; here $\sum f = |A_1||B_1|$. If $q \geq 2$, then $\sum f^q \leq (\sum f^2) \|f\|_\infty^{q-2}$ and $\|f\|_\infty \leq |B_1|$. In both cases

$$E(A_1, B_1) \geq 2^{-3 \max(q, q') \varepsilon m} |A_1||B_1|^2.$$

Choose ε_* so that $3 \max(q, q') \varepsilon_* < \tau$; the displayed bounds imply (A.2.52). $\square$

Lemma A.2.20 (Bounded asymmetric BSG model). *For every $d \geq 1$ and $\kappa > 0$ there are $\tau > 0$ and $M = M(d, \kappa) < \infty$ such that, for all sufficiently large m , if nonempty 2^{-m} -sets $A_1, B_1 \subset [0, 1]^d$ obey $E(A_1, B_1) \geq 2^{-\tau m} |A_1||B_1|^2$, then there are finite $H_0, X \subset 2^{-m} \mathbb{Z}^d$ with*

$$|H_0 + H_0| \leq 2^{\kappa m} |H_0|, \quad |B_1 \cap H_0| \geq 2^{-\kappa m} |B_1|, \quad (\text{A.2.55})$$

$$|A_1 \cap (X + H_0)| \geq 2^{-\kappa m} |A_1|, \quad |X||H_0| \leq 2^{\kappa m} |A_1|, \quad (\text{A.2.56})$$

and $H_0 \subset [-M, M]^d$.

Proof. Use the fully proved imbalanced-energy proposition of Volume II.11 in the ambient abelian group $2^{-m} \mathbb{Z}^d$. In its notation take $L_0 = 2^{\tilde{d}m} \geq |A_1|/|B_1|$ and first choose its free exponent $\eta < \kappa/(100d)$. Its factors L_0^η then cost at most $2^{\kappa m/100}$, and every fixed power of $1 + \log |B_1|$ is $2^{o(m)}$. Take $\alpha = 2^{-\tau m-1}$ and then τ so small that all fixed powers of α^{-1} from that proposition cost less than $2^{\kappa m/4}$. For sufficiently large m the four conclusions of the proposition therefore imply (A.2.55)–(A.2.56), after translating its H and counter-translating X .

The bounded-support assertion follows from the construction proved there, rather than from an unmentioned compactness assumption. Its first difference set is contained in $[-1, 1]^d$ and each subsequent difference step at most doubles the bounding cube. The number $J = J(\eta)$ of steps is fixed before m , so its H lies in $[-2^J, 2^J]^d$. The translation placing a point of H in $B_1 \subset [0, 1]^d$ has coordinates bounded by $2^J + 1$. Thus the translated H_0 lies in a cube $[-M, M]^d$ with $M = 2^{J+1} + 2$, independent of m . $\square$

The geometric step in the transfer is a product count. A plane supporting one factor supplies k -dimensional branching; an orthogonal projection of the other factor supplies the complementary branching. Their sum separates both kinds of children at once.

Lemma A.2.21 (Multiscale sumset product). *Let $H, Y \subset [0, 1]^d$ be (L, S) -uniform 2^{-SL} -sets. Suppose H has branching numbers R_s and, for each $J \in \mathcal{D}_{sL}(H)$, an affine k_s -plane P_J satisfies $H \cap J \subset P_J(2^{-(s+1)L})$. For each $I \in \mathcal{D}_{sL}(Y)$ let*

$$F_s(Y_I^{(L)}) = \left[\log_2 \inf_{W \in \text{Gr}(d, d-k_s)} N(\pi_W(Y_I^{(L)}), 2^{-L}) \right].$$

Suppose this integer has the same value M_s for all occupied level- sL parents I of Y . Then, for a constant $c_d > 0$,

$$|Y + H| \geq c_d^S \prod_{s=0}^{S-1} R_s 2^{M_s}. \quad (\text{A.2.57})$$

Furthermore, if R_s^Y denotes the branching number of Y , then

$$\log_2 R_s^Y \leq k_s L + M_s + C_d \quad (\text{A.2.58})$$

for a constant C_d depending only on d .

Proof. Fix a block s , an occupied parent I of Y , and an occupied parent J of H . Set $r = 2^{-L}$ and normalize both parents to $[0, 1]^d$. Let U and Z be the lower-left corners of their occupied r -children. Thus $|Z| = R_s$, and because each child contains a point of H lying within r of P_J , all of Z lies within $(1 + \sqrt{d})r$ of the normalized plane. Write V for its direction and $W = V^\perp$.

The definition of M_s implies $N(\pi_W U, r) \geq 2^{M_s-1}$. By the doubling bound for Euclidean balls, a maximal $C_d r$ -separated subset of $\pi_W U$ has at least $c_d 2^{M_s}$ members, after enlarging the fixed constant C_d . Likewise the projection $\pi_V Z$ contains a $C_d r$ -separated subset of at least $c_d R_s$ members: every $C_d r$ -ball in V , intersected with the fixed $(1 + \sqrt{d})r$ tube about the plane, contains at most $O_d(1)$ grid corners of spacing r . Choose corresponding representatives u_i, z_j . For distinct i, i' the W -components of $(u_i + z_j) - (u_{i'} + z_{j'})$ are separated by at least $C_d r - 2(1 + \sqrt{d})r$; for equal i and distinct j, j' the V -components are separated by $C_d r$. Choose C_d large enough that all $u_i + z_j$ are separated by more than $10\sqrt{d}r$. Therefore the sum boxes $I'_i + J'_j$ of the corresponding child cubes are pairwise disjoint, and there are at least $c_d R_s 2^{M_s}$ such pairs.

Begin with the one root pair and perform this selection at each block inside every previously selected pair of occupied parents. The child sum boxes are contained in their parent sum box; they are disjoint inside each parent by the preceding argument and across different parents by induction. At level SL each selected pair contains one point from Y and one from H , and the disjoint boxes force distinct sums. Multiplying the uniform lower bound over all S blocks proves (A.2.57).

For the upper bound, choose $W \in \text{Gr}(d, d - k_s)$ with $N(\pi_W U, r) \leq 2^{M_s+1}$; such a W exists by the definition of the infimum and the ceiling. Over each radius- r ball of its projected cover, the grid corners U lie in a radius- r tube about an affine k_s -plane. Intersecting that tube with $[0, 1]^d$ and covering its plane direction by $O_d(r^{-k_s})$ balls of radius r shows that it contains at most $C_d r^{-k_s}$ corners. Hence $R_s^Y = |U| \leq C_d 2^{k_s L + M_s}$, as asserted. $\square$

Theorem A.2.22 (Shmerkin's inverse L^q theorem). *Fix $d \geq 1$, $q \in (1, \infty)$ and $\delta_0 > 0$. For every sufficiently large integer $L = L(d, \delta_0)$ there is $\varepsilon_0 = \varepsilon_0(d, q, L) > 0$ such that, for $0 < \varepsilon < \varepsilon_0$ and every sufficiently large $S = S(d, q, \delta_0, L, \varepsilon)$, the following holds. Put $m = SL$. If μ, ν are probabilities supported on $2^{-m}\mathbb{Z}^d \cap [0, 1]^d$ and*

$$\|\mu * \nu\|_q \geq 2^{-\varepsilon m} \|\mu\|_q, \quad (\text{A.2.59})$$

then there are nonempty 2^{-m} -sets $A \subset \text{spt } \mu$ and $B \subset \text{spt } \nu$, and integers $k_s \in \{0, \dots, d\}$, $0 \leq s < S$, such that:

$$\begin{aligned} \|\mu|_A\|_q &\geq 2^{-\delta_0 m} \|\mu\|_q, & \max_{x, y \in A} \frac{\mu(\{x\})}{\mu(\{y\})} &\leq 2, \\ \nu(B) &\geq 2^{-\delta_0 m}, & \max_{x, y \in B} \frac{\nu(\{x\})}{\nu(\{y\})} &\leq 2. \end{aligned}$$

Both A and B are (L, S) -uniform. Let R_s be the branching number of A at level s . For every $I \in \mathcal{D}_{SL}(A)$ there is $W_I \in \text{Gr}(d, d - k_s)$ such that

$$R_s \geq 2^{(k_s - \delta_0)L} \# \mathcal{D}_{(s+1)L}(\pi_{W_I}(A \cap I)). \quad (\text{A.2.60})$$

For every $J \in \mathcal{D}_{sL}(B)$ there is an affine k_s -plane V_J such that every point of $B \cap J$ lies in a level- $(s+1)L$ dyadic cube meeting V_J . When $A \cup B \subset [1/3, 2/3]^d$, the sets may additionally be translated by vectors in $[-1/3, 1/3]^d$ so that their points stay in the central third of every ancestral level- sL cube.

The source is Shmerkin, *Inverse theorems for discretized sums and L^q norms of convolutions in $\mathbb{R}^d$* , arXiv:2308.09846v3 (23 September 2024), Theorem 1.2. The proof below uses the internally proved imbalanced-energy proposition of Volume II.11, rather than importing the source's citation to an external asymmetric BSG theorem. It also uses our separately proved [Theorem A.2.18](#); hence the false literal fixed-window source formulation is not a premise.

Proof. We spell out the parameter dependencies before doing the two dimension balances. Replacing δ_0 by a smaller number if necessary, suppose $0 < \delta_0 < 1/4$, and put $\rho = \delta_0/4$. We first construct the sets with error ρ ; the remaining margin is reserved for the optional central-cell conclusion. Choose a small $\gamma > 0$, then choose L sufficiently large for [Theorem A.2.18](#) with tolerance γ . Enlarge L again so that every fixed-dimensional constant divided by L and

$$a_L := \frac{\log_2[2(dL+1)(dL+C_d+2)]}{L} \quad (\text{A.2.61})$$

are small compared with ρ^2/d . We also require

$$\omega_L \left(1 + \frac{4d}{\delta_0}\right) < \frac{\delta_0}{4}, \quad \omega_L := \frac{2d + \log_2[2(dL+1)]}{L},$$

for the final central-cell refinement. Let $\sigma_0 = \sigma_0(d, \gamma, L)$ be the small-doubling threshold in [Theorem A.2.18](#). Choose $\kappa > 0$ so small that $2\kappa < \sigma_0$ and $\kappa \ll \rho^2/d$. Use [Theorem A.2.20](#) with this κ to obtain τ , shrinking τ so that $\tau < \rho/20$. Finally choose $\varepsilon_0^{(\delta_0)} \leq \varepsilon_*(d, q, \tau)$ from [Theorem A.2.19](#), and for this direct argument assume $0 < \varepsilon < \varepsilon_0^{(\delta_0)}$. The requirements on S are only that it be large enough for those results and to absorb fixed constants into $2^{o(m)}$. The chain is

$$\delta_0 \longrightarrow \gamma \longrightarrow L \longrightarrow \sigma_0 \longrightarrow \kappa \longrightarrow \tau \longrightarrow \varepsilon_0^{(\delta_0)} \longrightarrow S.$$

Apply [Theorems A.2.19](#) and [A.2.20](#). We obtain A_1, B_1, H_0, X with [\(A.2.52\)](#) and [\(A.2.55\)–\(A.2.56\)](#). The set H_0 may not be contained in $[0, 1]^d$; this small localization point matters. Partition its fixed bounding cube $[-M, M]^d$ into at most $Q = Q(d, \kappa)$ integer translates of $[0, 1]^d$. Some cell $n + [0, 1]^d$, $n \in \mathbb{Z}^d$, contains a subset H'_0 with $|H'_0| \geq Q^{-1}|H_0|$. Thus

$$|(H'_0 - n) + (H'_0 - n)| \leq |H_0 + H_0| \leq Q2^{\kappa m}|H'_0| \leq 2^{2\kappa m}|H'_0|$$

for large m . Apply [Theorem A.2.18](#) to $H'_0 - n$, and denote its uniform output by $H \subset H'_0 - n$. Writing R_s^H for its branching numbers and ℓ_s for its plane dimensions, we have

$$|H| \geq 2^{-2\gamma m}|H_0|, \quad (\ell_s - \gamma)L \leq \log_2 R_s^H \leq \ell_s L + c_d, \quad (\text{A.2.62})$$

and in each level- sL parent the points of H lie within $2^{-(s+1)L}$ of an affine ℓ_s -plane. The first inequality absorbs the fixed factor Q^{-1} by taking S large. Note that $H + n \subset H_0$; consequently every subset upper bound below remains valid even though we have translated the localized set.

Put $A_1^* = A_1 \cap (X + H_0)$ and $B_1^* = B_1 \cap H_0$. We do not translate or center them now: keeping the original lattice grid is essential for the two sumset comparisons. The optional central-cell refinement will be made only after both balances.

For a 2^{-L} -set Y and each s , define

$$F_s(Y) = \left\lceil \log_2 \inf_{W \in \text{Gr}(d, d - \ell_s)} N(\pi_W Y, 2^{-L}) \right\rceil.$$

It takes at most $dL + C_d + 2$ values. Apply [Theorem A.2.2](#) with these statistics to the retained A_1^* , obtaining an (L, S) -uniform set A_2 with $F_s((A_2)_I^{(L)}) = M_s^A$ independent of I . Do the same for the retained B_1^* , obtaining B_2 and M_s^B . With $\lambda = \kappa + a_L$ the two retention estimates are

$$|A_2| \geq 2^{-\lambda m} |A_1|, \quad |B_2| \geq 2^{-\lambda m} |B_1|. \quad (\text{A.2.63})$$

Both sets still lie in $[0, 1)^d$ and on the original grid; no translations have been applied.

We first balance the dimensions for A_2 . Since $A_2 \subset X + H_0$ and $H + n \subset H_0$,

$$|A_2 + H| \leq |X| |H_0 + H_0| \leq 2^{2\kappa m} |A_1| \leq 2^{(2\kappa + \lambda)m} |A_2|. \quad (\text{A.2.64})$$

By [Theorem A.2.21](#) and [Equation \(A.2.62\)](#), if R_s^A are the branching numbers of A_2 , then

$$\sum_s (\ell_s L + M_s^A - \log_2 R_s^A) \leq (2\kappa + \lambda + \gamma + C_d/L)m. \quad (\text{A.2.65})$$

The individual summands need not be nonnegative. The local upper bound [\(A.2.58\)](#) supplies a fixed C_d for which

$$D_s^A := \ell_s L + M_s^A + C_d - \log_2 R_s^A \geq 0, \quad \sum_s D_s^A \leq D_A m,$$

where we may take

$$D_A = 2\kappa + \lambda + \gamma + 2C_d/L, \quad D_B = \kappa + 2\gamma + C_d/L. \quad (\text{A.2.66})$$

Let $E = \{s : D_s^A > \rho L/4\}$. Then

$$|E|/S \leq 4D_A/\rho. \quad (\text{A.2.67})$$

Use [Theorem A.2.4](#) to reduce A_2 to one occupied child at every scale in E , calling the result A . Thus

$$|A| \geq 2^{-dL|E|} |A_2| \geq 2^{-(\lambda + 4dD_A/\rho)m} |A_1|. \quad (\text{A.2.68})$$

At $s \notin E$, the branching number of A is R_s^A . For each occupied parent I choose $W_I \in \text{Gr}(d, d - \ell_s)$ whose projected child-corner set has covering number at most $2^{M_s^A + 1}$. A radius- 2^{-L} ball in the normalized projection meets at most C_d ambient dyadic projected cells; replacing child corners by the points of $A \cap I$ changes the covering radius by at most $\sqrt{d} 2^{-L}$. Hence

$$\#\mathcal{D}_{(s+1)L}(\pi_{W_I}(A \cap I)) \leq C_d 2^{M_s^A}. \quad (\text{A.2.69})$$

Since $D_s^A \leq \rho L/4$, taking L large enough that $3\rho L/4 \geq \log_2 C_d$ yields

$$R_s^A \geq 2^{(\ell_s - \rho)L} \#\mathcal{D}_{(s+1)L}(\pi_{W_I}(A \cap I)).$$

At $s \in E$ put the final $k_s = 0$ and choose $W_I = \mathbb{R}^d$: there is only one occupied child, so the projected count is one and (A4) holds. At $s \notin E$ put $k_s = \ell_s$.

Now perform the complementary balance for B_2 , still using the *original* ℓ_s in the statistic M_s^B . Since $B_2 \subset H_0$ and $H + n \subset H_0$,

$$|B_2 + H| \leq |H_0 + H_0| \leq 2^{(\kappa+2\gamma)m}|H|. \quad (\text{A.2.70})$$

The product lower bound gives $|B_2 + H| \geq c_d^S |H| \prod_s 2^{M_s^B}$, because $|H| = \prod_s R_s^H$. Therefore

$$\sum_s M_s^B \leq D_B m. \quad (\text{A.2.71})$$

For each level- sL parent J of B_2 choose $W_J \in \text{Gr}(d, d - \ell_s)$ with $N(\pi_{W_J}(B_2)_J^{(L)}, 2^{-L}) \leq 2^{M_s^B+1}$. Cover every ball of that radius by at most C_d balls of radius $2^{-L}/8$. One of the resulting at most $C_d 2^{M_s^B}$ small balls contains at least $R_s^B / (C_d 2^{M_s^B})$ occupied child corners. The projections of the centers of those child cubes differ by at most $2^{-L}/4$. Since the projection of a half-open cube of side 2^{-L} contains the open radius- $2^{-L}/3$ ball about the projection of its center, a common point of W_J belongs to the projections of *all* selected child cubes. The affine ℓ_s -plane orthogonal to W_J through that point therefore meets every one of them.

Choose in each parent exactly

$$R_s^{B,\text{new}} = \max \left(1, \left\lfloor \frac{R_s^B}{C_d 2^{M_s^B}} \right\rfloor \right)$$

of these children, with the same count for all parents at scale s . The ratio $R_s^{B,\text{new}}/R_s^B$ is at least $(2C_d 2^{M_s^B})^{-1}$. The prescribed pruning lemma yields a uniform $B_3 \subset B_2$ for which the ℓ_s -plane incidence holds at every scale and

$$|B_3| \geq 2^{-D_B m} (2C_d)^{-S} |B_2|. \quad (\text{A.2.72})$$

Finally collapse B_3 to one child at each $s \in E$, obtaining $B \subset B_3$. These collapses cost at most $2^{dL|E|}$. On $s \notin E$, the chosen ℓ_s -plane still meets every occupied child cube. On $s \in E$, its one surviving child cube meets an affine 0-plane (choose any point of the cube). Thus (B4) uses the same final k_s as (A4), and

$$|B| \geq 2^{-[\lambda + D_B + C_d/L + 4dD_A/\rho]m} |B_1|. \quad (\text{A.2.73})$$

The masses on A_1 and on B_1 were each within a factor two. Consequently (A.2.52), (A.2.68), and (A.2.73) give

$$\begin{aligned} \|\mu|_A\|_q &\geq \frac{1}{2} 2^{-[\tau + (\lambda + 4dD_A/\rho)/q]m} \|\mu\|_q, \\ v(B) &\geq \frac{1}{2} 2^{-[\tau + \lambda + D_B + C_d/L + 4dD_A/\rho]m}. \end{aligned}$$

By taking γ small, then L large, then κ and τ small in the stated order, both exponents including the fixed factor $1/2$ are less than ρ for sufficiently large S . The same parameter choices ensure D_A is small enough for (A.2.67); there is no circular dependence on ε . The

near-constant mass conditions (A2), (B2) are hereditary, and uniformity (A3), (B3) follows from the two prunings. Thus all conclusions except the optional (C1) already hold with ρ in place of δ_0 .

It remains to prove (C1) with its exact conditional hypothesis. If $A \cup B$ is not contained in $[1/3, 2/3]^d$, there is nothing to show. Otherwise apply [Theorem A.2.5](#) separately to these *already constructed* sets. It gives subsets $A_c \subset A$, $B_c \subset B$ of cardinalities at least $4^{-dS}|A|$, $4^{-dS}|B|$ and translation vectors y_A, y_B that satisfy (C1) on those subsets. Do not translate the points while making further selections. Uniformize A_c, B_c by [Theorem A.2.2](#) with $V = 1$, obtaining uniform $\tilde{A}, \tilde{B}$ and

$$|\tilde{A}| \geq 2^{-\omega_L m} |A|, \quad |\tilde{B}| \geq 2^{-\omega_L m} |B|. \quad (\text{A.2.74})$$

Let $\tilde{R}_s^A, R_s^A$ be the branch numbers of $\tilde{A}, A$. Since $\tilde{A} \subset A$, $\tilde{R}_s^A \leq R_s^A$ for every s ; using the product formula for uniform trees and (A.2.74) gives

$$\sum_s \log_2 \frac{R_s^A}{\tilde{R}_s^A} = \log_2 \frac{|A|}{|\tilde{A}|} \leq \omega_L m.$$

Define E_c by $\log_2(R_s^A/\tilde{R}_s^A) > \delta_0 L/4$. Then $|E_c|/S \leq 4\omega_L/\delta_0$. Collapse both $\tilde{A}$ and $\tilde{B}$ to one child at every scale in E_c by [Theorem A.2.4](#), and reset the common k_s to zero there. Each collapse costs at most $2^{dL|E_c|}$, so each final set has at least

$$2^{-\omega_L(1+4d/\delta_0)m}$$

times the cardinality of its corresponding pre-centering set. Their L^q norm and mass therefore still exceed $2^{-\delta_0 m} \|\mu\|_q$ and $2^{-\delta_0 m}$, respectively, by the already obtained ρ -bounds and the chosen inequality for ω_L .

At $s \notin E_c$, the original (A4) inequality with error ρ and the loss bound $\tilde{R}_s^A \geq 2^{-\delta_0 L/4} R_s^A$ give (A4) with error $\rho + \delta_0/4 < \delta_0$; the projected child count only decreases on restriction. At $s \in E_c$, one child and $k_s = 0$ make (A4) immediate. The original (B4) plane incidence is hereditary at uncollapsed scales, and a single child meets a 0-plane at a collapsed scale. The near-constant mass conditions are hereditary, and the translations y_A, y_B continue to satisfy (C1) because all final points are subsets of A_c, B_c . Relabel these final sets as A, B . This proves the optional conclusion for a temporarily fixed δ_0 , with positive admissible bound $\varepsilon_0^{(\delta_0)}$.

To obtain the stated dependence on L alone, choose dyadic target accuracies $\delta_j = 2^{-j}$, $j \geq 2$, and make their block thresholds L_j increasing with j and at least j . For each fixed integer L , only finitely many j satisfy $L \geq L_j$; take $\varepsilon_0(d, q, L)$ to be the minimum of the positive bounds $\varepsilon_0^{(\delta_j)}(d, q, L)$ just proved for those finitely many j (and any positive number if the set is empty). Given an arbitrary $\delta_0 > 0$, choose $j \geq 2$ with $\delta_j \leq \delta_0$ and require $L \geq L_j$. The theorem at accuracy δ_j has stronger mass and projection conclusions than the assertions at accuracy δ_0 , while its hypotheses are unchanged. Its admissible ε -range contains every $0 < \varepsilon < \varepsilon_0(d, q, L)$. This establishes the L -only threshold and completes the proof. $\square$

Chapter A.3

He–Bourgain projections of arbitrary rank

Volume III.6 proves a rank-one discretized projection theorem and the rank-one He regularity and slicing lemmas. When the target plane has rank $m > 1$, the relevant transversality is against every complementary $(n-m)$ -plane, not merely against hyperplanes described by one normal vector. The chapter moves through three connected tasks. Schubert-angle estimates transfer nonconcentration to the directions seen by slices. A matrix sum–product argument supplies the half-dimensional base case; its algebraic and semialgebraic steps are proved before they are used in the rank induction. Finally the finite-set theorem is passed to analytic sets by a compactness and Frostman-bin argument. The main statement is given early for orientation, but each implication is proved at its point of use below.

A.3.1 Schubert angles and transverse direction measures

Definition A.3.1. Let $1 \leq m < n$. The Grassmannian $\text{Gr}(n, m)$ is the set of m -dimensional linear subspaces of $\mathbb{R}^n$, equipped with the metric $d(V, V') = \|\pi_V - \pi_{V'}\|_{\text{op}}$. For $V \in \text{Gr}(n, m)$ and $W \in \text{Gr}(n, n-m)$, choose orthonormal bases $v_1, \dots, v_m$ and $w_1, \dots, w_{n-m}$ and define

$$\Delta(V, W) = |\det(v_1, \dots, v_m, w_1, \dots, w_{n-m})|. \quad (\text{A.3.1})$$

The absolute value makes this independent of the basis choices. The same notation $\Delta(V, W)$ is used for every complementary pair of dimensions $r, n-r$ when the working value of r changes. The ρ -Schubert tube about W in the currently specified Grassmannian $\text{Gr}(n, r)$ is $\mathcal{S}(W, \rho) = \{V \in \text{Gr}(n, r) : \Delta(V, W) \leq \rho\}$. For a bounded subset E of a Euclidean space, $\mathcal{N}_\delta(E)$ is the least number of closed radius- δ balls needed to cover E .

Lemma A.3.2 (The determinant angle is a projection Jacobian). *The number $\Delta(V, W)$ is the product of the singular values of $\pi_{W^\perp}|_V : V \rightarrow W^\perp$. In particular, $\Delta(V, W) = 0$ if and only if $V \cap W \neq \{0\}$. Moreover, for fixed W the map $V \mapsto \Delta(V, W)$ is continuous.*

Proof. Complete the orthonormal basis $w_1, \dots, w_{n-m}$ of W by an orthonormal basis $u_1, \dots, u_m$ of $W^\perp$. In these coordinates the determinant in (A.3.1) has block form

$$\det \begin{pmatrix} U^\top Q & 0 \\ W^\top Q & I_{n-m} \end{pmatrix} = \det(U^\top Q),$$

where the columns of Q, U, W are respectively the v_i, u_i, w_i . The matrix $U^T Q$ represents $\pi_{W^\perp}|_V$ in orthonormal coordinates. Its determinant modulus is the product of its singular values. Its kernel is $V \cap W$, giving the zero criterion. Locally one can choose continuous orthonormal frames on the Grassmannian; the absolute determinant is frame-independent, hence continuous globally. $\square$

The zero locus of $\Delta(\cdot, W)$ is a Schubert variety. It is generally *not* a smooth hypersurface: planes with intersection with W of dimension at least two are singular points. The following direct parameterization supplies the covering estimate needed for Frostman measures without using smoothness of that zero locus.

Lemma A.3.3 (Uniform covering of the zero-angle locus). *Put $D = m(n - m)$ and $Z(W) = \{V \in \text{Gr}(n, m) : V \cap W \neq \{0\}\}$. There is $C = C(n, m)$ such that, for $W \in \text{Gr}(n, n - m)$ and $0 < r < 1$, $Z(W)$ can be covered by at most $Cr^{-(D-1)}$ Grassmannian balls of radius r . The bound is uniform in W .*

Proof. Set $q = n - m$. A plane $V \in Z(W)$ contains a line $\ell \subset W$. The lines of W form the real projective space $\mathbb{P}(W)$ of dimension $q - 1$; it is covered by finitely many coordinate charts obtained by fixing a nonzero coordinate of a unit representative. For a fixed line ℓ , every m -plane containing it is uniquely $\ell \oplus E$ with $E \in \text{Gr}(\ell^\perp, m - 1)$. By the graph-chart calculation of Volume II.7, L-II.7.25, this Grassmannian has dimension $(m - 1)(n - m) = (m - 1)q$ and has a finite Lipschitz-chart cover at each fixed scale. Over a small projective chart, a smooth orthogonal change of coordinates identifies the varying spaces $\ell^\perp$ with one fixed $(n - 1)$ -space. The map $(\ell, E) \mapsto \ell \oplus E$ is smooth on the resulting compact coordinate pieces, hence Lipschitz there.

Thus $Z(W)$ is the image of finitely many bounded parameter pieces of dimension $(q - 1) + (m - 1)q = mq - 1 = D - 1$ under maps with bounded Lipschitz constants. Each piece is covered by $O_{n,m}(r^{-(D-1)})$ parameter balls of radius comparable to r , and their images give the asserted cover. A rotation taking one W to another preserves the Grassmannian metric, so the constant is uniform. This argument also covers $m = 1$ or $q = 1$, when the appropriate zero-dimensional factor consists of finitely many points. $\square$

Proposition A.3.4 (From determinant tubes to ordinary Frostman balls). *Let $D = m(n - m)$. There is $C = C(n, m)$ such that*

$$\mathcal{S}(W, \rho) \subset \{V : d(V, Z(W)) \leq C\rho^{1/m}\} \quad (0 < \rho < 1/2). \quad (\text{A.3.2})$$

Consequently, if a finite measure σ on $\text{Gr}(n, m)$ satisfies $\sigma(B(V, r)) \leq Mr^t$ for every V and $0 < r < 1$, where $t > D - 1$, then

$$\sigma(\mathcal{S}(W, \rho)) \leq C_{n,m,t} M \rho^{(t-D+1)/m} \quad (\text{A.3.3})$$

uniformly in W and $0 < \rho < 1/2$.

Proof. Let $V \in \mathcal{S}(W, \rho)$. By [Theorem A.3.2](#), the product of the m singular values of $\pi_{W^\perp}|_V$ is at most ρ . Each singular value is at most one, so the least is at most $\rho^{1/m}$. Choose a unit $v \in V$ for which $\|\pi_{W^\perp} v\| \leq \rho^{1/m}$. If $\rho^{1/m} \leq 1/2$, the unit vector $w = \pi_W v / \|\pi_W v\|$ belongs to W and satisfies $\|v - w\| \leq C\rho^{1/m}$. An orthogonal transformation T rotating v to w in

their two-dimensional span and fixing its orthogonal complement has $\|T - I\|_{\text{op}} \leq C\rho^{1/m}$. The plane TV belongs to $Z(W)$, because it contains w , and

$$d(V, TV) = \|\pi_V - T\pi_V T^{-1}\|_{\text{op}} \leq 2\|T - I\|_{\text{op}} \leq C\rho^{1/m}.$$

For the remaining bounded range of ρ , enlarge C and choose any element of the nonempty compact set $Z(W)$. This proves (A.3.2).

Put $r = C\rho^{1/m}$, decreasing the range of ρ if necessary so $r < 1$. Cover $Z(W)$ by at most $C'r^{-(D-1)}$ radius- r balls using Theorem A.3.3. Their radius- $2r$ enlargements cover $\mathcal{S}(W, \rho)$ by (A.3.2). The Frostman bound therefore gives $\sigma(\mathcal{S}(W, \rho)) \leq C'Mr^{t-D+1}$, which is (A.3.3). Enlarging the constant handles the bounded remaining range. $\square$

He repeatedly passes from a distribution of m -planes in $\mathbb{R}^n$ to a distribution of their images inside a fixed lower-dimensional slice. The following lemma proves that Schubert nonconcentration survives that operation. For complementary planes L, U inside a Euclidean subspace W , write $\Delta_W(L, U)$ for the determinant angle computed in W , and let $\mathcal{S}_W(U, \rho)$ be its sublevel set. For $V \in \text{Gr}(n, m)$, write $J_W(V)$ for the product of the m singular values of $\pi_W|_V : V \rightarrow W$, with $J_W(V) = 0$ when this map is not injective.

Lemma A.3.5 (Schubert nonconcentration under restriction to a slice). *Let $m \leq n' < n$, $W \in \text{Gr}(n, n')$, $\kappa > \varepsilon > 0$, and $0 < \delta < 1$. When $m = n'$, use $\text{Gr}(W, 0) = \{\{0\}\}$ and $\Delta_W(W, \{0\}) = 1$; the induced Schubert condition is then vacuous below radius 1. Suppose a probability σ on $\text{Gr}(n, m)$ satisfies*

$$\sigma(\mathcal{S}(T, \rho)) \leq \delta^{-\varepsilon} \rho^\kappa \quad (T \in \text{Gr}(n, n-m), \rho \geq \delta).$$

Put $G = \{V \in \text{Gr}(n, m) : \dim(\pi_W V) = m\}$. Then $\sigma\{V : J_W(V) \leq \rho\} \leq \delta^{-\varepsilon} \rho^\kappa$ for every $\rho \geq \delta$, and in particular $\sigma(G) \geq 1 - \delta^{\kappa-\varepsilon}$. For sufficiently small δ , the conditional pushforward $\sigma_W = (V \mapsto \pi_W V)_\#(\sigma|_G/\sigma(G))$ satisfies

$$\sigma_W(\mathcal{S}_W(U, \rho)) \leq \delta^{-2\varepsilon} \rho^\kappa \quad (U \in \text{Gr}(W, n' - m), \rho \geq \delta). \quad (\text{A.3.4})$$

Proof. Choose a fixed $(n-m)$ -plane T_0 containing $W^\perp$. Then $T_0^\perp \subset W$, so $\pi_{T_0^\perp}|_V$ factors through $\pi_W|_V$. The m -Jacobian of the latter is $J_W(V)$, while the intervening projection has operator norm at most one. By Theorem A.3.2, $\Delta(V, T_0) \leq J_W(V)$. Therefore

$$\sigma\{V : J_W(V) \leq \rho\} \leq \sigma(\mathcal{S}(T_0, \rho)) \leq \delta^{-\varepsilon} \rho^\kappa.$$

Taking $\rho = \delta$ bounds $G^c = \{J_W = 0\}$ as asserted.

Fix $U \in \text{Gr}(W, n' - m)$ and set $T = U \oplus W^\perp \in \text{Gr}(n, n-m)$. For $V \in G$, the linear map $A = \pi_W|_V : V \rightarrow W$ is injective and has image $V' = \pi_W V$. Its polar decomposition is $A = JP$, where $J : V \rightarrow V'$ is an isometry and P is positive definite on V with $0 < \det P \leq 1$. Because $T^\perp = W \cap U^\perp$, the determinant computation of Theorem A.3.2, first in $\mathbb{R}^n$ and then inside W , gives

$$\Delta(V, T) = |\det(\pi_{W \cap U^\perp} \circ A)| = (\det P) \Delta_W(V', U) \leq \Delta_W(V', U).$$

It follows that

$$\{V \in G : \pi_W V \in \mathcal{S}_W(U, \rho)\} \subset \mathcal{S}(T, \rho).$$

The assumed Schubert bound and $\sigma(G) \geq 1 - \delta^{\kappa-\varepsilon}$ therefore imply

$$\sigma_W(\mathcal{S}_W(U, \rho)) \leq \frac{\delta^{-\varepsilon} \rho^\kappa}{1 - \delta^{\kappa-\varepsilon}} \leq \delta^{-2\varepsilon} \rho^\kappa$$

once δ is small enough. The map $V \mapsto \pi_W V$ is continuous on G in Grassmannian graph coordinates, so the conditional pushforward is well defined. $\square$

The other elementary direction change in He's induction replaces one random m -plane by the sum of several independent random m -planes. For an ordered family of subspaces $L_0, \dots, L_j$ whose total dimension is at most n , let $\text{Vol}(L_0, \dots, L_j)$ be the Euclidean volume of the parallelepiped formed by concatenating orthonormal bases of the L_i . Equivalently, it is the square root of the Gram determinant of those basis vectors; it is zero when they are dependent and is independent of the chosen orthonormal bases.

Lemma A.3.6 (Schubert nonconcentration for independent plane sums). *Let $q \geq 2$, $qm < n$, $\kappa > \varepsilon > 0$, and let $V_1, \dots, V_q$ be independent random elements of $\text{Gr}(n, m)$, each with law σ_j satisfying*

$$\sigma_j(\mathcal{S}(T, \rho)) \leq \delta^{-\varepsilon} \rho^\kappa \quad (T \in \text{Gr}(n, n-m), \rho \geq \delta).$$

Put $G = \{\dim(V_1 + \dots + V_q) = qm\}$. Then

$$\mathbb{P}(G) \geq 1 - (q-1)\delta^{\kappa-\varepsilon}.$$

For sufficiently small δ , the conditional law Σ of $V_1 + \dots + V_q$ on G satisfies

$$\Sigma(\mathcal{S}(W, \rho)) \leq \delta^{-2\varepsilon} \rho^{\kappa/q} \quad (W \in \text{Gr}(n, n-qm), \rho \geq \delta). \quad (\text{A.3.5})$$

Proof. For $2 \leq j \leq q$, condition on $V_1, \dots, V_{j-1}$ and extend their sum to a fixed $(n-m)$ -plane T_j ; this is possible because $(j-1)m \leq n-m$. If V_j meets the preceding sum nontrivially, then $\Delta(V_j, T_j) = 0$. The conditional probability of this event is at most $\delta^{\kappa-\varepsilon}$, by applying the hypothesis at $\rho = \delta$. A union bound over j gives the asserted bound for $\mathbb{P}(G)$.

Fix $W \in \text{Gr}(n, n-qm)$. For each j , put $S_j = \text{span}(W, V_1, \dots, V_{j-1})$, so $\dim S_j \leq n-m$. Let D_j be the m -Jacobian of the projection $\pi_{S_j^\perp}|_{V_j}$; thus $0 \leq D_j \leq 1$. Gram–Schmidt applied successively to orthonormal frames gives the exact identity

$$\text{Vol}(W, V_1, \dots, V_q) = \prod_{j=1}^q D_j. \quad (\text{A.3.6})$$

For every realization of the preceding planes, extend S_j to an $(n-m)$ -plane T_j . Since $T_j^\perp \subset S_j^\perp$, the projection Jacobian formula in [Theorem A.3.2](#) gives $\Delta(V_j, T_j) \leq D_j$. Independence therefore yields, for $u \geq \delta$,

$$\mathbb{P}(D_j \leq u \mid V_1, \dots, V_{j-1}) \leq \delta^{-\varepsilon} u^\kappa. \quad (\text{A.3.7})$$

On G , let $L = V_1 + \dots + V_q$ and let J be the volume of the concatenated orthonormal frames of the V_j . Then $0 < J \leq 1$, and a second Gram determinant calculation gives

$$\text{Vol}(W, V_1, \dots, V_q) = J \Delta(L, W) \leq \Delta(L, W).$$

Thus $G \cap \{\Delta(L, W) \leq \rho\}$ implies that the product in (A.3.6) is at most ρ , and hence some $D_j \leq \rho^{1/q}$. Since $\rho^{1/q} \geq \delta$ when $\rho \geq \delta$, (A.3.7) and a union bound give

$$\mathbb{P}(G \cap \{\Delta(L, W) \leq \rho\}) \leq q\delta^{-\varepsilon} \rho^{\kappa/q}.$$

Divide by $\mathbb{P}(G)$; for small δ the fixed factor $q/[1 - (q-1)\delta^{\kappa-\varepsilon}]$ is at most $\delta^{-\varepsilon}$. This proves (A.3.5). $\square$

One logical step in He's proof is independent of the higher-rank sum-product theorem: after finding many disjoint good pieces, a direction that is bad for a substantial part of their union must be bad for a substantial weighted collection of pieces. The finite formulation keeps that counting argument separate from the geometric covering losses.

Lemma A.3.7 (Weighted aggregation of bad directions). *Let A be a nonempty finite subset of $\mathbb{R}^n$, written as a disjoint union $A = A_1 \sqcup \dots \sqcup A_J$ with every $A_i \neq \emptyset$. For $0 < \theta \leq 1$ and $M > 0$ define*

$$\mathcal{B}_\delta(A; \theta, M) = \{V \in \text{Gr}(n, m) : \text{some } B \subset A \text{ has } \#B \geq \theta\#A \text{ and } \mathcal{N}_\delta(\pi_V B) \leq M\}.$$

If $0 < \eta < \theta \leq 1$ and σ is a probability on $\text{Gr}(n, m)$, then

$$\sigma(\mathcal{B}_\delta(A; \theta, M)) \leq \frac{1-\eta}{\theta-\eta} \sum_{i=1}^J \frac{\#A_i}{\#A} \sigma(\mathcal{B}_\delta(A_i; \eta, M)). \quad (\text{A.3.8})$$

Proof. Fix $V \in \mathcal{B}_\delta(A; \theta, M)$ and choose a witnessing B . Let $I(V) = \{i : \#(B \cap A_i) \geq \eta\#A_i\}$. For $i \in I(V)$ the set $B \cap A_i$ witnesses $V \in \mathcal{B}_\delta(A_i; \eta, M)$, since its projected set is contained in $\pi_V B$. Put $w = \sum_{i \in I(V)} \#A_i / \#A$. Then

$$\theta\#A \leq \#B \leq \sum_{i \in I(V)} \#A_i + \eta \sum_{i \notin I(V)} \#A_i = [\eta + (1-\eta)w]\#A,$$

so $w \geq (\theta - \eta)/(1 - \eta)$. Consequently, pointwise in V ,

$$\mathbf{1}_{\mathcal{B}_\delta(A; \theta, M)}(V) \leq \frac{1-\eta}{\theta-\eta} \sum_i \frac{\#A_i}{\#A} \mathbf{1}_{\mathcal{B}_\delta(A_i; \eta, M)}(V).$$

Integrating proves (A.3.8). Each bad-direction set is Borel for finite A : it is a finite union over $B \subset A$ of the closed conditions $\mathcal{N}_\delta(\pi_V B) \leq M$ (when M is not an integer, replace it by $\lfloor M \rfloor$). To see closedness, take convergent V_j and, for each j , at most $\lfloor M \rfloor$ covering centers in the bounded projection of A enlarged by a radius- δ ball. Pass to convergent subsequences of those centers; their limiting radius- δ balls cover the limiting finite projected set. $\square$

The two hypotheses in the next theorem have different jobs: a Frostman-type bound prevents A from hiding in one small ball, while the Schubert-tube bound prevents the distribution of target planes from being concentrated near planes that kill a common complementary direction. Neither condition may be omitted from the statement.

A.3.2 The projection theorem and geometric reductions

Theorem A.3.8 (He’s discretized projection theorem). *Let $1 \leq m < n$, $0 < \alpha < n$, and $\kappa > 0$. There is $\varepsilon = \varepsilon(n, m, \alpha, \kappa) > 0$ such that for every sufficiently small $\delta > 0$ the following holds. Suppose $A \subset B(0, 1) \subset \mathbb{R}^n$ and σ is a probability on $\text{Gr}(n, m)$ satisfying*

$$\mathcal{N}_\delta(A) \geq \delta^{-\alpha+\varepsilon}, \quad (\text{A.3.9})$$

$$\mathcal{N}_\delta(A \cap B(x, \rho)) \leq \delta^{-\varepsilon} \rho^\kappa \mathcal{N}_\delta(A) \quad (x \in \mathbb{R}^n, \rho \geq \delta), \quad (\text{A.3.10})$$

$$\sigma(\mathcal{S}(W, \rho)) \leq \delta^{-\varepsilon} \rho^\kappa \quad (W \in \text{Gr}(n, n-m), \rho \geq \delta). \quad (\text{A.3.11})$$

Then a set $D \subset \text{Gr}(n, m)$ has $\sigma(D) \geq 1 - \delta^\varepsilon$ and, for every $V \in D$ and every $A' \subset A$ with $\mathcal{N}_\delta(A') \geq \delta^\varepsilon \mathcal{N}_\delta(A)$,

$$\mathcal{N}_\delta(\pi_V A') \geq \delta^{-(m/n)\alpha-\varepsilon}. \quad (\text{A.3.12})$$

Source and proof route. Weikun He, *Orthogonal projections of discretized sets*, arXiv:1710.00795v2 (9 May 2018), Theorem 1, printed pp. 1–2. The proof below reconstructs the rank- m matrix sum–product base case, the half-dimensional reduction, the quotient, sum, and intersection rank arrows, and the all-rank large-set normalization. [Theorem A.3.73](#) assembles the induction, using the matrix theorem proved later in this chapter. Volume III.6 supplies the rank-one case.

It is useful to separate that difficult core from the final logical upgrade. Put $b = (m/n)\alpha$, and, for a bounded nonempty set $T \subset \mathbb{R}^n$, define

$$\mathcal{E}_\delta(T; \varepsilon) = \{V \in \text{Gr}(n, m) : \text{some } T' \subset T \text{ satisfies } \mathcal{N}_\delta(T') \geq \delta^\varepsilon \mathcal{N}_\delta(T), \\ \mathcal{N}_\delta(\pi_V T') < \delta^{-b-\varepsilon}\}.$$

For finite T , this is Borel by the covering-center argument in the proof of [Theorem A.3.7](#). The higher-rank base case uses an independent sum–product theorem, stated after one more elementary slice comparison.

Before stating that input, we finish the elementary slice comparison. In the next lemma $\mathcal{E}_\delta^W(B; u)$ means the displayed bad-direction definition with V ranging over $\text{Gr}(W, m)$, while retaining the threshold $b = (m/n)\alpha$. This agrees with He’s induction when a slice in W is assigned dimension parameter $(\dim W/n)\alpha$.

Lemma A.3.9 (Transferring a bad projection to a slice). *Let $m \leq n' < n$, $W \in \text{Gr}(n, n')$, and $B \subset W$ be bounded. If $V \in \text{Gr}(n, m)$ has $J_W(V) > 0$ and $V' = \pi_W V$, then*

$$\mathcal{N}_\delta(\pi_{V'} B) \leq C_m J_W(V)^{-m} \mathcal{N}_\delta(\pi_V B). \quad (\text{A.3.13})$$

Consequently, if $\varepsilon, h > 0$, $V \in \mathcal{E}_\delta(B; \varepsilon)$, and $J_W(V) \geq \delta^h$, then for all sufficiently small δ ,

$$V' \in \mathcal{E}_\delta^W(B; 2\varepsilon + mh).$$

Proof. Write $A = \pi_W|_V : V \rightarrow V'$. The adjoint $A^* = \pi_V|_W : W \rightarrow V$ vanishes on $W \cap (V')^\perp$ and restricts to an isomorphism $F = \pi_V|_{V'} : V' \rightarrow V$. The singular values of F equal those of A ; they lie in $(0, 1]$ and their product is $J_W(V)$. Hence the least is at least $J_W(V)$, so F^{-1} is $J_W(V)^{-1}$ -Lipschitz. For $x \in W$, $\pi_V x = F(\pi_{V'} x)$, and therefore $\pi_V B = F^{-1}(\pi_{V'} B)$. The

inverse image under F of a radius- δ ball in V lies in a radius- $J_W(V)^{-1}\delta$ ball in V' . A ball of this radius in an m -plane can be covered by at most $C_m J_W(V)^{-m}$ radius- δ balls. This proves (A.3.13).

Now choose $B' \subset B$ witnessing $V \in \mathcal{E}_\delta(B; \varepsilon)$. Its size still satisfies $\mathcal{N}_\delta(B') \geq \delta^\varepsilon \mathcal{N}_\delta(B)$, while (A.3.13) gives

$$\mathcal{N}_\delta(\pi_V B') < C_m \delta^{-b-\varepsilon-mh} < \delta^{-b-2\varepsilon-mh}$$

for small δ . Since $\delta^{2\varepsilon+mh} \leq \delta^\varepsilon$, the same B' witnesses the asserted W -internal bad event. $\square$

The next regularization input is used twice in He's proof: a direction with a small projection can be chosen so that the small projection is itself nonconcentrated. The proof is geometric rather than a formal property of Frostman measures. When many candidate directions fail, their dense inverse-image cylinders intersect in a ball that violates the original nonconcentration bound.

Lemma A.3.10 (Regularizing a bad projection). *Let $n = qm$ with $q \geq 2$, $0 < \alpha < n$, $\kappa > 0$, and $0 < \varepsilon < \kappa/(100q^2)$. Put $b = \alpha/q$ and $\kappa_1 = \kappa/(2q^2)$. Suppose a finite $A \subset B(0, 1) \subset \mathbb{R}^n$ and a probability σ on $\text{Gr}(n, m)$ satisfy (A.3.9)–(A.3.11) at exponent ε . Define $\mathcal{E}_\delta^{\text{reg}}(A; \varepsilon)$ to consist of the $V \in \mathcal{E}_\delta(A; \varepsilon)$ for which there is $B \subset A$ with*

$$\begin{aligned} \mathcal{N}_\delta(B) &\geq \delta^{2\varepsilon} \mathcal{N}_\delta(A), & \mathcal{N}_\delta(\pi_V B) &< \delta^{-b-\varepsilon}, \\ \mathcal{N}_\delta(\pi_V B \cap B_V(x, \rho)) &\leq \delta^{-b-10\varepsilon} \rho^{\kappa_1} \quad (x \in V, \rho \geq \delta). \end{aligned}$$

Then, for sufficiently small δ depending on the displayed parameters,

$$\sigma(\mathcal{E}_\delta(A; \varepsilon) \setminus \mathcal{E}_\delta^{\text{reg}}(A; \varepsilon)) \leq \delta^{2\varepsilon}. \quad (\text{A.3.14})$$

The event in (A.3.14) is understood in the completion of σ ; for finite A it is universally measurable.

Proof. Write $N = \mathcal{N}_\delta(A)$ and let V belong to the event on the left of (A.3.14). Choose $A' \subset A$ witnessing $V \in \mathcal{E}_\delta(A; \varepsilon)$, so $\mathcal{N}_\delta(A') \geq \delta^\varepsilon N$ and $\mathcal{N}_\delta(\pi_V A') < \delta^{-b-\varepsilon}$. Partition V into half-open side- δ cubes. At most $C_m \delta^{-b-\varepsilon}$ of them meet $\pi_V A'$. Call such a cube Q heavy when

$$\mathcal{N}_\delta(A' \cap \pi_V^{-1} Q) \geq \delta^{b+3\varepsilon} N,$$

and let B be the union of the parts above heavy cubes. Subadditivity of covering numbers gives

$$\mathcal{N}_\delta(A' \setminus B) \leq C_m \delta^{2\varepsilon} N, \quad \mathcal{N}_\delta(B) \geq \frac{1}{2} \delta^\varepsilon N \geq \delta^{2\varepsilon} N$$

for sufficiently small δ . Also $\mathcal{N}_\delta(\pi_V B) < \delta^{-b-\varepsilon}$. Since $V \notin \mathcal{E}_\delta^{\text{reg}}(A; \varepsilon)$, some $x \in V$ and $r \geq \delta$ satisfy

$$\mathcal{N}_\delta(\pi_V B \cap B_V(x, r)) > r^{\kappa_1} \delta^{-b-10\varepsilon}. \quad (\text{A.3.15})$$

The left side is at most $\delta^{-b-\varepsilon}$, so $r^{\kappa_1} < \delta^{9\varepsilon}$ and in particular $r < 1$. At least $c_m r^{\kappa_1} \delta^{-b-10\varepsilon}$ heavy projection cubes meet $B_V(x, r)$. Color the grid cubes by their indices modulo 4 in each coordinate. One color retains at least a fixed fraction of these cubes, and a radius- δ

ball in $\mathbb{R}^n$ meets the lifted parts over at most one cube of that color. Adding the covering lower bounds of those lifted parts therefore gives, for $R = r + C_m \delta$,

$$\mathcal{N}_\delta(A \cap \pi_V^{-1}(B_V(x, R))) \geq c_m r^{\kappa_1} \delta^{-7\varepsilon} N \geq R^{\kappa_1} \delta^{-6\varepsilon} N \quad (\text{A.3.16})$$

after decreasing δ . In particular the left side is at most N , so $R \leq \delta^{5\varepsilon/\kappa_1}$ for small δ .

Suppose for contradiction that the probability in (A.3.14) is at least $\delta^{2\varepsilon}$. Divide the interval of possible radii $[\delta, \delta^{5\varepsilon/\kappa_1}]$ into intervals $[2^j \delta, 2^{j+1} \delta)$. There are at most $C \log(1/\delta) \leq \delta^{-\varepsilon}$ such intervals for small δ . Hence one interval $[\rho, 2\rho)$ supports a measurable set D of directions with $\sigma(D) \geq \delta^{3\varepsilon}$. For each $V \in D$, enlarge the cylinder in (A.3.16) to radius 3ρ and absorb fixed constants. The resulting lower bound is

$$\mathcal{N}_\delta(A \cap \pi_V^{-1}(B_V(x_V, 3\rho))) \geq \rho^{2\kappa_1} \delta^{-5\varepsilon} N \quad (\text{A.3.17})$$

for a suitable center $x_V \in V$. This deliberately weaker power absorbs both the radius change and fixed constants.

Let $P \subset A$ be a maximal δ -separated set, and put $S = \bigcup_{p \in P} B(p, \delta/10)$. The balls in this union are disjoint, so $\lambda(S) = c_n \delta^n \#P$, and Euclidean packing gives $N \leq \#P \leq C_n N$. Since the δ -balls centered at P cover A , (A.3.17) ensures that at least $\rho^{2\kappa_1} \delta^{-5\varepsilon} N$ points of P lie in a slightly enlarged cylinder. Increasing its radius to 12ρ and absorbing constants, we obtain a measurable set

$$S_V = S \cap \pi_V^{-1}(B_V(x_V, 12\rho)) \quad \text{with} \quad \lambda(S_V) \geq t \lambda(S), \quad t = \rho^{2\kappa_1} \delta^{-4\varepsilon}.$$

The centers may be chosen measurably. Enumerate P once and take $x_V = \pi_V p$ for the first p for which the radius- 12ρ cylinder contains at least $t\#P$ points of P . Such a p exists: any net point in the earlier radius- 3ρ cylinder recenters that cylinder inside the radius- 12ρ cylinder, and the fixed power loss in t absorbs the net-covering constants. The qualifying condition is Borel in V for each of finitely many p . If the displayed t exceeds one, the preceding mass estimate is already impossible; hence we may assume $0 < t \leq 1$ below.

Normalize $\sigma|_D$ to a probability and choose $V_1, \dots, V_q$ independently from it. Fubini and Jensen imply that the expected relative λ -measure of $S_{V_1} \cap \dots \cap S_{V_q}$ is at least t^q . Since this relative measure is at most one, the probability that it is at least $t^q/2$ is at least $t^q/2$. Under the original product law $\sigma^{\otimes q}$, the event therefore has probability at least

$$\frac{1}{2} \sigma(D)^q t^q \geq \frac{1}{2} \rho^{2q\kappa_1} \delta^{-q\varepsilon} = \frac{1}{2} \rho^{\kappa/q} \delta^{-q\varepsilon}. \quad (\text{A.3.18})$$

Let $J = \text{Vol}(V_1, \dots, V_q)$. By the same successive Gram-determinant calculation as in Theorem A.3.6, and by (A.3.11),

$$\mathbb{P}_{\sigma^{\otimes q}}\{J < \rho^{(q-1)/q}\} \leq (q-1)\delta^{-\varepsilon} \rho^{\kappa/q}. \quad (\text{A.3.19})$$

Indeed, if a product of the $q-1$ successive projection Jacobians is below $\rho^{(q-1)/q}$, one factor is below $\rho^{1/q}$; extend the preceding sum to an $(n-m)$ -plane and apply the Schubert hypothesis. The threshold $\rho^{1/q}$ is at least δ . Since $q \geq 2$, the lower bound in (A.3.18) eventually exceeds (A.3.19). Thus some tuple has both a large intersection of the S_{V_j} and $J \geq \rho^{(q-1)/q}$.

For such a tuple, the map $z \mapsto (\pi_{V_1} z, \dots, \pi_{V_q} z)$ is invertible and has inverse norm at most $C_n J^{-1}$. Hence the intersection of its q radius- 12ρ cylinders lies in a ball of radius $C_n \rho / J \leq C_n \rho^{1/q}$. The nonconcentration hypothesis (A.3.10), applied to the net P and its disjoint balls, gives the upper bound

$$\frac{\lambda(S \cap B(z, C_n \rho^{1/q}))}{\lambda(S)} \leq C_{n,\kappa} \delta^{-\varepsilon} \rho^{\kappa/q}$$

for every $z \in \mathbb{R}^n$: a radius- δ ball contains only $O_n(1)$ points of the δ -separated net, and every ball forming $S \cap B(z, C_n \rho^{1/q})$ has its center in the $O_n(\delta)$ -enlargement of that ball. A fixed factor is harmless after reducing δ . But the intersection event supplies the lower bound $t^q/2 = \frac{1}{2} \rho^{\kappa/q} \delta^{-4q\varepsilon}$. These inequalities are incompatible as $\delta \downarrow 0$. This proves (A.3.14).

For completeness, all direction events used above are universally measurable. For finite A , a covering-number condition is Borel as a function of the projected finite configuration; failure of the regularity bound is its projection after adjoining the center and radius variables, hence analytic. Finite unions over $B \subset A$ and Boolean operations preserve universal measurability. The dyadic radius pigeonhole and the finite choice of x_V can therefore be carried out in the completed probability space. $\square$

We can now extract the regular slice needed for the dimension-reduction step. The hypothesis below is the precise negation of the finite good-piece conclusion, not the negation of the full all-subsets theorem.

Lemma A.3.11 (Regular affine slice from persistent bad projections). *Let $n = qm$ with $q \geq 2$, $0 < \alpha < n$, $\kappa > 0$, and $0 < \varepsilon < \kappa/(100q^2)$. Put $\beta = (q-1)\alpha/q$ and $\kappa_1 = \kappa/(2q^2)$. There is $C = C(n, q, \kappa) > 0$ such that the following holds for sufficiently small δ . Suppose a finite $A \subset B(0, 1)$ and a probability σ on $\text{Gr}(n, m)$ satisfy (A.3.9)–(A.3.11), and*

$$\sigma(\mathcal{E}_\delta(T; \varepsilon)) > \delta^\varepsilon \quad \text{for every nonempty } T \subset A. \quad (\text{A.3.20})$$

Then an affine $(n-m)$ -plane $y + W$ and a finite $B \subset A^{(\delta)} \cap (y + W)$ exist such that

$$\mathcal{N}_\delta(B) \geq \delta^{-\beta+C\varepsilon}, \quad (\text{A.3.21})$$

$$\mathcal{N}_\delta(B \cap B_W(x, \rho)) \leq \delta^{-\beta-C\varepsilon} \rho^{\kappa_1} \quad (x \in y + W, \rho \geq \delta). \quad (\text{A.3.22})$$

Here $A^{(\delta)}$ is the open δ -neighborhood of A ; changing its radius by a fixed factor only changes the admissible δ threshold and the constant C .

Proof. The assumption (A.3.20) for $T = A$, together with Theorem A.3.10, gives $V_1 \in \mathcal{E}_\delta^{\text{reg}}(A; \varepsilon)$. Choose its witnessing $A_1 \subset A$. Since $\mathcal{N}_\delta(A) \geq \delta^{-\alpha+\varepsilon}$,

$$\mathcal{N}_\delta(A_1) \geq \delta^{-\alpha+3\varepsilon}, \quad \mathcal{N}_\delta(\pi_{V_1} A_1) < \delta^{-\alpha/q-\varepsilon}, \quad (\text{A.3.23})$$

and, for all $x \in V_1$ and $\rho \geq \delta$,

$$\mathcal{N}_\delta(\pi_{V_1} A_1 \cap B_{V_1}(x, \rho)) \leq \delta^{-\alpha/q-10\varepsilon} \rho^{\kappa_1}. \quad (\text{A.3.24})$$

Inductively choose $V_j \in \text{Gr}(n, m)$ and $A_j \subset A_{j-1}$ for $2 \leq j \leq q$. Let $S_{j-1} = V_1 + \dots + V_{j-1}$, extend it to a fixed $(n-m)$ -plane T_j , and set $h = 3\varepsilon/\kappa$. By (A.3.20), the bad-direction event for A_{j-1} has σ -mass greater than δ^ε ; the Schubert tube $\mathcal{S}(T_j, \delta^h)$ has mass at most $\delta^{2\varepsilon}$. We may therefore choose $V_j \in \mathcal{E}_\delta(A_{j-1}; \varepsilon)$ outside that tube and let A_j be a witnessing subset. Thus

$$\Delta(V_j, T_j) > \delta^h, \quad \mathcal{N}_\delta(A_j) \geq \delta^\varepsilon \mathcal{N}_\delta(A_{j-1}), \quad \mathcal{N}_\delta(\pi_{V_j} A_j) < \delta^{-\alpha/q-\varepsilon}. \quad (\text{A.3.25})$$

The successive Gram-determinant formula gives

$$J := \text{Vol}(V_1, \dots, V_q) \geq \delta^{(q-1)h} > 0. \quad (\text{A.3.26})$$

In particular the planes are a direct sum equal to $\mathbb{R}^n$. From (A.3.23) and (A.3.25),

$$\mathcal{N}_\delta(A_q) \geq \delta^{-\alpha+(q+2)\varepsilon}. \quad (\text{A.3.27})$$

Cover $\pi_{V_q} A_q$ by at most $C_m \delta^{-\alpha/q-\varepsilon}$ radius- $\delta/2$ balls: each radius- δ ball takes only C_m such smaller balls. The inverse images of these balls cover A_q , so one fiber tube $B_0 = A_q \cap \pi_{V_q}^{-1}(B_{V_q}(y, \delta/2))$ obeys

$$\mathcal{N}_\delta(B_0) \geq \delta^{-\beta+(q+4)\varepsilon} \quad (\text{A.3.28})$$

for small δ , after absorbing a fixed covering constant. Put $W = V_q^\perp$ and let B be the orthogonal projection of B_0 onto the affine plane $y + W$. Because every point of B_0 is within 2δ of that plane, $\mathcal{N}_\delta(B) \geq c_n \mathcal{N}_\delta(B_0)$; choose C large enough to obtain (A.3.21). The same construction gives $B \subset A^{(\delta)} \cap (y + W)$, since every projected point moves by at most $\delta/2$.

It remains to prove the ball estimate on B . Write $U = V_1 + \dots + V_{q-1}$, so $\dim U = n-m = \dim W$. The volume $\Delta(U, V_q)$ is at least J by (A.3.26); hence $\pi_U|_W : W \rightarrow U$ is invertible with inverse norm at most $C_n J^{-1}$. A covering comparison for this inverse, followed by the fact that B lies within 2δ of A_q , gives

$$\mathcal{N}_\delta(B \cap B_W(x, \rho)) \leq C_n J^{-n} \mathcal{N}_\delta(\pi_U A_q \cap B_U(\pi_U x, C_n \rho)) \quad (\text{A.3.29})$$

for $\rho \geq \delta$, with a harmless fixed enlargement of the projected ball. To justify the covering comparison, pull a radius- δ cover of the right-hand projected set back through $(\pi_U|_W)^{-1}$; each resulting radius- $C_n J^{-1} \delta$ ball in W takes at most $C_n J^{-n}$ radius- δ balls.

The coordinate map $z \in U \mapsto (\pi_{V_1} z, \dots, \pi_{V_{q-1}} z)$ is invertible with inverse norm at most $C_n J^{-1}$. Indeed the volume of the first $q-1$ planes is at least J . Product covers, with the first factor restricted to a $C_n \rho$ -ball, therefore yield

$$\begin{aligned} & \mathcal{N}_\delta(\pi_U A_q \cap B_U(\pi_U x, C_n \rho)) \\ & \leq C_n J^{-n} \mathcal{N}_\delta(\pi_{V_1} A_q \cap B_{V_1}(\pi_{V_1} x, C'_n \rho)) \prod_{j=2}^{q-1} \mathcal{N}_\delta(\pi_{V_j} A_q). \end{aligned}$$

Since $A_q \subset A_1$, apply (A.3.24) to the first factor and (A.3.25) to the others. The product has exponent $(q-1)\alpha/q = \beta$ and a finite total loss in ε . Combining with (A.3.29) and $J^{-2n} \leq \delta^{-6n(q-1)\varepsilon/\kappa}$ gives

$$\mathcal{N}_\delta(B \cap B_W(x, \rho)) \leq C_{n,q,\kappa} \delta^{-\beta-C_0\varepsilon} \rho^{\kappa_1}$$

for an explicit choice such as $C_0 = q + 20 + 6n(q - 1)/\kappa$. Taking $C = C_0 + 10$ and decreasing δ absorbs the fixed coefficient, proving (A.3.22). $\square$

In the sum-product theorem, $\text{End}(\mathbb{R}^m)$ is the algebra of linear maps $\mathbb{R}^m \rightarrow \mathbb{R}^m$ with operator norm, $aX = \{ax : x \in X\}$, and $\mathcal{N}_\rho$ counts Euclidean radius- ρ balls. The irreducibility hypothesis is part of the theorem, not an implicit convention.

The geometric step in the half-dimensional case can be isolated before invoking sum-product. It is useful to state it for oblique coordinates: this avoids hiding a quantitative change of variables behind the phrase “we may assume the two selected planes are orthogonal.”

Lemma A.3.12 (A matrix tube pulls back to a Schubert tube). *Let E_1, E_2 be Euclidean m -spaces, let $F : E_1 \oplus E_2 \rightarrow \mathbb{R}^{2m}$ be an invertible linear map, and put $L = \max(1, \|F\|, \|F^{-1}\|)$. Suppose $V \in \text{Gr}(2m, m)$ is such that $P_V = \pi_V F|_{E_1}$ is invertible, and define $\phi_V = P_V^{-1} \pi_V F|_{E_2} : E_2 \rightarrow E_1$. For $0 \neq v \in E_2$ and every proper linear subspace $W \subset E_1$ there is an m -plane $Q = Q(F, v, W)$, independent of V , for which*

$$\Delta(V, Q) \leq L^2 \|v\|^{-1} \text{dist}(\phi_V v, W). \quad (\text{A.3.30})$$

The assertion includes $W = \{0\}$; when $m = 1$ the right side with $W = \{0\}$ is understood literally.

Proof. First suppose $m \geq 2$. Enlarge W to a hyperplane $H \subset E_1$. The m -plane $M = F(H \oplus \mathbb{R}v)$ is well defined, and set $Q = M^\perp$. For $h \in H$, the vector $z = F(h, -v)$ belongs to M and $\|z\| \geq L^{-1} \|v\|$. Since $\pi_V F(\phi_V v, -v) = 0$ and $\|\pi_V F\| \leq L$,

$$\frac{\|\pi_V z\|}{\|z\|} \leq L^2 \|v\|^{-1} \|h - \phi_V v\|.$$

The least singular value of $\pi_V|_M$ is no larger than the left-hand side. All its other singular values are at most one, so its determinant modulus is no larger either. By Theorem A.3.2 and equality of the nonzero singular values of $\pi_V|_M$ and $\pi_M|_V$, this determinant modulus is $\Delta(V, Q)$. Take the infimum over $h \in H$ and use $\text{dist}(\phi_V v, H) \leq \text{dist}(\phi_V v, W)$. For $m = 1$ take $H = W = \{0\}$; the same argument applies. $\square$

A.3.3 The Cartesian complex-ring bridge

Definition A.3.13 (Bounded algebra words). *For a subset A of an associative algebra and an integer $s \geq 1$, let $\mathcal{W}_s(A)$ consist of signed expressions with at most s leaves from A , using addition, subtraction and ordered multiplication. The zero expression is permitted. Thus a fixed number of additions or multiplications of elements from $\mathcal{W}_s(A)$ lies in $\mathcal{W}_S(A)$ for another fixed integer S . This notation counts leaves, not operation depth.*

The matrix sum-product argument needs a segment theorem for the *coordinatewise* ring $\mathbb{C}^r$, not merely the scalar complex ring. We give the dimension-induction step, including its quantitative gain. Its scalar starting point is the homogeneous form of the one-scale segment theorem reconstructed in the current canonical Volume VI.12. Its Pass-98 Bourgain–Gamburd chain is locally proved there; the bridge below therefore has an

internal scalar supplier. The separate algebraic escape step is supplied below by [Theorems A.3.36](#) and [A.3.43](#).

Definition A.3.14 (Homogeneous product–difference words). *If A lies in an associative ring and $p, q \geq 1$, write*

$$H_{p,q}(A) = qA^{(p)} - qA^{(p)}, \quad A^{(p)} = \{a_1 \cdots a_p : a_j \in A\}.$$

Here qB is the q -fold sumset, repetitions are permitted, matrix products retain their displayed order, and the product in $\mathbb{C}^r$ is coordinatewise. Cancellation gives $0 \in H_{p,q}(A)$. For bounded fixed indices, sums, differences and products of elements of $H_{p,q}(A)$ lie in some $H_{P,Q}(A)$ with fixed P, Q , provided all summands being added have the same product degree P . The latter qualification matters under rescaling.

Lemma A.3.15 (Scalar homogeneous segment interface). *By the locally proved scalar complex segment theorem of Volume VI.12 in its homogeneous form, for every $\sigma > 0$ and fixed $M \geq 1$ there are $p, q, C \geq 1$, $b > 0$ and $\eta_0 > 0$ such that, if $F \subset B_{\mathbb{C}}(0, M)$, $0 < \eta < \eta_0$, and $\mathcal{N}_\eta(F) \geq \eta^{-\sigma}$, then some unit $u \in \mathbb{C}$ and $J \in [\eta^C, 1]$ satisfy*

$$[0, J]u \subset H_{p,q}(F) + B(0, J\eta^b). \quad (\text{A.3.31})$$

The indices and exponents are independent of F and η .

Proof. For $M = 1$, the proof of Volume VI.12 theorem `thm:p91-bg-complex-segment-root` starts from its homogeneous Proposition-5 interface $sF^{(s)} - sF^{(s)}$. Its affine normalization uses factors $(f - f_0)/\lambda$; multiplication by λ^s and expansion of the differences leave every monomial with exactly s leaves of F . Thus its segment is contained in $H_{p,q}(F)$, with fixed p, q , up to its stated error. Subtract an approximant to one endpoint; this centers the segment and doubles q , without changing p . The variable segment exponent is bounded, and a fixed portion of the $\eta^{1/2}$ relative gain survives absorption of all fixed constants. This gives (A.3.31) with some $b > 0$.

For fixed $M > 1$, apply this result with exponent $\sigma/2$ to F/M at scale η/M ; the required covering lower bound holds for sufficiently small η . Multiply every resulting homogeneous word by M^P : it becomes a word in F of the same degree. This changes the segment length and error by fixed factors depending on M, p only. Shortening a bounded segment to length at most one and slightly decreasing b absorbs those factors for small η . $\square$

Proposition A.3.16 (Cartesian complex segment from the scalar theorem). *Fix $r \geq 1$ and $\sigma > 0$. By [Theorem A.3.15](#), there are $p, q, C \geq 1$, $b > 0$ and $\delta_0 > 0$ such that, for every $A \subset B_{\mathbb{C}^r}(0, 1)$ and $0 < \delta < \delta_0$ with $\mathcal{N}_\delta(A) \geq \delta^{-\sigma}$, there are a unit $v \in \mathbb{C}^r$ and $L \in [\delta^C, 1]$ for which*

$$[0, L]v \subset H_{p,q}(A) + B(0, L\delta^b). \quad (\text{A.3.32})$$

The ball and norm may be Euclidean or coordinate supremum balls; their fixed-dimensional comparison changes only the constants.

Proof. We induct on r . The scalar lemma is the case $r = 1$. Suppose the result is known in $\mathbb{C}^d$, with constants C, b, p, q for the smaller positive exponent $\sigma/3$. For $A \subset \mathbb{C}^{d+1}$, the discrete Loomis–Whitney inequality gives, after permuting coordinates,

$$\mathcal{N}_\delta(\pi_{1,\dots,d}A) \geq \delta^{-\sigma/3} \quad (\text{A.3.33})$$

when δ is small. Indeed, replace A by its occupied coordinatewise δ -cubes. If Q is the resulting finite set of grid cells, repeated Hölder gives $|Q|^d \leq \prod_{i=1}^{d+1} |\pi_i Q|$. The grid-cell counts and covering numbers differ by constants depending only on d , which the exponent slack absorbs.

Apply the induction hypothesis to the projection. Write its length as $L = \delta^\gamma$, $0 \leq \gamma \leq C$, and choose, for every $t \in [0, L]$, a lift $\phi(t) \in H_{p,q}(A)$ with

$$\|\pi_{1,\dots,d}\phi(t) - tv\| \leq e := L\delta^b. \quad (\text{A.3.34})$$

Every $\phi(t)$ has norm at most a fixed $M = M(p, q)$. Put $S = \{\phi_{d+1}(t) : 0 \leq t \leq L\}$ and $h = L^2\delta^{2b}$. We consider two alternatives.

If $\mathcal{N}_h(S) < \delta^{-b/2}$, sample more than $3\delta^{-b/2}$ equally spaced parameters in $[0, L]$. Two samples t_1, t_2 at distance at least $cL\delta^{b/2}$ have last coordinates at distance at most $2h$. Set $a = t_1 - t_2$ and $D = \phi(t_1) - \phi(t_2)$. Then $|a| \geq cL\delta^{b/2}$, $\|\pi_{1,\dots,d}D - av\| \leq 2e$, and $|D_{d+1}| \leq 2h$. Coordinatewise multiplication and (A.3.34) yield, uniformly for $t \in [0, L]$,

$$\|\phi(t)D - (ta(v_1^2, \dots, v_d^2), 0)\| \leq C_M(Le + h).$$

Since $\sum_j |v_j|^2 = 1$, Cauchy–Schwarz gives $\|(v_1^2, \dots, v_d^2)\| \geq d^{-1/2}$. The target segment therefore has length $L' = L|a|\|(v_j^2)_j\| \geq c_d L^2 \delta^{b/2}$, whereas its error divided by L' is $O_{M,d}(\delta^{b/2})$. The products $\phi(t)D$ belong to one fixed $H_{P,Q}(A)$ of product degree $2p$. Decrease the new gain to $b/4$ and the scale threshold to absorb the constants. This proves (A.3.32) in this case.

Now suppose $\mathcal{N}_h(S) \geq \delta^{-b/2}$. Because $h = \delta^{2\gamma+2b}$ and $\gamma \leq C$, a fixed $\sigma_* > 0$, depending only on C, b , satisfies $\mathcal{N}_h(S) \geq h^{-\sigma_*}$. Apply the scalar interface to the bounded set S at scale h . It gives a centered segment $[0, J]u \subset \mathbb{C}$, where $J \geq h^{C_*}$, and for every point of the segment an approximant in $H_{p,q_*}(S)$ with error Jh^{b_*} . Every scalar approximant has a vector lift in a fixed homogeneous $H_{P,Q_*}(A)$, since each element of S is the last coordinate of a word of product degree p .

First assume that, whenever $x, y, x + y \in [0, L]$,

$$\|\phi(x + y) - \phi(x) - \phi(y)\| \leq L\delta^{b/2}. \quad (\text{A.3.35})$$

Take $m = \lceil \delta^{-b/4} \rceil$ and $t_j = jL/m$. Iterating (A.3.35) gives $\|\phi(t_j) - j\phi(t_1)\| \leq mL\delta^{b/2} \leq 2L\delta^{b/4}$. The first d coordinates in (A.3.34) show $\|\phi(t_1)\| \geq cL/m$. Taking $j = m$ also shows $m\|\phi(t_1)\| \leq M + 2L\delta^{b/4}$. Consequently the arithmetic progression $\{j\phi(t_1) : 0 \leq j \leq m\}$ covers the segment from zero of length $L' = \min\{m\|\phi(t_1)\|, 1\} \geq cL$ within $O_M(L'\delta^{b/4})$; the point $j = 0$ is in $H_{p,q}(A)$ by cancellation. Replace $j\phi(t_1)$ by $\phi(t_j)$ and decrease the gain to $b/8$. This proves the conclusion in the additive case.

Otherwise choose $x, y, x + y \in [0, L]$ for which the defect $z = \phi(x + y) - \phi(x) - \phi(y)$ has norm $> L\delta^{b/2}$. By (A.3.34), $\|\pi_{1,\dots,d}z\| \leq 3e$, so for small δ

$$|z_{d+1}| \geq \frac{1}{2}L\delta^{b/2}. \quad (\text{A.3.36})$$

Fix an integer k large enough that $kb/2 > 2(C + b)C_* + 1$. Multiply each lifted scalar approximant by z^k coordinatewise. Its first d coordinates have norm at most $C(3e)^k$, while its

last coordinate approximates $z_{d+1}^k[0, J]u$ with error at most $|z_{d+1}|^k J h^{b_*}$. The new segment length $L' = \min\{|z_{d+1}|^k J, 1\}$ is at least a fixed positive power of δ by (A.3.36) and $J \geq h^{C_*}$. When $|z_{d+1}|^k J \leq 1$, the ratio of the first-coordinate error to this length is at most

$$C \frac{(3e/|z_{d+1}|)^k}{J} \leq C \delta^{kb/2 - (2C+2b)C_*}.$$

When $|z_{d+1}|^k J > 1$, the error itself is $O(\delta^{kb})$, after enlarging a fixed constant. The last-coordinate relative error is at most Ch^{b_*} in either case. Thus a fixed positive gain survives, and all approximants have the same homogeneous product degree $p(k + p_*)$ in A . This is (A.3.32) in the final case. All exponents, word indices and thresholds depend only on d, σ , so the induction closes. $\square$

Proposition A.3.17 (Near-diagonal matrix segment). *Fix $n \geq 1$ and $\sigma > 0$. Assume Theorem A.3.16. There are fixed integers $p, q \geq 1$, constants $C \geq 1, \beta > 0$, and $\delta_0 > 0$ such that, if $0 < \delta < \delta_0$ and $A \subset M_n(\mathbb{C})$ satisfies*

$$A \subset B(0, 1), \quad \mathcal{N}_\delta(A) \geq \delta^{-\sigma}, \quad \text{dist}(a, \Delta) \leq \delta \quad (a \in A),$$

where Δ is the diagonal algebra, then some unit $v \in \Delta$ and $L \in [\delta^C, 1]$ satisfy

$$[0, L]v \subset H_{p,q}(A) + B(0, L\delta^\beta). \quad (\text{A.3.37})$$

Thus this is the quantitative segment conclusion of the Bourgain–Gamburd diagonal-matrix theorem used by He, with its scalar ancestor supplied internally by Volume VI.12.

Proof. Let $\pi : M_n(\mathbb{C}) \rightarrow \Delta$ be diagonal-coordinate projection. Since $\|a - \pi a\| \leq C_n \delta$, a δ -cover of πA lifts to a $C_n \delta$ -cover of A . Fixed-dimensional covering comparison yields $\mathcal{N}_\delta(\pi A) \geq c_n \delta^{-\sigma}$. For sufficiently small δ , this forces $\sigma \leq 2n$; otherwise the hypotheses are empty. Choose $a = 1 - \sigma/(4n) \in [1/2, 1)$ and put $\rho = \delta^a$. A complex n -dimensional ball of radius ρ needs at most $C_n(\rho/\delta)^{2n}$ balls of radius δ . Hence

$$\mathcal{N}_\rho(\pi A) \geq c_n \delta^{-\sigma/2}. \quad (\text{A.3.38})$$

Fix the constants C_*, b_*, p_*, q_* of Theorem A.3.16 for dimension n and covering exponent $\sigma/(4a)$. Choose an integer k so large that

$$\frac{k}{a} - (k-1) > C_* + b_* + 2, \quad (\text{A.3.39})$$

and put $h = \delta^{a/k}$, so $h^k = \rho$. The elementary covering telescoping inequality

$$\mathcal{N}_{h^k}(\pi A) \leq C_n^k \prod_{j=0}^{k-1} \sup_z \mathcal{N}_{h^{j+1}}(\pi A \cap B(z, h^j))$$

follows by successively covering each occupied ball at the next scale. By (A.3.38), for some $0 \leq j < k$ and some $z \in \Delta$ the local factor is at least $h^{-\sigma/(3a)}$ when δ is small. Let

$E = \pi A \cap B(z, h^j)$ and choose $\zeta \in E$. Set $t = 2h^j$ and $B = t^{-1}(E - \zeta) \subset B_\Delta(0, 1)$. The covering comparison under this rescaling gives

$$\mathcal{N}_h(B) = \mathcal{N}_{th}(E) \geq h^{-\sigma/(4a)}$$

after absorbing the factor 2 and the fixed covering constants into the exponent slack. Apply [Theorem A.3.16](#) to B at scale h . It gives a unit $v \in \Delta$, a $J \in [h^{C_*}, 1]$, and

$$[0, J]v \subset H_{p_*, q_*}(B) + B(0, Jh^{b_*}). \quad (\text{A.3.40})$$

Choose for each $b = t^{-1}(\pi a - \zeta) \in B$ one $a \in A$ projecting to $\zeta + tb$, and choose a fixed $a_0 \in A$ with $\pi a_0 = \zeta$. Put $\tilde{b} = t^{-1}(a - a_0)$. Then $\pi \tilde{b} = b$ and $\|\tilde{b} - b\| \leq C_n \delta/t$. Every $\tilde{b}$ has bounded norm. Therefore each homogeneous word of degree p_* over B lifts to the same ordered matrix word over the $\tilde{b}$'s with error $O_{p_*, q_*, n}(\delta/t)$. Multiplication by t^{p_*} clears all denominators, and expansion of the p_* factors $(a - a_0)$ leaves only products of exactly p_* members of A . Hence for one fixed Q ,

$$[0, t^{p_*} J]v \subset H_{p_*, Q}(A) + B(0, Ct^{p_*} Jh^{b_*} + Ct^{p_*-1} \delta). \quad (\text{A.3.41})$$

Since $t \geq 2h^{k-1}$ and $J \geq h^{C_*}$, [\(A.3.39\)](#) implies

$$\frac{\delta}{tJ} \leq Ch^{k/a-(k-1)-C_*} \leq Ch^{b_*+2}.$$

Thus the second error in [\(A.3.41\)](#) is also at most a fixed multiple of $t^{p_*} Jh^{b_*}$. Let $L = \min\{t^{p_*} J, 1\}$. Because $j < k$, $t \geq 2h^{k-1}$, so $L \geq ch^{p_*(k-1)+C_*}$; because $t \leq 2$, the ratio $(t^{p_*} J)/L$ is bounded by 2^{p_*} . Restrict [\(A.3.41\)](#) to $[0, L]v$ and absorb fixed constants by taking $\beta = ab_*/(2k) > 0$ and enlarging C . This is [\(A.3.37\)](#). $\square$

Corollary A.3.18 (Direction inside the generated real algebra). *In [Theorem A.3.17](#), suppose additionally that A lies in a real linear subalgebra $E \subset M_n(\mathbb{C})$. After decreasing β by a fixed factor, the direction can be chosen as a unit $v_E \in E$ with $\text{dist}(v_E, \Delta) \leq \delta^\beta$. The same segment length and homogeneous-word inclusion hold. In particular one may take E to be the real algebra generated by A .*

Proof. Apply [Theorem A.3.17](#) with gain 2β . The approximant to the endpoint Lv lies in $H_{p, q}(A) \subset E$, so $\text{dist}(v, E) \leq \delta^{2\beta}$. Using the Frobenius inner product on the real vector space of complex matrices, orthogonally project v to E and normalize the projection. Norm equivalence in fixed dimension gives a unit $v_E \in E$ with $\|v_E - v\| \leq C_n \delta^{2\beta}$. Consequently every tv_E , $0 \leq t \leq L$, differs from tv by at most $C_n L \delta^{2\beta}$. For sufficiently small δ this is absorbed into $L\delta^\beta$ along with the original approximation error. Since $v \in \Delta$, the same bound controls $\text{dist}(v_E, \Delta)$. $\square$

Corollary A.3.19 (A fixed diagonal segment exponent). *Fix an integer $n \geq 1$ and $\sigma > 0$. By [Theorem A.3.15](#), there are fixed $\alpha, \beta > 0$, an integer $S \geq 1$, and $\delta_0 > 0$ such that every set A satisfying [Theorem A.3.17](#) has a unit diagonal $v \in \Delta$ for which*

$$[0, \delta^\alpha]v \subset \mathcal{W}_S(A) + B(0, \delta^{\alpha+\beta}). \quad (\text{A.3.42})$$

More precisely, the approximants can be chosen in $H_{P,Q}(A)$, where the pair (P, Q) belongs to a fixed finite list depending only on n, σ ; one pair is used for the whole resulting segment for each A . If A is contained in a real linear subalgebra E , one may instead take a unit $v_E \in E$ with $\text{dist}(v_E, \Delta) \leq \delta^\beta$, at the cost of decreasing β . All constants are independent of A .

Proof. Let C_0, b_0, p, q be the constants of [Theorem A.3.17](#). Decrease b_0 to $b = \min(b_0, 1)$ and write the segment length as $L = \delta^\gamma$, $0 \leq \gamma \leq C_0$. Its direction v is diagonal and unit.

For any $u \in [\gamma, \gamma + b/2]$, the point tv , $t = \delta^u \in [L\delta^{b/2}, L]$, has an approximant $a_t \in H_{p,q}(A)$ with $\|a_t - tv\| \leq L\delta^b \leq t\delta^{b/2}$. Choose an integer $m \geq 2$ and numbers $u_1, \dots, u_{m-1}$ in this interval. Multiply approximants to tv for these $m-1$ fixed factors by the approximant to xv for a variable $x \in [0, L]$. The elementary telescoping formula for a difference of two m -fold matrix products, together with $\|v\| = 1$, shows that every point of

$$[0, \delta^{\gamma + \sum_i u_i}]v^m \quad (\text{A.3.43})$$

is within $C_m \delta^{\gamma + \sum_i u_i + b/2}$ of one ordered word in $\mathcal{W}_{S_m}(A)$. The bound is uniform even at $x = 0$: the first factor's absolute error is $L\delta^b$, which after multiplication by the fixed factors is the segment length times δ^b . The other factors have relative error at most $\delta^{b/2}$. For bounded m , S_m is bounded in terms of m, p, q . Expanding the product of the m homogeneous differences shows that all these approximants belong to one $H_{mp, Q_m}(A)$ with Q_m bounded in terms of m, p, q ; pad the positive and negative sums by the same monomial to give them the same number of terms.

The available exponents in (A.3.43) form the interval

$$I_m(\gamma) = [m\gamma, m\gamma + (m-1)b/2].$$

Choose $m_0 \geq 2$ so that $(m_0 - 1)b/2 \geq C_0$. For every $m \geq m_0$, $I_m(\gamma)$ meets $I_{m+1}(\gamma)$, since $\gamma \leq C_0$. Their union is $[m_0\gamma, \infty)$; this remains true when $\gamma = 0$ because the right endpoints tend to infinity. Fix $\alpha > m_0 C_0 + b/4$, for instance $\alpha = m_0 C_0 + 1$. Then the exponent $\alpha - b/4$ belongs to some $I_m(\gamma)$ for every $\gamma \in [0, C_0]$. It can be obtained with $m \leq M_0$ for a fixed $M_0 = M_0(C_0, b, \alpha)$: the intervals grow to the right by at least $b/2$ in their right endpoints at each step.

Choose the corresponding m, u_i . The raw segment in (A.3.43) has scalar length $\delta^{\alpha - b/4}$. Since a unit diagonal v has a coordinate of modulus at least $n^{-1/2}$, $\|v^m\| \geq n^{-m/2} \geq n^{-M_0/2}$. Thus for sufficiently small δ its geometric length is at least δ^α . Restrict to that length and normalize the direction v^m . The raw error is $C_{M_0} \delta^{\alpha + b/4}$, which is at most $\delta^{\alpha + b/8}$ after decreasing δ_0 . Taking $S = \max_{m \leq M_0} S_m$ proves (A.3.42). The possible pairs (mp, Q_m) form the asserted finite list.

If $A \subset E$, every approximant lies in E . The endpoint $\delta^\alpha v$ of the final segment therefore has distance at most $\delta^{\alpha + b/8}$ from E , so $\text{dist}(v, E) \leq \delta^{b/8}$. Project v to E and normalize as in [Theorem A.3.18](#); the resulting direction stays within $O(\delta^{b/8})$ of Δ and the segment error remains a positive power better than δ^α . The approximating words are unchanged by this adjustment of direction, so their homogeneous degree remains (mp, Q_m) . $\square$

Corollary A.3.20 (Relaxed near-diagonal segment). *Fix $n \geq 1$ and $\sigma > 0$. By [Theorem A.3.15](#), there are $\alpha, \beta > 0$, an integer S , and $e_0 > 0$ with the following property. Let*

$E \subset M_n(\mathbb{C})$ be a real linear subalgebra, let $0 < e < e_0$, and let $\delta > 0$ be sufficiently small. If $B \subset E$ satisfies

$$B \subset B(0, \delta^{-e}), \quad \mathcal{N}_\delta(B) \geq \delta^{-\sigma}, \quad \text{dist}(b, \Delta) \leq \delta^{1-e} \quad (b \in B), \quad (\text{A.3.44})$$

then some unit $v \in E$ satisfies

$$[0, \delta^\alpha]v \subset \mathcal{W}_S(B) + B(0, \delta^{\alpha+\beta}). \quad (\text{A.3.45})$$

The constants are independent of E, B, e .

Proof. Put $t = \delta^e$ and $B_0 = tB$. Since Δ is a linear subspace, $B_0 \subset B(0, 1)$ and $\text{dist}(B_0, \Delta) \leq \delta$. In the real ambient dimension $D = 2n^2$, a ball of radius δ^{1-e} is covered by at most $C_n \delta^{-De}$ balls of radius δ . Hence

$$\mathcal{N}_\delta(B_0) = \mathcal{N}_{\delta^{1-e}}(B) \geq c_n \delta^{De} \mathcal{N}_\delta(B) \geq \delta^{-\sigma/2}$$

when $De < \sigma/4$ and δ is small. Apply the algebra-direction version of [Theorem A.3.19](#) to B_0 with covering exponent $\sigma/2$. Write its constants as α_0, β_0, S_0 , and let $P_{\max}$ be the largest product degree in its finite homogeneous-degree list. It gives a unit $v \in E$ and, for one fixed (P, Q) from that list,

$$[0, \delta^{\alpha_0}]v \subset H_{P,Q}(B_0) + B(0, \delta^{\alpha_0+\beta_0}).$$

Every term of $H_{P,Q}(B_0)$ has exactly P factors, so $H_{P,Q}(B_0) = t^P H_{P,Q}(B)$. Multiplying the inclusion by t^{-P} yields

$$[0, \delta^{\alpha_0-Pe}]v \subset H_{P,Q}(B) + B(0, \delta^{\alpha_0+\beta_0-Pe}).$$

Set $\alpha = \alpha_0 + \beta_0/4$ and $\beta = \beta_0/2$. If e_0 is small enough that $P_{\max}e_0 < \beta_0/4$, the last segment contains $[0, \delta^\alpha]v$ and its error is at most $\delta^{\alpha+\beta}$. Since the finite list has bounded leaf counts, one S contains every possible $H_{P,Q}(B)$ in $\mathcal{W}_S(B)$. $\square$

A.3.4 Matrix sum-product and random linear actions

Theorem A.3.21 (He's matrix-algebra sum-product theorem). *Fix $m \geq 1, \kappa > 0$, and $0 < \varsigma < m$. There is $\varepsilon > 0$ such that, for all sufficiently small $\delta > 0$, the following holds. Let $\mathcal{A} \subset \text{End}(\mathbb{R}^m)$ and $X \subset \mathbb{R}^m$ satisfy*

$$\begin{aligned} \mathcal{A}, X &\subset B(0, \delta^{-\varepsilon}), \\ \mathcal{N}_\rho(\mathcal{A}), \mathcal{N}_\rho(X) &\geq \delta^\varepsilon \rho^{-\kappa} \quad (\rho \geq \delta), \\ \mathcal{N}_\delta(X) &\leq \delta^{-\varsigma-\varepsilon}. \end{aligned}$$

Assume also that, for every proper nonzero linear subspace $W \subset \mathbb{R}^m$, there are $a \in \mathcal{A}$ and $w \in W \cap B(0, 1)$ with $\text{dist}(aw, W) \geq \delta^\varepsilon$. Then

$$\mathcal{N}_\delta(X + X) + \max_{a \in \mathcal{A}} \mathcal{N}_\delta(X + aX) \geq \delta^{-\varepsilon} \mathcal{N}_\delta(X). \quad (\text{A.3.46})$$

Source and proof route. He, *Discretized sum-product estimates in matrix algebras*, arXiv:1611.09639v2, Theorem 3, printed pp. 3–4; it is quoted as Theorem 8 in his projection paper. The preceding Cartesian complex-ring and near-diagonal propositions reconstruct Bourgain–Gamburd Proposition 7 and Corollary 8 from the scalar segment interface, which canonical Volume VI.12 proves through its Pass-98 chain. Below we prove the probabilistic action lemma, effective real Wedderburn classification, almost-generated model-algebra reduction, approximate-ring calculus, the one-direction segment and ball-generation implications, and the final multiscale implication from He Theorem 5 to Theorem 3. The formerly external avoidance-to-escape step follows from the self-contained polynomial error bound [Theorem A.3.43](#); the general real-analytic Łojasiewicz theorem is stated only for source context.

Proposition A.3.22 (Growth under a nonconcentrated random linear action). *Let $m \geq 1$, $\kappa > 0$, and $0 < \varsigma < \tau$. Put*

$$g = \frac{\kappa(\tau - \varsigma)}{\kappa + \tau}, \quad c = \frac{g}{4}.$$

There is $e_0 = e_0(m, \kappa, \varsigma, \tau) > 0$ such that for $0 < e < e_0$ and sufficiently small $\delta > 0$ the following holds. Let $X \subset B(0, \delta^{-e}) \subset \mathbb{R}^m$ be bounded and let λ be a probability on $\text{End}(\mathbb{R}^m)$, supported in the operator-norm ball of radius δ^{-e} . Assume

$$\mathcal{N}_r(X) \geq \delta^e r^{-\kappa} \quad (r \geq \delta), \quad (\text{A.3.47})$$

$$\mathcal{N}_\delta(X) \leq \delta^{-\varsigma - e}, \quad \lambda\{T : \|Tv - w\| \leq r\} \leq \delta^{-e} r^\tau \quad (\|v\| = 1, w \in \mathbb{R}^m, r \geq \delta). \quad (\text{A.3.48})$$

Then

$$\mathcal{N}_\delta(X + X) + \sup_{T \in \text{spt } \lambda} \mathcal{N}_\delta(X + TX) \geq \delta^{-c} \mathcal{N}_\delta(X). \quad (\text{A.3.49})$$

Proof. Write $N = \mathcal{N}_\delta(X)$. If $\mathcal{N}_\delta(X + X) \geq \delta^{-c} N$, there is nothing to prove. Otherwise put

$$a = \frac{\tau - \varsigma}{\tau + \kappa} \in (0, 1), \quad \rho = \delta^a, \quad R = \delta^{1-2e}.$$

We first turn small doubling into a local counting bound. Choose a 4ρ -separated subset $Z \subset X$ maximal at that separation, and choose a ball $B(w, \rho)$ for which $\mathcal{N}_\delta(X \cap B(w, \rho))$ is maximal. The translates $z + (X \cap B(w, \rho))$, $z \in Z$, are pairwise separated by at least 2ρ . Hence, up to a fixed dimensional covering constant,

$$\mathcal{N}_\delta(X + X) \geq c_m \mathcal{N}_{4\rho}(X) \max_w \mathcal{N}_\delta(X \cap B(w, \rho)).$$

Since Euclidean covering is doubling and $\mathcal{N}_{4\rho}(X) \geq c_m \mathcal{N}_\rho(X)$, the first inequality in [\(A.3.47\)](#) yields

$$\max_w \mathcal{N}_\delta(X \cap B(w, \rho)) \leq C_m \delta^{-c-e} \rho^\kappa N. \quad (\text{A.3.50})$$

Let $Z_\delta \subset X$ be a maximal 2δ -separated set; write $M = \#Z_\delta$. Packing gives $C_m^{-1} N \leq M \leq C_m N$. For $T \in \text{spt } \lambda$, define its coarse additive energy

$$E_R(T) = \#\{(x, y, x', y') \in Z_\delta^4 : \|x + Ty - x' - Ty'\| \leq R\}.$$

Assign every sum $x + Ty$ to one radius- δ ball in a minimal cover of $X + TX$. Pairs assigned to the same ball differ by at most $2\delta \leq R$. Cauchy–Schwarz therefore gives

$$\mathcal{N}_\delta(X + TX) \geq \frac{M^4}{E_R(T)}. \quad (\text{A.3.51})$$

Split the expectation of $E_R(T)$ according to $\|y - y'\| \geq \rho$ or $< \rho$. In the first case set $v = (y - y')/\|y - y'\|$ and apply (A.3.48) with radius $R/\|y - y'\|$. This radius is at least δ : indeed $\|y - y'\| \leq 2\delta^{-e}$, so it is at least $\frac{1}{2}\delta^{1-e} > \delta$ for small δ . Thus the contribution of the far pairs is at most

$$\delta^{-e} R^\tau \rho^{-\tau} M^4 \leq C_m \delta^{\tau-\zeta-C_1 e} \rho^{-\tau} M^3,$$

where (A.3.48) supplies $M \leq C_m \delta^{-\zeta-e}$.

For each $y \in Z_\delta$, the number of y' with $\|y - y'\| < \rho$ is, by (A.3.50) and packing, at most $C_m \delta^{-c-e} \rho^\kappa M$. For fixed x, y, y' and T , the event in the definition of $E_R(T)$ confines x' to a ball of radius R ; the 2δ -separation of Z_δ gives at most $C_m(R/\delta)^m = C_m \delta^{-2me}$ choices. Hence the contribution of the near pairs is at most

$$C_m \delta^{-c-C_2 e} \rho^\kappa M^3.$$

With the chosen $\rho = \delta^a$, both $\tau - \zeta - a\tau$ and $a\kappa$ equal g . Consequently

$$\mathbb{E}_\lambda E_R(T) \leq C_m \delta^{g-c-C_3 e} M^3. \quad (\text{A.3.52})$$

Since $E_R(T)$ is a finite sum of Borel indicators, some $T \in \text{spt } \lambda$ satisfies $E_R(T) \leq \mathbb{E}_\lambda E_R(T)$. For this T , (A.3.51) and (A.3.52) give

$$\mathcal{N}_\delta(X + TX) \geq \frac{M^4}{\mathbb{E}_\lambda E_R(T)} \geq c_m \delta^{-g+c+C_3 e} M.$$

Since $g = 4c$, choose e_0 so that $C_3 e_0 < c$. For sufficiently small δ , the last lower bound exceeds $\delta^{-c} N$, proving (A.3.49). $\square$

The action hypothesis is automatic for the unit ball in each matrix algebra that occurs in the finite-dimensional irreducible-algebra classification. The formulation below states the exact linear-algebra property needed. The classification itself is proved below.

Corollary A.3.23 (Unit-ball action growth). *Let E be a linear subspace of $\text{End}(\mathbb{R}^m)$. Assume that for every unit $v \in \mathbb{R}^m$ the evaluation map*

$$L_v : E \longrightarrow \mathbb{R}^m, \quad T \longmapsto Tv$$

is onto and has a linear right inverse of norm at most a fixed C_E . For $\kappa > 0$ and $0 < \zeta < m$ there are $e, c > 0$ such that, for all sufficiently small δ , every bounded $X \subset B(0, \delta^{-e})$ with

$$\mathcal{N}_r(X) \geq \delta^e r^{-\kappa} \quad (r \geq \delta), \quad \mathcal{N}_\delta(X) \leq \delta^{-\zeta-e}$$

satisfies

$$\mathcal{N}_\delta(X + X) + \sup_{T \in E, \|T\| \leq 1} \mathcal{N}_\delta(X + TX) \geq \delta^{-c} \mathcal{N}_\delta(X).$$

In particular the assertion applies to $E = \text{End}(\mathbb{R}^m)$.

Proof. Equip E with its Euclidean Lebesgue measure and normalize its restriction to its operator-norm unit ball to a probability λ . Put $D = \dim E$. For each unit v , decompose $E = \ker L_v \oplus (\ker L_v)^\perp$. The right-inverse bound makes each singular value of $L_v|_{(\ker L_v)^\perp}$ at least C_E^{-1} . The $(D-m)$ -dimensional volume of a section of the unit ball parallel to $\ker L_v$ is uniformly bounded in v . Fubini and the linear change-of-variables formula therefore give

$$\lambda\{T : \|Tv - w\| \leq r\} \leq C_{E,m} r^m \quad (w \in \mathbb{R}^m, r > 0).$$

For sufficiently small δ , the fixed coefficient is at most δ^{-e} . Apply [Theorem A.3.22](#) with $\tau = m$, then decrease its output exponent if necessary. For $E = \text{End}(\mathbb{R}^m)$, the map $w \mapsto ww^T$ is a right inverse of L_v with norm one, proving the last assertion. $\square$

Remark A.3.24. *The other two model algebras in He’s matrix theorem also satisfy the corollary’s hypothesis. Let $\mathbb{H} = \mathbb{R} \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k$ be the quaternion division algebra, with $i^2 = j^2 = k^2 = -1$, $ij = k = -ji$, $jk = i = -kj$, $ki = j = -ik$, and its Euclidean norm. For $\mathbb{F} = \mathbb{C}$ or $\mathbb{H}$, regard $\mathbb{F}^r$ as a real Euclidean space and let $E = M_r(\mathbb{F})$ act by left matrix multiplication. If $\|v\| = 1$, some coordinate v_j has $|v_j| \geq r^{-1/2}$. For prescribed $w \in \mathbb{F}^r$, the matrix with its j th column $(w_i v_j^{-1})_{i=1}^r$ and all other columns zero sends v to w and has norm at most $C_r \|w\|$. Thus evaluation has a uniformly bounded right inverse, both for the complex and quaternionic realifications.*

A.3.5 Effective real matrix-algebra structure

The source’s effective Wedderburn step has an algebraic and a metric part. We prove both, and also give the extension in which a subalgebra need not initially contain the identity. He explicitly requires subalgebras to contain the identity, so this extension is not needed to cover his statement. The algebraic part identifies the commutant; quantitative conjugacy then follows from a singular-value argument and invariant averaging.

Definition A.3.25 (Real matrix algebras and commutants). *A real subalgebra of $\text{End}_{\mathbb{R}}(V)$ is a real-linear subspace closed under composition; it is unital if it contains I_V . For a subset $S \subset \text{End}_{\mathbb{R}}(V)$, its commutant or centralizer is*

$$C(S) = \{T \in \text{End}_{\mathbb{R}}(V) : Ts = sT \text{ for every } s \in S\}.$$

A finite-dimensional real division algebra is a nonzero associative unital real algebra in which every nonzero element has a two-sided inverse. For $0 < \rho \leq 1$, a set $A \subset \text{End}_{\mathbb{R}}(V)$ acts ρ -irreducibly on V if for every nonzero proper real subspace $W \subset V$ there are $a \in A$ and $w \in W$ with $\|w\| \leq 1$ and $\text{dist}(aw, W) \geq \rho$. A subalgebra E acts ρ -irreducibly if $E \cap B(0, 1)$ acts so in this sense.

Lemma A.3.26 (Centralizer norm bound). *Let $0 < \rho \leq \frac{1}{2}$, and let E be a subalgebra of $\text{End}(\mathbb{R}^m)$ such that for every nonzero proper subspace $W \subset \mathbb{R}^m$ there are $a \in E$, $w \in W$ with $\|a\|, \|w\| \leq 1$ and $\text{dist}(aw, W) \geq \rho$. If $g \in \text{GL}_m(\mathbb{R})$ commutes with E and $|\det g| = 1$, then*

$$\|g\|, \|g^{-1}\| \leq \rho^{-m}.$$

Proof. Let $s_1 \geq \dots \geq s_m > 0$ be the singular values of g . If every ratio s_{j+1}/s_j were at least ρ , then

$$1 = \prod_{j=1}^m s_j \geq s_1^m \rho^{m(m-1)/2},$$

so $s_1 \leq \rho^{-(m-1)/2} < \rho^{-m}$. If instead $s_1 > \rho^{-m}$, some $s_{p+1}/s_p < \rho$, with $1 \leq p < m$. Write $g = k \operatorname{diag}(s_1, \dots, s_m) \ell$ with orthogonal k, ℓ , and put $W = k \operatorname{span}(e_1, \dots, e_p)$. For $w \in W$, $\|g^{-1}w\| \leq s_p^{-1}\|w\|$. For every z , the component of gz perpendicular to W has norm at most $s_{p+1}\|z\|$. If $a \in E$ commutes with g and $\|a\|, \|w\| \leq 1$, it follows that

$$\operatorname{dist}(aw, W) = \operatorname{dist}(gag^{-1}w, W) \leq s_{p+1}s_p^{-1} < \rho,$$

contrary to the hypothesis on E . Thus $\|g\| \leq \rho^{-m}$. Apply the same argument to g^{-1} , which also commutes with E and has determinant modulus one. $\square$

Lemma A.3.27 (Finite-dimensional real division algebras). *Every finite-dimensional associative unital real division algebra D is isomorphic, as a real algebra, to $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$.*

Proof. Every $x \in D$ is algebraic over $\mathbb{R}$. Its minimal polynomial is irreducible: if it factored as fg , then $f(x)g(x) = 0$ in a division algebra, so one factor would annihilate x . Every irreducible real polynomial has degree one or two, by the fundamental theorem of algebra. If every element has degree one, then $D = \mathbb{R}$. Otherwise a degree-two element can be translated and scaled to an element i with $i^2 = -1$; it generates a copy $C = \mathbb{R}[i] \cong \mathbb{C}$.

The centralizer $C_D(i)$ is exactly C . Indeed, if y commutes with i , finite dimensionality gives a nonzero polynomial over C annihilating y . Factor that polynomial into linear factors over the algebraically closed field C . The absence of zero divisors forces y to equal one of its roots in C .

The real-linear involution $x \mapsto xix$ decomposes D into the commuting part C and the anticommuting part $D_- = \{x : xi = -ix\}$. If $D_- = \{0\}$, then $D = C$. Otherwise choose $0 \neq j \in D_-$. Its square commutes with both i and j . Thus $j^2 = a + bi \in C$, and $j(a + bi) = (a - bi)j$ forces $b = 0$. The real number a is nonzero; it cannot be positive, since then $(j - \sqrt{a})(j + \sqrt{a}) = 0$ would make j real. Rescale j to have $j^2 = -1$. For any $x \in D_-$, the product xj^{-1} commutes with i , hence lies in C . Therefore $D_- = Cj$ and

$$D = C \oplus Cj, \quad jz = \bar{z}j \ (z \in C), \quad j^2 = -1.$$

These are precisely the quaternion multiplication rules, proving $D \cong \mathbb{H}$. $\square$

Lemma A.3.28 (Finite-dimensional double centralizer). *Let $E \subset \operatorname{End}_{\mathbb{R}}(V)$ be a nonzero, not necessarily unital algebra acting irreducibly on a finite-dimensional real vector space V , and put $K = C(E)$. Then $I_V \in E$, K is a real division algebra, and $E = C(K)$.*

Proof. First consider the unitization $E^+ = E + \mathbb{R}I_V$. It acts irreducibly and has the same commutant as E . We first prove the assertion for E^+ ; to simplify notation write E for this unital algebra until the last paragraph of the proof. For $0 \neq k \in K$, both $\ker k$ and $\operatorname{im} k$ are E -invariant. Thus $\ker k = 0$ and $\operatorname{im} k = V$; also $k^{-1} \in K$. This proves that K is a division algebra.

We prove the following finite-dimensional density claim by induction on s : for every K -linearly independent $v_1, \dots, v_s \in V$, the map

$$E \longrightarrow V^s, \quad a \longmapsto (av_1, \dots, av_s)$$

is surjective. For $s = 1$, the image Ev_1 is a nonzero E -invariant subspace and hence equals V . Assume the claim for $s - 1$, and let $M \subset V^s$ be the image for s . Its projection to the first $s - 1$ coordinates is all of V^{s-1} . The kernel of that projection is an E -invariant subspace of the last copy of V , hence either $\{0\}$ or V . If it is V , then $M = V^s$. If it is zero, M is the graph of an E -linear map $\phi : V^{s-1} \rightarrow V$. Each coordinate component of ϕ belongs to $\text{End}_E(V) = K$, so $\phi(x_1, \dots, x_{s-1}) = \sum_{j < s} k_j x_j$ for some $k_j \in K$. But the identity element of E puts $(v_1, \dots, v_s) \in M$, forcing $v_s = \sum_{j < s} k_j v_j$, contrary to K -independence. The claim follows.

Choose a left K -basis of V . Every K -linear endomorphism of V is prescribed by its values on that basis; the density claim realizes those values by an element of E . Since each member of E already commutes with K , $E = \text{End}_K(V) = C(K)$.

Return to the original algebra E_{old} . It is a nonzero two-sided ideal in $E^+ = \text{End}_K(V)$. This endomorphism algebra is simple: in a K -basis, multiplication of any nonzero matrix on the left and right by elementary matrices isolates a nonzero elementary matrix. Multiplication by a diagonal matrix whose selected entry is the inverse of its nonzero K -entry, followed by further elementary multiplications, generates every elementary matrix. Hence $E_{\text{old}} = E^+$, in particular $I_V \in E_{\text{old}}$, and the asserted equality holds for the original algebra. $\square$

Proposition A.3.29 (Conditioned conjugacy after algebraic classification). *Let $E \subset \text{End}(\mathbb{R}^m)$ satisfy the hypotheses of [Theorem A.3.26](#), and put $K = C(E)$. Assume, as an explicit algebraic input, that $E = C(K)$ and that K is isomorphic as a unital real algebra to $\mathbb{C}$ or $\mathbb{H}$. Write $d = \dim_{\mathbb{R}} K$, so $d \in \{2, 4\}$. Then $d \mid m$, and there is a real change of basis $g \in \text{GL}_m(\mathbb{R})$ with*

$$\|g\| + \|g^{-1}\| \leq 2\rho^{-m}$$

such that $g^{-1}Kg$ is the standard block action of K on $K^{m/d}$, and $g^{-1}Eg$ is its standard matrix-algebra commutant.

Proof. Fix an algebra isomorphism $\psi : \mathbb{F} \rightarrow K$, with $\mathbb{F} = \mathbb{C}$ or $\mathbb{H}$, and let $\mathbb{U} = \{q \in \mathbb{F} : |q| = 1\}$. This compact group is connected. Since $\psi(q)$ is invertible and $\det \psi(1) = 1$, continuity gives $\det \psi(q) = 1$ for every $q \in \mathbb{U}$. Each $\psi(q)$ commutes with E ; hence [Theorem A.3.26](#) gives

$$\|\psi(q)\|, \|\psi(q)^{-1}\| \leq \rho^{-m} \quad (q \in \mathbb{U}).$$

Average the Euclidean inner product over the rotation-invariant probability on the unit circle or unit three-sphere:

$$\langle v, w \rangle_* = \int_{\mathbb{U}} \langle \psi(q)v, \psi(q)w \rangle dq.$$

Left multiplication preserves that probability, so every $\psi(q)$ is orthogonal for $\langle \cdot, \cdot \rangle_*$. Moreover,

$$\rho^{2m} \|v\|^2 \leq \|v\|_*^2 \leq \rho^{-2m} \|v\|^2.$$

Choose a $*$ -unit vector v_1 . In the complex case, $v_1, \psi(i)v_1$ are $*$ -orthonormal: $\psi(i)$ is orthogonal, its inverse is $-\psi(i)$, and thus it is skew-adjoint. In the quaternionic case,

$$v_1, \psi(i)v_1, \psi(j)v_1, \psi(k)v_1$$

are $*$ -orthonormal by the same calculation and the quaternion multiplication rules. Their real span is a copy of $\mathbb{F}$; its $*$ -orthogonal complement is $\psi(\mathbb{F})$ -invariant because every unit $\psi(q)$ is $*$ -orthogonal. Repeat on that complement. This produces m/d mutually orthogonal $\mathbb{F}$ -blocks and proves $d \mid m$. Let g send the standard real basis of these blocks to the constructed basis. It is an isometry from standard Euclidean space to $(\mathbb{R}^m, \langle \cdot, \cdot \rangle_*)$, whence the norm comparison gives $\|g\|, \|g^{-1}\| \leq \rho^{-m}$. The block basis makes $g^{-1}Kg$ the standard left $\mathbb{F}$ -action. Taking commutants and using the explicit hypothesis $E = C(K)$ gives the last assertion. For the conventional embedding by left multiplication of matrices in $M_{m/d}(\mathbb{F})$, compose g with coordinatewise complex or quaternionic conjugation. This map is real orthogonal and changes the commutant of the left scalar action into that conventional matrix action, without changing the norm bounds. $\square$

Corollary A.3.30 (Effective real Wedderburn classification). *Let $0 < \rho \leq \frac{1}{2}$, and let a subalgebra $E \subset \text{End}(\mathbb{R}^m)$ with $E \neq \{0\}$ act ρ -irreducibly as in Theorem A.3.26. Then E is either $\text{End}(\mathbb{R}^m)$ or is conjugate to the standard realification of $M_{m/2}(\mathbb{C})$ or $M_{m/4}(\mathbb{H})$. In the latter cases the conjugating real matrix g satisfies $\|g\| + \|g^{-1}\| \leq 2\rho^{-m}$.*

Proof. Quantitative irreducibility implies ordinary irreducibility. By Theorem A.3.28, $K = C(E)$ is a real division algebra and $E = C(K)$. Theorem A.3.27 gives $K \cong \mathbb{R}, \mathbb{C}$, or $\mathbb{H}$. If $K = \mathbb{R}$, then $E = C(K) = \text{End}(\mathbb{R}^m)$. In the other two cases apply Theorem A.3.29. $\square$

Remark A.3.31 (The one-dimensional convention). *For $m \geq 2$, quantitative irreducibility already implies $E \neq \{0\}$. If “subalgebra” is permitted to mean a nonunital algebra and $m = 1$, the zero algebra satisfies the irreducibility condition vacuously but is not $M_1(\mathbb{R})$. Thus the nonzero condition is indispensable in precisely this extended convention. He’s paper defines all subalgebras to contain the identity, so its Proposition 21 has no such exception.*

A.3.6 Avoidance, semialgebraic projection, and escape

Definition A.3.32 (Quantitative avoidance of subalgebras). *Let A be a subset of a finite-dimensional normed real algebra E , and let $t > 0$. We say that A is t -away from proper subalgebras of E if for every proper real subalgebra $F \subsetneq E$, including a nonunital one, there is $a \in A$ such that $\text{dist}(a, F) > t$. This is slightly stronger than He’s convention, which tests only unital subalgebras.*

Lemma A.3.33 (A finite witness for algebra avoidance). *Let E be a unital finite-dimensional normed real algebra of dimension $D \geq 2$. There is a constant $C_D \geq 1$, depending also on the fixed norm of E , with the following property. If $0 < \rho \leq \frac{1}{2}$, $A \subset B_E(0, \rho^{-1})$, and for every proper unital subalgebra $F \subsetneq E$ there is $a \in A$ with $\text{dist}(a, F) \geq \rho$, then some $A_0 \subset A$ of cardinality at most D is c -away from every proper unital subalgebra, where $c = \rho^{C_D}$.*

Proof. Fix $c = \rho^{C_D}$, with C_D chosen large below. Begin with the empty list. If the selected list $b_1, \dots, b_k$ is not c -away, there is a proper unital subalgebra F with $\text{dist}(b_j, F) \leq c$ for every $j \leq k$. Choose $b_{k+1} \in A$ with $\text{dist}(b_{k+1}, F) \geq \rho$. We prove inductively that

$$\text{dist}(b_j, \text{span}(b_1, \dots, b_{j-1})) \geq \rho/2 \quad (1 \leq j \leq k). \quad (\text{A.3.53})$$

For the first vector this follows because $0 \in F$. Suppose it holds through k . Equivalence with a Euclidean norm and the Gram–Schmidt determinant formula show that, whenever $x = \sum_{j \leq k} t_j b_j$ and $\|x\| \leq 2\rho^{-1}$, then

$$\sum_{j \leq k} |t_j| \leq C_E \rho^{-2D-2}. \quad (\text{A.3.54})$$

Indeed, the determinant of the column map on its range is at least a fixed multiple of $(\rho/2)^k$, while every $(k-1)$ -column minor is at most $C_E \rho^{-(k-1)}$; Cramer’s rule gives the displayed deliberately loose bound. Let x be a nearest point in the span to b_{k+1} . Since 0 belongs to the span, $\|x\| \leq 2\|b_{k+1}\| \leq 2\rho^{-1}$. Approximating each b_j by a point of F and using (A.3.54) gives

$$\text{dist}(x, F) \leq C_E \rho^{-2D-2} c.$$

Take $C_D > 2D + 5$, and if necessary enlarge it to absorb C_E uniformly for $\rho \leq \frac{1}{2}$. Then this bound is at most $\rho/2$, so

$$\text{dist}(b_{k+1}, \text{span}(b_1, \dots, b_k)) \geq \text{dist}(b_{k+1}, F) - \text{dist}(x, F) \geq \rho/2.$$

Thus every step increases the span dimension. After at most D steps no further step is possible, so the selected list is c -away from every proper unital subalgebra. $\square$

Definition A.3.34 (Semialgebraic set and function). *A subset of $\mathbb{R}^n$ is semialgebraic if it is a finite Boolean combination of sets $\{x : P(x) = 0\}$ and $\{x : P(x) > 0\}$ for real polynomials P . A function on a semialgebraic domain is semialgebraic when its graph is semialgebraic.*

He’s source proof uses the following general analytic inequality in passing from avoidance of subalgebras to escape from arbitrary linear subspaces. We state it for context, but the proof here uses only its polynomial instance, proved below from elementary algebra and analysis.

Theorem A.3.35 (Compact real-analytic Łojasiewicz inequality: external input). *Let M be a real-analytic manifold with a Riemannian distance, $f : M \rightarrow \mathbb{R}$ real analytic, $Z = f^{-1}(0)$, and $K \subset M$ compact. There are $c, C > 0$, depending on f, K , such that*

$$|f(x)| \geq c \min\{\text{dist}(x, Z)^C, 1\} \quad (x \in K),$$

with the minimum interpreted as 1 when $Z = \emptyset$.

Source and scope. This general analytic theorem is not proved here and is listed as contextual Category 3, not as an input to any proof in this volume. The exact form is He [6, Theorem 13, p. 9], citing Łojasiewicz [7, Théorème 2, p. 62]. One may also derive the needed polynomial special case from Bierstone–Milman [8, Theorem 6.4 and Remark 6.5, p. 34], but that route requires their earlier semianalytic-set theory. Our independent polynomial proof is Theorem A.3.43.

Lemma A.3.36 (Quantitative two-sided escape from algebra avoidance). *Let E be a fixed finite-dimensional normed simple unital real algebra. There is $C_E \geq 1$ such that, for $0 < \rho \leq \frac{1}{2}$, if $A \subset B_E(0, \rho^{-1})$ is ρ -away from every proper unital subalgebra, then for every nonzero proper real subspace $W \subset E$ there exist $a \in A$ and $w \in W \cap B_E(0, 1)$ for which*

$$\max\{\text{dist}(aw, W), \text{dist}(wa, W)\} \geq \rho^{C_E}. \quad (\text{A.3.55})$$

The conclusion is vacuous for $E = \mathbb{R}$.

Proof. Put $D = \dim_{\mathbb{R}} E$ and suppose $D \geq 2$. By [Theorem A.3.33](#), choose $A_0 = \{a_1, \dots, a_N\} \subset A$, $N \leq D$, which is c_0 -away from every proper unital subalgebra, where $c_0 = \rho^{C_0}$ and C_0 depends only on E . Equip E temporarily with a fixed Euclidean norm. Equivalence of norms changes all bounds below by fixed constants, which are absorbed by enlarging exponents for $\rho \leq \frac{1}{2}$.

Suppose that [\(A.3.55\)](#) fails for a k -plane W , $1 \leq k < D$. Set $x_i = \rho a_i$; then $\|x_i\| \leq C_E^{(0)}$. For $U \in \text{Gr}(E, k)$, let P_U be Euclidean orthogonal projection onto U and define

$$F_{k,N}(U, y_1, \dots, y_N) = \sum_{i=1}^N \left(\|(I - P_U)L_{y_i}P_U\|_{\text{HS}}^2 + \|(I - P_U)R_{y_i}P_U\|_{\text{HS}}^2 \right),$$

where $L_y z = yz$ and $R_y z = zy$. In local analytic coordinates on the Grassmannian this is a real-analytic nonnegative function. It vanishes exactly when every y_i preserves U under both left and right multiplication. The failed escape bound, applied to a Euclidean orthonormal basis of W and using $x_i = \rho a_i$, gives

$$F_{k,N}(W, x_1, \dots, x_N) \leq C_E^{(1)} \rho^{2(C_E+1)}.$$

Identify each k -plane with its orthogonal projector. Let $\mathcal{P}_k = \{P \in \text{End}(E) : P^T = P, P^2 = P, \text{tr } P = k\}$ and let $\mathcal{Z}_{k,N}$ be the zero set of $F_{k,N}$ in $\mathcal{P}_k \times E^N$. Apply [Theorem A.3.43](#), proved below, to $F_{k,N}$ on the compact semialgebraic set $\mathcal{P}_k \times \overline{B(0, C_E^{(0)})}^N$. There are constants $c_{k,N}, \theta_{k,N} > 0$ independent of ρ, A, W such that

$$F_{k,N}(P, y) \geq c_{k,N} \min\{\text{dist}((P, y), \mathcal{Z}_{k,N} \cap K_R)^{\theta_{k,N}}, 1\}, \quad K_R = \overline{\mathcal{P}_k \times B(0, C_E^{(0)})}^N.$$

If the displayed upper bound on $F_{k,N}(W, x)$ is below $c_{k,N}$, this puts $(W, x_1, \dots, x_N)$ within $C_{k,N} \rho^{2(C_E+1)/\theta_{k,N}}$ of the zero set. Choose the exponent C_E in the escape assertion so large that this distance is less than $\rho c_0/2$ for all $1 \leq k < D$ and $1 \leq N \leq D$. The zero set is nonempty, since every tuple with all $y_i = 0$ belongs to it. Thus there are $U \in \text{Gr}(E, k)$ and $y_i \in E$ with $F_{k,N}(U, y_1, \dots, y_N) = 0$ and $\|x_i - y_i\| < \rho c_0/2$ for every i .

The common two-sided stabilizer

$$E_U = \{z \in E : zU \subset U \text{ and } Uz \subset U\}$$

is a unital subalgebra. It is proper: if $E_U = E$, then U would be a nonzero proper two-sided ideal of the simple algebra E . Since every $y_i \in E_U$, also $y_i/\rho \in E_U$, and consequently $\text{dist}(a_i, E_U) < c_0/2$ for every i . This contradicts the c_0 -avoidance of A_0 and proves [\(A.3.55\)](#). $\square$

Remark A.3.37 (Why the plane is allowed to move). *The needed distance is to $\mathcal{L}_{k,N}$ in the joint plane–element variables. A tempting fixed-plane substitute is false. Indeed, give $E = M_2(\mathbb{R})$ its Frobenius inner product, put $v_t = \text{diag}(1, t)$, $U_t = \mathbb{R}v_t$, and $a = e_{11}$. For $t > 0$, the two-sided stabilizer $E_{U_t} = \{b : bU_t \subset U_t, U_t b \subset U_t\}$ equals $\mathbb{R}I$, because v_t is invertible, whereas E_{U_0} is the algebra of diagonal matrices. Hence $\text{dist}(a, E_{U_t}) = 1/\sqrt{2}$ for every $t > 0$. But orthogonal projection onto U_t gives the exact defect*

$$\left\| (I - P_{U_t}) \frac{e_{11}}{\sqrt{1+t^2}} \right\|_F^2 = \frac{1}{1+t^2} - \frac{1}{(1+t^2)^2} = \frac{t^2}{(1+t^2)^2}, \quad F_{1,1}(U_t, a) = \frac{2t^2}{(1+t^2)^2} \rightarrow 0.$$

Thus no uniform inequality $\text{dist}(a, E_U) \leq CF_{1,1}(U, a)^{1/2}$ can replace the joint error bound. Moving U_t to U_0 removes the apparent contradiction: (U_0, a) belongs to the zero set.

We now prove the algebraic error bound invoked above. The first lemma extracts a power law from a semialgebraic minimum-error profile. The trace-form argument that follows establishes semialgebraicity of that profile without importing elimination theory.

Lemma A.3.38 (Power lower bound for a semialgebraic germ). *Let $h : (0, \varepsilon) \rightarrow (0, \infty)$ be semialgebraic and bounded on some interval $(0, \varepsilon_1)$ with $\varepsilon_1 \leq \varepsilon$. Then there are $c > 0$, an integer $N \geq 1$, and $\varepsilon_0 \in (0, \varepsilon)$ such that*

$$h(t) \geq ct^N \quad (0 < t < \varepsilon_0).$$

Proof. Choose a finite Boolean formula in polynomial equalities and strict inequalities describing the graph of h , and discard identically zero polynomials. At each graph point at least one polynomial occurring in the formula vanishes. Otherwise every sign in the formula is locally constant, so the graph would contain a nonempty open subset of $\mathbb{R}^2$, impossible for the graph of a function. The product $P(t, s)$ of the remaining, finitely many nonzero polynomials is therefore nonzero and satisfies $P(t, h(t)) = 0$ for every t in the domain.

Factor $P(t, s) = s^q Q(t, s)$ with $q \geq 0$ maximal. Then $a_0(t) := Q(t, 0)$ is a nonzero polynomial, and positivity of h gives $Q(t, h(t)) = 0$. Write $Q(t, s) = a_0(t) + sR(t, s)$. Boundedness of h implies that $R(t, h(t))$ is bounded for sufficiently small t ; hence $|a_0(t)| \leq Ch(t)$. There are an integer $r \geq 0$ and a polynomial b with $b(0) \neq 0$ such that $a_0(t) = t^r b(t)$. Consequently $h(t) \geq ct^r$ for small positive t . Taking $N = \max\{1, r\}$ and shrinking ε_0 below $\min\{1, \varepsilon_1\}$ gives the claimed bound. $\square$

The remaining projection step can be proved without taking a general real-algebraic elimination theorem as an input. The finite-dimensional trace form below detects signs at the real roots of a polynomial, including repeated roots. We give the details because a bare quantifier-elimination citation would not meet the proof contract.

Lemma A.3.39 (Hermite trace form and real-root signs). *Let $p, q \in \mathbb{R}[T]$, with $\deg p = d \geq 1$. On the real vector space $A_p = \mathbb{R}[T]/(p)$ define the symmetric bilinear form*

$$B_{p,q}(\bar{u}, \bar{v}) = \text{Tr}_{A_p/\mathbb{R}}(M_{quv}),$$

where M_r denotes multiplication by the class of r in A_p . If $\text{sig } B$ is the number of positive eigenvalues minus the number of negative eigenvalues of a Gram matrix, with zero eigenvalues ignored, then

$$\text{sig } B_{p,q} = \sum_{\alpha \in \mathbb{R}: p(\alpha)=0} \text{sgn } q(\alpha). \quad (\text{A.3.56})$$

The sum is over distinct real roots, and $\text{sgn } 0 = 0$.

Proof. Multiplying p by a nonzero scalar does not change A_p . Factor it over $\mathbb{R}$ into powers of distinct real linear factors and distinct irreducible quadratic factors. The polynomial Chinese remainder theorem splits A_p as the direct product of the corresponding local quotient algebras. Products between different factors vanish, so $B_{p,q}$ is the orthogonal direct sum of their trace forms.

For a real factor $(T - \alpha)^m$, write $T = \alpha + \epsilon$ in $\mathbb{R}[\epsilon]/(\epsilon^m)$. Multiplication by any r has an upper-triangular matrix with diagonal entries $r(\alpha)$. Consequently its trace is $mr(\alpha)$, and the local form is

$$B_{p,q}(\bar{u}, \bar{v}) = mq(\alpha)u(\alpha)v(\alpha).$$

It has one positive, one negative, or no nonzero eigenvalue according as $q(\alpha)$ is positive, negative, or zero.

For a quadratic factor $((T - a)^2 + b^2)^m$, $b > 0$, put $\zeta = a + ib$. After complexification the local quotient splits into the two length- m quotients at ζ and $\bar{\zeta}$. Thus the trace of multiplication by r is $mr(\zeta) + mr(\bar{\zeta}) = 2m \text{Re } r(\zeta)$. The trace form factors through evaluation in $\mathbb{C}$, where it is

$$(z, w) \mapsto 2m \text{Re}(q(\zeta)zw).$$

In the real basis $(1, i)$ its matrix is $2m \begin{pmatrix} \text{Re } q(\zeta) & -\text{Im } q(\zeta) \\ -\text{Im } q(\zeta) & -\text{Re } q(\zeta) \end{pmatrix}$. Its two nonzero eigenvalues, when present, have opposite signs. Every quadratic factor therefore contributes signature zero. Adding the real-factor contributions proves (A.3.56). $\square$

Lemma A.3.40 (Algebraic dependence of a trace-form signature). *Suppose the coefficients of $p_x, q_x \in \mathbb{R}[T]$ are polynomials in $x \in \mathbb{R}^m$, and let $C \subset \mathbb{R}^m$ be semialgebraic. If $\deg p_x = d \geq 1$ is constant on C , then, for every integer j ,*

$$\{x \in C : \text{sig } B_{p_x, q_x} = j\}$$

is semialgebraic.

Proof. In the basis $1, T, \dots, T^{d-1}$ of A_{p_x} , reduction of powers of T modulo p_x shows that the entries of the Gram matrix $H(x)$ are rational functions of the coefficients of p_x, q_x . Their denominators are nonzero on C because the leading coefficient of p_x is nonzero there.

For completeness, the signature of any symmetric matrix with rational-function entries is determined by finitely many polynomial sign tests. If a diagonal entry a is nonzero, a congruence using that entry reduces the matrix to $[a] \oplus H'$; the entries of the Schur complement H' are again rational functions. If every diagonal entry vanishes but an off-diagonal entry H_{ij} does not, replace the basis vector e_i by $e_i + e_j$, obtaining the nonzero diagonal entry $2H_{ij}$, and proceed. If all entries vanish, the signature is zero. Choose the first available pivot at each step. This finite recursion has depth at most d ; every branch

condition is the sign or vanishing of a rational function with nonzero denominator on that branch, equivalently a finite Boolean combination of polynomial sign conditions. On each branch the signature is the sum of the pivot signs. Its level sets are therefore semialgebraic. $\square$

Lemma A.3.41 (Semialgebraic count of real-root sign patterns). *Let $p_x, f_{1,x}, \dots, f_{r,x} \in \mathbb{R}[T]$ have coefficients that are polynomials in $x \in \mathbb{R}^m$. On a semialgebraic set $C \subset \mathbb{R}^m$ suppose $\deg p_x = d \geq 1$ is constant. For $\sigma = (\sigma_1, \dots, \sigma_r) \in \{-1, 0, 1\}^r$, put*

$$n_\sigma(x) = \#\{\alpha \in \mathbb{R} : p_x(\alpha) = 0, \operatorname{sgn} f_{j,x}(\alpha) = \sigma_j \text{ for all } j\}.$$

Then each level set $\{x \in C : n_\sigma(x) = j\}$ is semialgebraic.

Proof. On $\{-1, 0, 1\}$ the three indicator functions are the polynomials

$$\chi_+(s) = \frac{s^2 + s}{2}, \quad \chi_0(s) = 1 - s^2, \quad \chi_-(s) = \frac{s^2 - s}{2}.$$

For each real root α of p_x , the indicator of its prescribed sign vector is $\prod_j \chi_{\sigma_j}(\operatorname{sgn} f_{j,x}(\alpha))$. Expand this product into monomials with exponents $e_j \in \{0, 1, 2\}$. For each such exponent vector,

$$\prod_j (\operatorname{sgn} f_{j,x}(\alpha))^{e_j} = \operatorname{sgn} \left(\prod_j f_{j,x}(\alpha)^{e_j} \right),$$

where the zeroth power means the constant polynomial 1, including when $f_{j,x}(\alpha) = 0$. Summing over the distinct real roots of p_x and applying [Theorem A.3.39](#) expresses $n_\sigma(x)$ as a fixed rational linear combination of signatures

$$\operatorname{sig}_{p_x, \prod_j f_{j,x}^{e_j}} B.$$

By [Theorem A.3.40](#), every signature in this finite list has semialgebraic level sets. Each takes one of the finitely many integer values between $-d$ and d , so the same is true of n_σ . $\square$

Theorem A.3.42 (Real-polynomial projection theorem; full local proof). *If $S \subset \mathbb{R}^{m+1}$ is semialgebraic and $\pi(x, t) = x$, then $\pi(S) \subset \mathbb{R}^m$ is semialgebraic. Consequently every set defined by a finite first-order formula using polynomial equalities and inequalities over $\mathbb{R}$, Boolean operations, and quantifiers over real variables is semialgebraic in its free variables.*

Proof. Choose polynomials $f_1(x, t), \dots, f_r(x, t)$ whose signs determine membership in S . Partition the x -space into finitely many semialgebraic pieces according to which coefficient of each $f_i(x, T)$ is its highest nonzero coefficient, allowing the polynomial to be identically zero. Fix one such piece C . The indices I for which $f_i(x, T)$ is not identically zero and their degrees in T are now constant on C .

If every active f_i is constant in T , the sign vector of all $f_i(x, t)$ is independent of t ; evaluating it at $t = 0$ gives a polynomial sign description of $\pi(S) \cap C$. Otherwise put

$$p_x(T) = \prod_{i \in I} f_i(x, T).$$

Its positive degree is constant on C . For fixed $x \in C$, the sign vector of the $f_i(x, T)$ is constant on each connected component of $\mathbb{R} \setminus \{T : p_x(T) = 0\}$. If a bounded component is bordered by two consecutive distinct real roots of p_x , Rolle's theorem supplies a root of p'_x inside that component. The two unbounded components are sampled by the sign vectors as $T \rightarrow +\infty$ and $T \rightarrow -\infty$. Hence a sign condition is attained at some real T if and only if it is attained at a real root of p_x , at a real root of p'_x , or at one of these two limiting sign vectors. The possible overlap between the first two cases causes no difficulty.

For each prescribed sign vector, the condition that it occurs at a real root of p_x or p'_x is semialgebraic by [Theorem A.3.41](#); both polynomials have fixed positive degree on C when used (and a constant derivative contributes no roots). At either infinity, the sign of a nonzero $f_i(x, T)$ is the sign of its leading coefficient times 1 or $(-1)^{\deg_T f_i}$, respectively; identically zero f_i have sign zero. The conditions for the limiting sign vectors are thus polynomial sign conditions on C . The original Boolean formula accepts a finite collection of sign vectors, so the union of these finite semialgebraic conditions equals $\pi(S) \cap C$. Finally take the union over the finitely many pieces C .

Repeated application proves closure under projection of any finite set of real coordinates. The complement of a semialgebraic set is semialgebraic by definition, and $\forall t \Phi$ is equivalent to $\neg \exists t \neg \Phi$. Induction on the number of quantifiers gives the final assertion. $\square$

Theorem A.3.43 (Compact semialgebraic polynomial error bound; full local proof). *Let $K \subset \mathbb{R}^M$ be nonempty, compact, and semialgebraic, and let F be a real polynomial that is nonnegative on K . Put $Z = \{x \in K : F(x) = 0\}$. If $Z \neq \emptyset$, there are $c > 0$ and an integer $N \geq 1$ such that*

$$F(x) \geq c \min\{\text{dist}(x, Z)^N, 1\} \quad (x \in K). \quad (\text{A.3.57})$$

If $Z = \emptyset$, then $\min_K F > 0$.

Proof. The empty-zero-set assertion is compactness. If $Z = K$, the displayed bound is immediate, so suppose Z and $K \setminus Z$ are both nonempty. For $0 < t < \max_{x \in K} \text{dist}(x, Z)$ set

$$K_t = \{x \in K : \text{dist}(x, Z) \geq t\}, \quad h(t) = \min_{x \in K_t} F(x).$$

Each K_t is nonempty and compact, disjoint from Z ; hence $0 < h(t) \leq \max_K F$. We show that the graph of h is semialgebraic rather than importing this property.

For $t > 0$ write $D_t(x) \iff (\forall z \in Z) \|x - z\|^2 \geq t^2$. The following formula says exactly that u is the attained minimum of F on K_t :

$$(t, u) \in \text{graph}(h) \iff (\exists x \in K)[D_t(x) \wedge F(x) = u \wedge (\forall y \in K)(D_t(y) \Rightarrow F(y) \geq u)].$$

Membership in K and Z is a finite Boolean combination of polynomial signs; D_t uses only a further quantified polynomial inequality. The fully proved [Theorem A.3.42](#) therefore makes this graph semialgebraic. By [Theorem A.3.38](#), there are $c_0 > 0$, $N \geq 1$, and $t_0 \in (0, 1)$ with $h(t) \geq c_0 t^N$ for $0 < t < t_0$.

For $x \in K \setminus Z$ let $s = \text{dist}(x, Z)$. Since $x \in K_s$, $F(x) \geq h(s)$ when $s < t_0$. On the compact set $\{x \in K : s \geq t_0\}$, F has a positive minimum; reducing c_0 to at most that minimum proves (A.3.57) there as well. On Z both sides of (A.3.57) are zero. $\square$

For the He application, identify each k -plane with its orthogonal projector P . Such projectors satisfy $P^T = P$, $P^2 = P$, and $\text{tr } P = k$, so their parameter space $\mathcal{A}_k$ is compact semialgebraic. In fixed orthonormal coordinates the function $F_{k,N}$ defined in the proof of [Theorem A.3.36](#) is a polynomial of degree at most six in $(P, y_1, \dots, y_N)$. The set $K_R = \mathcal{A}_k \times \overline{B(0, R)^N}$ is compact and semialgebraic, and $F_{k,N}$ is nonnegative and polynomial of degree at most six there. Its zero set inside K_R contains every tuple with $y_1 = \dots = y_N = 0$. Thus [Theorem A.3.43](#) gives the exact power-law joint-plane error bound required in the proof of [Theorem A.3.36](#). No general real-analytic Łojasiewicz theorem is needed for that implication.

Lemma A.3.44 (Polynomial escape from linear subspace avoidance). *Let V be a real normed vector space of dimension D , and let $P : V \rightarrow \mathbb{R}$ be a nonzero polynomial of total degree at most $d \geq 1$. There are constants $c_{V,P} > 0$ and $C_{D,d} > 0$ such that for $0 < \rho \leq \frac{1}{2}$ the following holds. Suppose $A \subset B_V(0, \rho^{-1})$ and for every proper linear subspace $W \subsetneq V$ there is $a \in A$ with $\text{dist}(a, W) \geq \rho$. Then some $x \in dD(A \cup \{0\})$, the sum of at most dD elements of $A \cup \{0\}$, satisfies*

$$|P(x)| \geq c_{V,P} \rho^{C_{D,d}}. \quad (\text{A.3.58})$$

Proof. Choose $a_1, \dots, a_D \in A$ recursively: at step j , apply the hypothesis to $W = \text{span}(a_1, \dots, a_{j-1})$. Thus $\|a_j\| \leq \rho^{-1}$ and its distance from the preceding span is at least ρ . After fixing equivalent Euclidean coordinates on V , the column map $T : \mathbb{R}^D \rightarrow V$, $Te_j = a_j$, satisfies

$$\|T^{-1}\| \leq C_V \rho^{-2D}. \quad (\text{A.3.59})$$

Indeed Gram–Schmidt bounds its determinant below by a fixed multiple of ρ^D , while each cofactor is at most $C_V \rho^{-(D-1)}$.

Let $\Lambda_d = \{0, 1, \dots, d\}^D$. Tensor-product Lagrange interpolation on this fixed grid gives, for every polynomial Q of total degree at most d and every $R \geq 1$,

$$\sup_{\|z\| \leq R} |Q(z)| \leq C'_{D,d} (1+R)^d \max_{u \in \Lambda_d} |Q(u)|. \quad (\text{A.3.60})$$

For completeness, applying the one-variable Lagrange formula successively in each coordinate expresses $Q(z)$ as $\sum_{u \in \Lambda_d} Q(u) L_u(z)$; the tensor polynomials L_u have degree at most d in each coordinate, but this naive bound would give R^{dD} . Instead first use the same interpolation to bound every coefficient of Q by $C'_{D,d} \max_{u \in \Lambda_d} |Q(u)|$; total degree at most d then gives [\(A.3.60\)](#).

Choose $y \in B_V(0, 1)$ with $c_P := |P(y)| > 0$. Put $Q = P \circ T$, $z = T^{-1}y$, and $R = \max(1, C_V \rho^{-2D})$. By [\(A.3.59\)](#), $\|z\| \leq R$. Hence [\(A.3.60\)](#) gives

$$\max_{u \in \Lambda_d} |P(Tu)| \geq c_{V,P} \rho^{2Dd},$$

after decreasing the fixed constant. Choose a maximizing u . Since each $0 \leq u_j \leq d$ is an integer, $Tu = \sum_j u_j a_j$ is the sum of at most dD elements of $A \cup \{0\}$. The displayed bound proves the lemma, for instance with $C_{D,d} = 2Dd$ after another fixed-constant adjustment. $\square$

A.3.7 Trace growth and commutator geometry

Lemma A.3.45 (Trace-type growth from a nondegenerate pairing). *Let V be a fixed real normed vector space of dimension $D \geq 1$, and let $\tau : V \times V \rightarrow \mathbb{R}$ be a fixed bilinear form whose map $x \mapsto \tau(x, \cdot)$ is invertible. There are $c_{V,\tau} > 0$ and $C_D > 0$ such that, for $0 < \rho \leq \frac{1}{2}$, $\delta > 0$, and a bounded $A \subset B_V(0, \rho^{-1})$ which is ρ -away from every proper linear subspace,*

$$\mathcal{N}_\delta\{\tau(a, b) : a, b \in A\} \geq c_{V,\tau} \rho^{C_D} \mathcal{N}_\delta(A)^{1/D}. \quad (\text{A.3.61})$$

In particular this applies to each model algebra $E = M_r(\mathbb{F})$, $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{H}$, with $\tau_E(x, y) = \text{tr}_{\mathbb{R}}(L_{xy})$, where $L_z : E \rightarrow E$ is left multiplication by z .

Proof. Choose a greedy basis $a_1, \dots, a_D \in A$ with $\|a_j\| \leq \rho^{-1}$ and $\text{dist}(a_j, \text{span}(a_1, \dots, a_{j-1})) \geq \rho$. The determinant and cofactor calculation in (A.3.59), followed by nondegeneracy of τ , shows that the map

$$T : V \rightarrow \mathbb{R}^D, \quad T(x) = (\tau(a_1, x), \dots, \tau(a_D, x))$$

has $\|T^{-1}\| \leq C_{V,\tau} \rho^{-2D}$. For each j , let $M_j = \mathcal{N}_\delta(\tau(a_j, A))$ and choose a cover by M_j intervals of radius δ . Their Cartesian products cover $T(A)$ by $\prod_j M_j$ boxes whose preimages have diameter at most $C_{V,\tau} \rho^{-2D} \delta$. Each such preimage can be covered by at most $C_{V,\tau} \rho^{-2D^2}$ radius- δ balls in V . Therefore

$$\mathcal{N}_\delta(A) \leq C_{V,\tau} \rho^{-2D^2} \prod_{j=1}^D M_j.$$

Some j satisfies $M_j \geq c_{V,\tau} \rho^{2D} \mathcal{N}_\delta(A)^{1/D}$. Since $\tau(a_j, A) \subset \{\tau(a, b) : a, b \in A\}$, this proves (A.3.61).

For the model algebra, writing x^* for the transpose with entrywise real, complex or quaternionic conjugation, direct calculation gives

$$\tau_E(x, y) = r \dim_{\mathbb{R}} \mathbb{F} \text{Re tr}_{\mathbb{F}}(xy), \quad \tau_E(x, x^*) = r \dim_{\mathbb{R}} \mathbb{F} \sum_{i,j} |x_{ij}|^2.$$

The second expression is positive for $x \neq 0$, so the pairing is nondegenerate and the first part applies. $\square$

Lemma A.3.46 (Trace-separated commutator translates). *Let E be a fixed finite-dimensional normed real algebra and $\tau : E \rightarrow \mathbb{R}$ a nonzero linear functional satisfying $\tau(xy) = \tau(yx)$ for all $x, y \in E$. For $a \in E$ put $\phi_a(x) = ax - xa$. There is a constant $C_{E,\tau}$ such that, whenever $A, B \subset E$ are bounded and $T \subset B$ has pairwise $C_{E,\tau} \delta$ -separated τ -values,*

$$\mathcal{N}_\delta(\phi_a(A) + B) \geq c_{E,\tau} \#T \mathcal{N}_\delta(\phi_a(A)). \quad (\text{A.3.62})$$

If a and B are bounded words in A , the set on the left is a bounded word set over A .

Proof. The cyclic identity gives $\tau(\phi_a(x)) = 0$. Thus $\tau(\phi_a(A) + t) = \{\tau(t)\}$ for each $t \in T$. Choose $C_{E,\tau} > 4\|\tau\|$. Distinct translates $\phi_a(A) + t$ are then separated by more than 4δ in E , so a radius- δ ball can meet at most one of them. Each translate has the same covering number as $\phi_a(A)$. Taking their union proves (A.3.62), up to the fixed comparison between open and closed covering balls. The word assertion follows from $\phi_a(A) = aA - Aa$ and closure of bounded words under addition. $\square$

Lemma A.3.47 (Conditioned diagonalization from a spectral gap). *Let $a \in M_n(\mathbb{C})$ have operator norm at most $R \geq 1$ and distinct eigenvalues $\lambda_1, \dots, \lambda_n$ with $|\lambda_i - \lambda_j| \geq g > 0$ for $i \neq j$. There is $V \in GL_n(\mathbb{C})$ such that $V^{-1}aV = \text{diag}(\lambda_1, \dots, \lambda_n)$ and*

$$\|V\| + \|V^{-1}\| \leq C_n(1 + R/g)^{n-1}. \quad (\text{A.3.63})$$

Consequently, if $\|ax - xa\| \leq \delta$, then

$$\text{dist}(V^{-1}xV, \Delta) \leq C_n(1 + R/g)^{2n-1}\delta. \quad (\text{A.3.64})$$

Proof. The spectral projector onto the λ_i -eigenspace is

$$P_i = \prod_{j \neq i} \frac{a - \lambda_j I}{\lambda_i - \lambda_j}.$$

Since $|\lambda_j| \leq R$, its norm is at most $(2R/g)^{n-1}$. Choose a unit eigenvector v_i spanning its range and form V with columns v_i . Then $\|V\| \leq \sqrt{n}$. Write $P_i = v_i \ell_i$, where the row functional ℓ_i has norm $\|P_i\|$. The functionals ℓ_i are the rows of V^{-1} , so $\|V^{-1}\| \leq \sqrt{n} \max_i \|P_i\|$. This proves (A.3.63).

Put $y = V^{-1}xV$ and $d = V^{-1}aV$. Then $\|[d, y]\| \leq \|V^{-1}\| \|V\| \delta$ and $[d, y]_{ij} = (\lambda_i - \lambda_j)y_{ij}$. Every off-diagonal entry of y therefore has modulus at most $g^{-1}\|V^{-1}\| \|V\| \delta$. There are only $n(n-1)$ such entries, so the fixed-dimensional norm comparison gives (A.3.64). $\square$

Lemma A.3.48 (A small commutator image yields a large diagonal packet). *Let $E \subset M_n(\mathbb{C})$ be a real linear subalgebra, let $A \subset E$ be bounded, and let $a \in E$ satisfy the spectral hypotheses of Theorem A.3.47. Put $\phi_a(x) = ax - xa$ and $K = \|V\| \|V^{-1}\|$ for the matrix V there. For any $\delta > 0$ and $\theta > 0$, if*

$$\mathcal{N}_\delta(\phi_a(A)) \leq \delta^\theta \mathcal{N}_\delta(A),$$

then some nonempty $F \subset A$ has $\mathcal{N}_\delta(F) \geq c_n \delta^{-\theta}$, and the difference packet $B = V^{-1}(F - F)V \subset V^{-1}EV$ satisfies

$$\text{dist}(b, \Delta) \leq C_n(1 + R/g)^{2n-1}\delta \quad (b \in B), \quad (\text{A.3.65})$$

$$\mathcal{N}_u(B) \geq c_n \left(\frac{\delta}{Ku} \right)^{2n^2} \delta^{-\theta} \quad (u \geq \delta/K). \quad (\text{A.3.66})$$

Here R and g are the norm and eigenvalue-gap parameters of Theorem A.3.47; when the right side of (A.3.66) is below one the bound is understood as a harmless weak estimate.

Proof. Cover $\phi_a(A)$ by $\mathcal{N}_\delta(\phi_a(A))$ radius- δ balls. The intersections of their inverse images with A cover A . Subadditivity of covering numbers and a fixed-dimensional comparison of radii show that one fiber $F = A \cap \phi_a^{-1}(B(y, \delta))$ has $\mathcal{N}_\delta(F) \geq c_n \delta^{-\theta}$. For $x = f - f' \in F - F$, $\|\phi_a(x)\| \leq 2\delta$. Apply (A.3.64) and absorb the factor two to obtain (A.3.65).

Fix $f_0 \in F$. The translate $F - f_0$ is contained in $F - F$ and has the same δ -covering number as F . Conjugation $x \mapsto V^{-1}xV$ and its inverse have Lipschitz constants at most K . If B is covered by N balls of radius u , then $F - f_0$ is covered by N balls of radius Ku . Each such ball in the real $2n^2$ -dimensional matrix space is covered by at most $C_n(Ku/\delta)^{2n^2}$ radius- δ balls when $u \geq \delta/K$. It follows that $\mathcal{N}_\delta(F) \leq C_n(Ku/\delta)^{2n^2} \mathcal{N}_u(B)$, which gives (A.3.66). $\square$

Proposition A.3.49 (A regular commutator fibre produces a segment). *Fix $n \geq 1$, $C \geq 1$, and $\theta > 0$. By Theorem A.3.15, there are $e_0, \alpha, \beta > 0$ and an integer S with the following property. Let $E \subset M_n(\mathbb{C})$ be a real linear subalgebra, $0 < e < e_0$, and $\delta > 0$ sufficiently small. Suppose $A \subset E \cap B(0, \delta^{-e})$ and $a \in E$ has*

$$\|a\| \leq \delta^{-Ce}, \quad |\lambda_i(a) - \lambda_j(a)| \geq \delta^{Ce} \quad (i \neq j),$$

where the n eigenvalues are distinct. If

$$\mathcal{N}_\delta(\{ax - xa : x \in A\}) \leq \delta^\theta \mathcal{N}_\delta(A), \quad (\text{A.3.67})$$

then there is a unit $\eta \in E$ such that

$$[0, \delta^\alpha] \eta \subset \mathcal{W}_S(A) + B(0, \delta^{\alpha+\beta}). \quad (\text{A.3.68})$$

The constants are independent of E, A, a, e .

Proof. Apply Theorem A.3.47 to a , with $R = \delta^{-Ce}$ and $g = \delta^{Ce}$, and write $K = \|V\| \|V^{-1}\|$. Fixed-dimensional constants are absorbed, for δ sufficiently small after e , in

$$K \leq \delta^{-H_0 e}, \quad C_n(1 + R/g)^{2n-1} \leq \delta^{-H_1 e}, \quad (\text{A.3.69})$$

where H_0, H_1 depend only on n, C . Apply Theorem A.3.48 with the parameter θ . It gives $F \subset A$ and $B = V^{-1}(F - F)V \subset E' := V^{-1}EV$. Put $D = 2n^2$ and $H = \max(H_0 + 2, H_1)$. From the norm bound on A , (A.3.65), and (A.3.69),

$$B \subset B(0, \delta^{-He}), \quad \sup_{b \in B} \text{dist}(b, \Delta) \leq \delta^{1-He}. \quad (\text{A.3.70})$$

Taking $u = \delta$ in (A.3.66) also gives

$$\mathcal{N}_\delta(B) \geq c_n K^{-D} \delta^{-\theta} \geq c_n \delta^{-\theta + DH_0 e} \geq \delta^{-\theta/2}$$

when $e_0 < \theta/(4DH_0)$ and δ is small. Thus Theorem A.3.20 applies to B inside the real algebra E' , with covering exponent $\theta/2$ and relaxation parameter He . After decreasing e_0 so that He_0 is below that corollary's threshold, we obtain fixed $\alpha_0, \beta_0 > 0$, S_0 , and a unit $v \in E'$ for which

$$[0, \delta^{\alpha_0}] v \subset \mathcal{W}_{S_0}(B) + B(0, \delta^{\alpha_0 + \beta_0}). \quad (\text{A.3.71})$$

Conjugate (A.3.71) by V . Conjugation respects addition and ordered multiplication, and $VBV^{-1} = F - F$; hence its word set is contained in $\mathcal{W}_{2S_0}(A)$. Put $w = VvV^{-1} \in E$ and $\eta = w/\|w\|$. Since $\|v\| = 1$, we have $\|w\| \geq K^{-1}$, while the conjugated error is at most $K\delta^{\alpha_0+\beta_0}$. Therefore the resulting segment has length at least $\delta^{\alpha_0+H_0e}$ and error at most $\delta^{\alpha_0+\beta_0-H_0e}$. Set $\alpha = \alpha_0 + \beta_0/4$ and $\beta = \beta_0/2$. Require additionally $H_0e_0 < \beta_0/4$. Then the segment contains $[0, \delta^\alpha]\eta$, and its error is at most $\delta^{\alpha+\beta}$. Taking $S = 2S_0$ proves (A.3.68). $\square$

A.3.8 From escape to segments and model-algebra balls

Proposition A.3.50 (Two-sided escape produces a word basis). *Let E be a unital finite-dimensional normed real algebra of dimension D , with a submultiplicative norm. Let $0 < \rho, \lambda \leq \frac{1}{2}$, and let $A \subset B_E(0, \rho^{-1})$. Assume that for every nonzero proper real subspace $W \subset E$ there are $a \in A$ and $w \in W \cap B_E(0, 1)$ with*

$$\max\{\text{dist}(aw, W), \text{dist}(wa, W)\} \geq \lambda. \quad (\text{A.3.72})$$

Put $A_1 = A \cup \{1_E\}$. For every $\eta \in E$ with $\rho \leq \|\eta\| \leq \rho^{-1}$, the set $A_1^D \eta A_1^D$ contains a basis $b_1, \dots, b_D$ of E satisfying

$$\|b_j\| \leq (\rho\lambda)^{-C_D}, \quad \text{dist}(b_j, \text{span}(b_1, \dots, b_{j-1})) \geq (\rho\lambda)^{C_D} \quad (\text{A.3.73})$$

for a constant C_D depending only on the fixed normed algebra.

Proof. Set $L = \max(2, \rho^{-1}, \lambda^{-1}, \|1_E\|)$ and $R = L^{2D+1}$. Every word in $A_1^D \eta A_1^D$ has norm at most R . Start with $b_1 = \eta$, and set $t_1 = \rho$. Suppose $b_1, \dots, b_k$ have been chosen with $k < D$, each of norm at most R , each of the required word form with at most $k-1$ nonidentity factors on each side, and with every successive distance at least t_k . Put $W_k = \text{span}(b_1, \dots, b_k)$. For every $w \in W_k$ with $\|w\| \leq 1$, write $w = \sum_{j=1}^k c_j b_j$. The determinant/cofactor estimate for this conditioned basis gives

$$\sum_{j=1}^k |c_j| \leq C_D^{(0)} R^{k-1} t_k^{-k} \leq C_D^{(0)} R^D t_k^{-D}. \quad (\text{A.3.74})$$

Choose a, w from (A.3.72). If the first distance in that maximum is at least λ , then

$$\lambda \leq \text{dist}(aw, W_k) \leq \sum_{j=1}^k |c_j| \text{dist}(ab_j, W_k).$$

Hence some j satisfies $\text{dist}(ab_j, W_k) \geq (C_D^{(0)})^{-1} \lambda R^{-D} t_k^D$. Choose $b_{k+1} = ab_j$. If instead the second distance in (A.3.72) is at least λ , use $b_{k+1} = b_j a$ and the same estimate. Either operation increases one word length by one, so after at most $D-1$ steps both lengths are at most D .

Define explicitly

$$t_{k+1} = (2C_D^{(0)})^{-1} \lambda R^{-D} t_k^D \quad (1 \leq k < D).$$

The preceding estimate makes the new successive distance at least t_{k+1} ; all previous distances are at least $t_k \geq t_{k+1}$. Because D is fixed and $R \leq C_E(\rho\lambda)^{-(2D+1)}$ for a constant depending only on the fixed normed algebra, this finite recurrence gives $t_D \geq (\rho\lambda)^{C_D}$ for a sufficiently large C_D . It also gives the stated norm bound, since every $\|b_j\| \leq R$. The positive successive distances make the D words a basis. $\square$

Proposition A.3.51 (Segment production from quantitative two-sided escape). *Fix $\kappa > 0$, $C_{\text{esc}} \geq 1$, and one of the standard simple real algebras $E = M_r(\mathbb{R}), M_r(\mathbb{C}), M_r(\mathbb{H})$, equipped with its fixed complex matrix embedding $E \subset M_n(\mathbb{C})$ and the inherited operator norm. By Theorem A.3.15, there are $e_0, \beta > 0$, $\alpha \geq 1$, and an integer S such that the following holds for $0 < e < e_0$ and sufficiently small $\delta > 0$. Suppose $A \subset E$ satisfies*

$$A \subset B(0, \delta^{-e}), \quad \mathcal{N}_\delta(A) \geq \delta^{-\kappa},$$

and for every nonzero proper real subspace $W \subset E$ there exist $a \in A$ and $w \in W \cap B(0, 1)$ with

$$\max\{\text{dist}(aw, W), \text{dist}(wa, W)\} \geq \delta^{C_{\text{esc}}e}. \quad (\text{A.3.75})$$

Then some unit $\eta \in E$ satisfies

$$[0, \delta^\alpha]\eta \subset \mathcal{W}_S(A) + B(0, \delta^{\alpha+\beta}). \quad (\text{A.3.76})$$

This is a full local implication. Passing from avoidance of proper unital subalgebras to (A.3.75) is proved in Theorem A.3.36, using the polynomial error bound Theorem A.3.43.

Proof. Put $D = \dim_{\mathbb{R}} E$. If $n = 1$, then E is $\mathbb{R}$ or $\mathbb{C}$, embedded in the diagonal algebra of $M_1(\mathbb{C})$. In that case Theorem A.3.20, with $\sigma = \kappa$, applies directly to A ; its diagonal proximity condition is exact. We therefore assume $n \geq 2$.

The fixed embedding contains an element with n distinct complex eigenvalues. For $M_r(\mathbb{R})$ and $M_r(\mathbb{C})$ take a diagonal matrix with distinct entries. For $M_r(\mathbb{H})$ use the standard block embedding

$$x + iy + jz + kw \mapsto \begin{pmatrix} x + iy & z + iw \\ -z + iw & x - iy \end{pmatrix}, \quad x, y, z, w \in \mathbb{R},$$

and choose a quaternionic diagonal matrix whose nonreal complex entries and their conjugates are pairwise distinct.

Choose $a_0 \in A$ with $\|a_0\| \geq \delta^{C_{\text{esc}}e}$; such an element exists by (A.3.75), since $D > 1$. Take $\rho = \delta^{C_0e}$ with $C_0 \geq \max(1, C_{\text{esc}})$ and $\lambda = \delta^{C_{\text{esc}}e}$. For small δ , the hypotheses of Theorem A.3.50 hold, with the same escape condition. Its basis elements have the form $u_i a_0 v_i$ with at most D factors of A on each side; identity factors may be deleted. Thus the basis lies in $A_{\text{bas}} := \mathcal{W}_{2D+1}(A)$, its norms are at most δ^{-C_1e} , and its successive distances are at least δ^{C_1e} for a fixed C_1 .

We record explicitly the resulting linear-subspace avoidance. If $T : \mathbb{R}^D \rightarrow E$ maps the standard basis to these basis elements, the determinant/cofactor bound gives $\|T^{-1}\| \leq \delta^{-C_2e}$ after enlarging C_2 . Given a proper real subspace $W \subset E$, choose a unit vector $z \perp W$ and write $z = \sum_i t_i b_i$. Then

$$1 = \text{dist}(z, W) \leq \sum_i |t_i| \text{dist}(b_i, W) \leq C_D \delta^{-C_2e} \max_i \text{dist}(b_i, W).$$

After enlarging C_2 once more, A_{bas} is δ^{C_2e} -away from every proper real linear subspace and lies in $B(0, \delta^{-C_2e})$.

Let P be the squared modulus of the discriminant of the complex characteristic polynomial, restricted to E . It is a nonzero real polynomial by the first paragraph. Apply [Theorem A.3.44](#) to A_{bas} with $\rho = \delta^{C_2e}$. It produces a matrix a_* in a bounded sum of $A_{\text{bas}} \cup \{0\}$, hence in $\mathcal{W}_{L_*}(A)$ for a fixed L_* , such that $|P(a_*)| \geq \delta^{C_3e}$ and $\|a_*\| \leq \delta^{-C_3e}$ after enlarging C_3 . Since every eigenvalue has modulus at most $\|a_*\|$ and the product of fourth powers of eigenvalue gaps equals $|P(a_*)|$, each individual gap obeys

$$|\lambda_i(a_*) - \lambda_j(a_*)| \geq \delta^{C_4e} \quad (i \neq j) \quad (\text{A.3.77})$$

for another fixed C_4 .

Set $\kappa' = \kappa/(9n^2)$ and $L_0 = \max(2D + 1, L_*)$. Define integers recursively by $L_{j+1} = 2L_* + 4L_j + 10$ and put $A_j = \mathcal{W}_{L_j}(A)$. Fix an integer J such that $\kappa + J\kappa' > D + 1$. All L_j for $j \leq J$ are fixed independently of A, e, δ . In particular there is $C_5 \geq \max(C_3, C_4)$ such that

$$A_j \subset B(0, \delta^{-C_5e}), \quad \mathcal{N}_\delta(A_j) \geq \delta^{-\kappa}, \quad A_j \text{ is } \delta^{C_5e}\text{-away from proper linear subspaces} \quad (\text{A.3.78})$$

for $0 \leq j \leq J$. These assertions follow from $A_j \supset A_{\text{bas}} \cup A$, the finite word-length norm bound, and the preceding conditioned basis.

For each $j < J$ put $\phi(x) = a_*x - xa_*$ and consider two cases. First suppose

$$\mathcal{N}_\delta(\phi(A_j)) \geq \delta^{2\kappa'} \mathcal{N}_\delta(A_j). \quad (\text{A.3.79})$$

Let $\tau(x) = \text{tr}_R(L_x)$ on E . It is cyclic, and the pairing $(x, y) \mapsto \tau(xy)$ is nondegenerate by [Theorem A.3.45](#). That lemma, (A.3.78), and $D \leq 2n^2$ give, after choosing e_0 small,

$$\mathcal{N}_\delta\{\tau(xy) : x, y \in A_j\} \geq \delta^{-4\kappa' - \kappa'/4}.$$

Choose a subset $T \subset A_j A_j$ whose τ -values are pairwise $C_{E,\tau}\delta$ -separated and satisfy $\#T \geq \delta^{-4\kappa'}$; the fixed separation factor is absorbed by the slack. The cyclic identity $\tau(\phi(x)) = 0$ and [Theorem A.3.46](#) now imply

$$\mathcal{N}_\delta(\phi(A_j) + A_j A_j) \geq c_E \delta^{-2\kappa'} \mathcal{N}_\delta(A_j).$$

The set on the left lies in A_{j+1} by the chosen recurrence for L_j . Shrinking δ absorbs the fixed constant, so

$$\mathcal{N}_\delta(A_{j+1}) \geq \delta^{-\kappa'} \mathcal{N}_\delta(A_j). \quad (\text{A.3.80})$$

If (A.3.79) fails, then (A.3.77) and (A.3.78) allow us to apply [Theorem A.3.49](#) to A_j with $\theta = 2\kappa'$, relaxation parameter C_5e , and spectral constant $C = 1$. Taking e_0 below that proposition's threshold gives a unit $\eta \in E$ and a segment of fixed exponents in $\mathcal{W}_{S_0}(A_j) \subset \mathcal{W}_{S_0 L_j}(A)$. Since $j < J$, the latter leaf count is bounded by one fixed S , proving (A.3.76).

Finally, the large-image case cannot occur for all $j = 0, \dots, J-1$. Indeed, iterating (A.3.80) would give $\mathcal{N}_\delta(A_J) \geq \delta^{-\kappa - J\kappa'}$. The first bound in (A.3.78) and $\dim_{\mathbb{R}} E = D$ instead give $\mathcal{N}_\delta(A_J) \leq C_E \delta^{-D(1+C_5e)}$. For e_0 small enough, this contradicts $\kappa + J\kappa' > D + 1$. Hence a small-image stage must occur, and the preceding paragraph completes the proof. The exponents supplied by the relaxed near-diagonal result are at least one by the choice in [Theorem A.3.19](#); hence the chosen α is at least one in both the $n = 1$ and packet cases. $\square$

Proposition A.3.52 (A one-direction segment fills a model-algebra ball). *Let E, A, ρ, λ satisfy the hypotheses of Theorem A.3.50, put $D = \dim_{\mathbb{R}} E$, and let $\eta \in E$ be a unit vector. Suppose, for $0 < L \leq 1$, $0 < \tau < 1$ and a fixed integer $s \geq 1$, that*

$$[0, L]\eta \subset \mathcal{W}_s(A) + B_E(0, L\tau). \quad (\text{A.3.81})$$

Then constants $c_E, C_E > 0$ and an integer $S = S(E, s)$ exist such that

$$B_E(0, c_E L(\rho\lambda)^{C_E}) \subset \mathcal{W}_S(A) + B_E(0, C_E L\tau(\rho\lambda)^{-C_E}). \quad (\text{A.3.82})$$

No scalar multiplication by arbitrary real numbers is hidden in the word set.

Proof. Apply Theorem A.3.50 to η ; its norm is one and $\rho \leq 1 \leq \rho^{-1}$. We obtain a basis $b_i = u_i \eta v_i$, where u_i, v_i are products of at most D factors of $A \cup \{1_E\}$ and the bounds (A.3.73) hold. In particular $\|u_i\|, \|v_i\| \leq C_E(\rho\lambda)^{-C_E}$. The conditioned-basis determinant and cofactor estimates give for $T : \mathbb{R}^D \rightarrow E$, $Te_i = b_i$,

$$\|T^{-1}\| \leq C_E(\rho\lambda)^{-C_E} \quad (\text{A.3.83})$$

after enlarging C_E ; indeed the determinant is bounded below by the product of the successive distances, while each cofactor is bounded by a fixed power of the basis-vector norms.

If $\|x\| \leq L/(2D\|T^{-1}\|)$, write $x = \sum_i t_i b_i$. Then $|t_i| \leq L/2$. For each i , (A.3.81) gives an approximant $w_i \in \mathcal{W}_s(A)$ to $|t_i|\eta$ with $\|w_i - |t_i|\eta\| \leq L\tau$. The signed sum $\sum_i \text{sgn}(t_i)u_i w_i v_i$ lies in $\mathcal{W}_S(A)$ for one fixed S , because each w_i is a bounded word and the products u_i, v_i have at most D factors each; their identity factors are simply deleted. Its distance from x is at most $C_E L\tau(\rho\lambda)^{-C_E}$. Use (A.3.83) to lower-bound the admissible radius, and enlarge C_E once more to obtain (A.3.82). $\square$

Lemma A.3.53 (Finite scale ladder for a ball estimate). *Fix $\alpha, \beta > 0$, integers $s, K \geq 1$, and $0 < \delta < 1$. Define $r_j = \delta^{\alpha^{-1}(\alpha/(\alpha+\beta))^j}$ for $0 \leq j \leq K$. Suppose A lies in an associative normed real algebra E and, for every $1 \leq j \leq K$,*

$$B_E(0, r_j^\alpha) \subset \mathcal{W}_s(A) + B_E(0, r_j^{\alpha+\beta}). \quad (\text{A.3.84})$$

Then

$$B_E(0, r_K^\alpha) \subset \mathcal{W}_{Ks}(A) + B_E(0, \delta). \quad (\text{A.3.85})$$

Proof. The radii satisfy $r_j^{\alpha+\beta} = r_{j-1}^\alpha$ and $r_0^\alpha = \delta$. For any x in the left side of (A.3.85), use the $j = K$ inclusion (A.3.84) to write $x = w_K + e_{K-1}$ with $w_K \in \mathcal{W}_s(A)$ and $\|e_{K-1}\| \leq r_{K-1}^\alpha$. Repeat with $j = K-1, \dots, 1$. The final residual has norm at most δ , while the sum of the K words has at most Ks leaves. This is (A.3.85). $\square$

Proposition A.3.54 (Ball generation from quantitative escape). *Fix one of the standard simple real algebras $E = M_r(\mathbb{R}), M_r(\mathbb{C}), M_r(\mathbb{H})$ with its inherited matrix norm. Use the internally supplied scalar segment Theorem A.3.15. Then the following model-algebra ball-generation assertion holds for E : for every $\kappa, \zeta > 0$ there are $b > 0$ and an integer s such that, for sufficiently small h , each $F \subset E \cap B(0, h^{-b})$ satisfying*

$$\mathcal{N}_r(F) \geq h^b r^{-\kappa} \quad (r \geq h), \quad F \text{ is } h^b\text{-away from proper unital subalgebras}$$

obeys

$$B_E(0, h^\zeta) \subset \mathcal{W}_s(F) + B_E(0, h). \quad (\text{A.3.86})$$

Thus this proposition proves the model-algebra ball-generation assertion of He Theorem 5; Theorem A.3.36 supplies the escape step without an external analytic input.

Proof. Fix $\kappa, \zeta > 0$. Apply Theorem A.3.36 and denote its exponent by C_{esc} . Apply Theorem A.3.51 with covering exponent $\kappa/2$ and the displayed C_{esc} . Write its constants as $e_0, \alpha_0, \beta_0, S_0$, where $\alpha_0 \geq 1$. Choose $e_1 \in (0, e_0)$, small enough for the inequalities below, and put $\rho_r = r^{C_0 e_1}$ and $\lambda_r = r^{C_{\text{esc}} e_1}$, where $C_0 \geq \max(1, C_{\text{esc}})$ is fixed.

Suppose at a scale r the set F satisfies the hypotheses of Theorem A.3.51 with exponent e_1 . That proposition gives a unit η and

$$[0, r^{\alpha_0}] \eta \subset \mathcal{W}_{S_0}(F) + B(0, r^{\alpha_0 + \beta_0}).$$

The same escape condition and the norm bound on F let us apply Theorem A.3.52 with $L = r^{\alpha_0}$ and $\tau = r^{\beta_0}$. Its constants, independent of r, F, e_1 , give

$$B_E(0, C_E r^{\alpha_0} (\rho_r \lambda_r)^{C_E}) \subset \mathcal{W}_{S_0}(F) + B_E(0, C_E r^{\alpha_0 + \beta_0} (\rho_r \lambda_r)^{-C_E}).$$

Take e_1 so small that $C_E(C_0 + C_{\text{esc}})e_1 < \beta_0/8$. After absorbing the fixed constants by requiring r small, put $\alpha = \alpha_0 + \beta_0/4 \geq 1$ and $\beta = \beta_0/2 > 0$. The last inclusion implies the uniform one-scale estimate

$$B_E(0, r^\alpha) \subset \mathcal{W}_{S_0}(F) + B_E(0, r^{\alpha + \beta}). \quad (\text{A.3.87})$$

Put $q = \alpha/(\alpha + \beta) < 1$, choose $K \geq 1$ with $q^K \leq \zeta$, and define $r_j = h^{q^j/\alpha}$ for $0 \leq j \leq K$. Because $\alpha \geq 1$, every $r_j \geq h$. Let $c_* = q^K/\alpha > 0$, and choose the input exponent $b > 0$ so small that

$$b < \frac{c_*}{2} \min(e_1, \kappa/2).$$

For every $1 \leq j \leq K$, write $c_j = q^j/\alpha \geq c_*$. The assumed norm and avoidance bounds at scale h imply $F \subset B(0, r_j^{-e_1})$ and that F is $r_j^{e_1}$ -away from proper unital subalgebras, since $h^b \geq r_j^{e_1}$. Applying Theorem A.3.36 at $\rho = r_j^{e_1}$ gives (A.3.75) at scale r_j . Also

$$\mathcal{N}_{r_j}(F) \geq h^b r_j^{-\kappa} = r_j^{-\kappa + b/c_j} \geq r_j^{-\kappa/2}.$$

Thus (A.3.87) holds at each of the K scales r_j , with the same exponents and word length. Apply Theorem A.3.53 to obtain

$$B_E(0, r_K^\alpha) \subset \mathcal{W}_{K S_0}(F) + B_E(0, h).$$

Finally $r_K^\alpha = h^{q^K} \geq h^\zeta$, so (A.3.86) follows with $s = K S_0$. □

Proposition A.3.55 (Almost-generated model algebra). *Fix $m \geq 2$ and $0 < \eta < 1$. Put $N = m^2$ and $c = \frac{2}{7}(\eta/4)^N$. There is $e_0 = e_0(m, \eta) > 0$ such that whenever $0 < e < e_0$, $0 < \delta < \delta_0(m, \eta, e)$, and $A \subset \text{End}(\mathbb{R}^m)$ satisfies*

$$A \subset B(0, \delta^{-e}) \quad \text{and} \quad A \text{ acts } \delta^e\text{-irreducibly on } \mathbb{R}^m,$$

there exist a scale $\delta_1 \in [\delta, \delta^c]$, a positive integer $k \leq 1/c$ with $\delta = \delta_1^k$, a real invertible matrix g , and one of the model algebras

$$E = M_m(\mathbb{R}), \quad M_{m/2}(\mathbb{C})_{\mathbb{R}}, \quad M_{m/4}(\mathbb{H})_{\mathbb{R}}$$

(only models with integral matrix size are allowed), such that

$$\begin{aligned} \|g\| + \|g^{-1}\| &\leq \delta^{-5me}, \\ gAg^{-1} &\subset E + B(0, \delta_1), \\ gAg^{-1} &\text{ is } \delta_1^\eta\text{-away from proper subalgebras of } E. \end{aligned}$$

Proof. Set $\theta = \eta/4$. Among integers $0 \leq \ell \leq N$, choose the largest ℓ for which a subalgebra $E_0 \subset \text{End}(\mathbb{R}^m)$ of codimension ℓ satisfies

$$A \subset E_0 + B(0, \delta^{\theta^\ell}). \quad (\text{A.3.88})$$

The full matrix algebra makes $\ell = 0$ admissible. Write $\alpha = \theta^\ell$, choose $k = \lceil 5/(2\alpha) \rceil$, and set $\delta_1 = \delta^{1/k}$. Since $0 < \alpha \leq 1$,

$$\frac{2\alpha}{7} \leq \frac{1}{k} \leq \frac{2\alpha}{5}, \quad k \leq \frac{7}{2\theta^N} = \frac{1}{c}.$$

Consequently $\delta \leq \delta_1 \leq \delta^c$ and $\delta = \delta_1^k$. Also $2/k \leq 4\alpha/5 < \alpha$, so (A.3.88) has error at most δ_1^2 .

We first show that E_0 acts δ^{4e} -irreducibly. Let W be a nonzero proper subspace, and choose $a \in A$ and $w \in W \cap B(0, 1)$ with $\text{dist}(aw, W) \geq \delta^e$. Choose $a_0 \in E_0$ with $\|a - a_0\| \leq \delta_1^2$. If $e_0 < \theta^N/4$, then for small δ ,

$$\text{dist}(a_0 w, W) \geq \delta^e - \delta_1^2 \geq \frac{1}{2}\delta^e, \quad \|a_0\| \leq 2\delta^{-e}.$$

Thus $a_0 / \max(1, \|a_0\|)$ belongs to $E_0 \cap B(0, 1)$ and sends w at distance at least $\frac{1}{4}\delta^{2e} > \delta^{4e}$ from W . In particular $E_0 \neq \{0\}$.

By Theorem A.3.30, there is $g \in \text{GL}_m(\mathbb{R})$ such that $gE_0g^{-1} = E$ is one of the displayed models and, for sufficiently small δ ,

$$\|g\| + \|g^{-1}\| \leq \delta^{-5me}, \quad \|g\|\|g^{-1}\| \leq \delta^{-10me}. \quad (\text{A.3.89})$$

Here we use the inverse of the conjugating matrix in that corollary when needed.

Let $F \subsetneq E$ be a proper subalgebra. Then $F_0 = g^{-1}Fg \subsetneq E_0$ has codimension at least $\ell + 1$ in the full matrix algebra. By maximality of ℓ , there is $a \in A$ with

$$\text{dist}(a, F_0) \geq \delta^{\theta^{\ell+1}} = \delta^{\eta\alpha/4}.$$

Indeed, if every a had distance strictly less than this, F_0 would satisfy (A.3.88) at its own, larger codimension, since $\delta^{\theta^{\ell+1}} \leq \delta^{\theta^{\text{codim } F_0}}$. Conjugation and (A.3.89) therefore give

$$\text{dist}(gag^{-1}, F) \geq \delta^{10me + \eta\alpha/4} > \delta_1^\eta$$

provided $10me < \eta\theta^N/28$, because $\eta/k - \eta\alpha/4 \geq \eta\alpha/28$. Finally, (A.3.88) gives

$$\text{dist}(gag^{-1}, E) \leq \delta^{-10me} \delta_1^2 \leq \delta_1$$

if $10me \leq c$. Choose e_0 to satisfy these displayed strict parameter requirements. $\square$

The next lemma is the other dimension-free part of the matrix sum–product proof. It is stated with a single large parameter K , rather than source $O(e)$ -notation, so its use at several scales is explicit. The finite Ruzsa covering and Petridis inequalities invoked in the proof are established in Volume II.11.

A.3.9 Approximate rings and the matrix action theorem

Lemma A.3.56 (Approximate ring of small-action matrices). *Fix $m \geq 1$. There is $C_m > 0$ such that for $K \geq 2$, $0 < \delta < K^{-2}$, and a nonempty bounded $X \subset B(0, K) \subset \mathbb{R}^m$, the following holds. Assume*

$$\mathcal{N}_\delta(X + X) \leq K \mathcal{N}_\delta(X)$$

and put

$$\mathcal{G}_K(X) = \{a \in \text{End}(\mathbb{R}^m) : \|a\| \leq K, \mathcal{N}_\delta(X + aX) \leq K \mathcal{N}_\delta(X)\}.$$

If $a, b, a_1, \dots, a_s \in \mathcal{G}_K(X)$, then

$$\mathcal{N}_\delta(X + (a \pm b)X), \mathcal{N}_\delta(X + abX) \leq K^{C_m} \mathcal{N}_\delta(X), \quad (\text{A.3.90})$$

$$\mathcal{N}_\delta(X + a_1X + \dots + a_sX) \leq K^{C_m(s+1)} \mathcal{N}_\delta(X), \quad (\text{A.3.91})$$

$$\mathcal{N}_r(X + aX) \leq K^{C_m} \mathcal{N}_r(X) \quad (r \geq \delta). \quad (\text{A.3.92})$$

If a is invertible with $\|a^{-1}\| \leq K$, then

$$\mathcal{N}_\delta(X + a^{-1}X) \leq K^{C_m} \mathcal{N}_\delta(X). \quad (\text{A.3.93})$$

More generally, if $\|a - b\| \leq h$, then

$$\mathcal{N}_\delta(X + bX) \leq C_m(1 + hK/\delta)^m K^{C_m} \mathcal{N}_\delta(X). \quad (\text{A.3.94})$$

In particular the same K^{C_m} -type bound holds when $h \leq \delta$, even without assuming $b \in \mathcal{G}_K(X)$. The bounds in (A.3.90) remain true if the operator norms of $a \pm b, ab$ exceed K ; membership in the particular set $\mathcal{G}_K(X)$ is not asserted.

Proof. Write $N = \mathcal{N}_\delta(X)$. Rounding bounded sets to the δ -lattice in $\mathbb{R}^m$, applying the finite-group Ruzsa covering lemma, and lifting back gives finite sets T_a, T_b with $\#T_a, \#T_b \leq C_m K$ such that

$$aX \subset X - X + T_a + B(0, C_m \delta), \quad bX \subset X - X + T_b + B(0, C_m \delta). \quad (\text{A.3.95})$$

The rounding changes the covering counts by at most C_m ; this is the same finite-lattice reduction used in Volume II.11. The Petridis–Ruzsa inequality there, applied to $\mathcal{N}_\delta(X + X) \leq KN$, gives for fixed p, q

$$\mathcal{N}_\delta(pX - qX) \leq K^{C_m(p+q)} N. \quad (\text{A.3.96})$$

Indeed the finite statement is dimension-free and each bounded rounding error costs at most a dimensional factor.

For addition and subtraction, (A.3.95) gives

$$X + (a \pm b)X \subset 3X - 2X + (T_a \pm T_b) + B(0, 2C_m\delta).$$

Cover this set using (A.3.96) and $\#T_a \#T_b \leq C_m^2 K^2$. For multiplication, apply a to the second inclusion of (A.3.95); since $\|a\| \leq K$,

$$abX \subset aX - aX + aT_b + B(0, C_m K \delta).$$

Apply the first inclusion of (A.3.95) to both copies of aX . The result lies in

$$2X - 2X + T_a - T_a + aT_b + B(0, C_m(K + 2)\delta).$$

Adding X , using (A.3.96), and subdividing radius- $C_m K \delta$ balls into radius- δ balls proves (A.3.90). The same covering argument applied to each $a_j X$ gives

$$X + \sum_{j=1}^s a_j X \subset (s+1)X - sX + \sum_{j=1}^s T_{a_j} + B(0, C_m s \delta),$$

which proves (A.3.91).

If a is invertible, then

$$X + a^{-1}X = a^{-1}(aX + X).$$

A K -Lipschitz image of a radius- δ ball is coverable by at most $C_m K^m$ radius- δ balls; hence (A.3.93) follows. If $\|a - b\| \leq h$, then $(b - a)X \subset B(0, hK)$, and hence $X + bX \subset X + aX + B(0, hK)$. Cover each enlarged radius- $(\delta + hK)$ ball by at most $C_m(1 + hK/\delta)^m$ radius- δ balls. This proves (A.3.94) and its special case.

It remains to verify (A.3.92). Fix $r \geq \delta$ and put $M_r = \sup_z \mathcal{N}_\delta(X \cap B(z, r))$. Covering X by radius- r balls gives

$$N \leq C_m \mathcal{N}_r(X) M_r. \quad (\text{A.3.97})$$

Choose a $4r$ -separated set of cardinality at least $c_m \mathcal{N}_r(X + aX)$ in $X + aX$ and a ball $B(z, r)$ with local covering number arbitrarily close to M_r . Its translates by those separated sums are contained in $X + X + aX$ and are $2r$ -separated. Thus

$$\mathcal{N}_\delta(X + X + aX) \geq c_m \mathcal{N}_r(X + aX) M_r. \quad (\text{A.3.98})$$

The hypothesis on $X + X$ says that $\text{Id} \in \mathcal{G}_K(X)$. Hence the multi-sum bound (A.3.91), applied to Id and a , bounds the left side by $K^{C_m} N$. Combine with (A.3.97)–(A.3.98) to obtain (A.3.92). $\square$

Corollary A.3.57 (Bounded-word and change-of-scale calculus). *Fix m and an integer $s \geq 1$. There is $C_{m,s}$ such that the following holds when $e > 0$ is sufficiently small in terms of m, s and δ is sufficiently small. Suppose $X \subset B(0, \delta^{-e})$ is nonempty, $A \subset B_{\text{End}(\mathbb{R}^m)}(0, \delta^{-e})$, and*

$$\mathcal{N}_\delta(X + X), \quad \sup_{a \in A} \mathcal{N}_\delta(X + aX) \leq \delta^{-e} \mathcal{N}_\delta(X).$$

For every $T \in \mathcal{W}_s(A \cup \{I\})$ and every $r \geq \delta$,

$$\|T\| \leq \delta^{-C_{m,s}e}, \quad \mathcal{N}_r(X + TX) \leq \delta^{-C_{m,s}e} \mathcal{N}_r(X). \quad (\text{A.3.99})$$

If $\|U - T\| \leq r$, the second bound holds with U in place of T after increasing $C_{m,s}$ by a constant depending only on m .

Proof. The norm bound follows by induction on the expression tree: each leaf has norm at most δ^{-e} , and a fixed number of additions and multiplications increases the exponent by a constant depending only on s . The identity is a good leaf by the doubling hypothesis. For each fixed expression tree, apply [Theorem A.3.56](#) recursively at scale δ . At an addition, subtraction, or multiplication node, enlarge K to a fixed power of the largest previously obtained bound; [\(A.3.90\)](#) then bounds the new covering number. There are finitely many tree types with at most s leaves, so one $C_{m,s}$ works for all of them. The condition $\delta < K^{-2}$ at each application follows by requiring $2C_{m,s}e < 1$, after the finite recursion has fixed the constant. Finally apply [\(A.3.92\)](#) to each resulting word, again enlarging $C_{m,s}$, to obtain the bound at every $r \geq \delta$.

For the last assertion, $(U - T)X \subset B(0, r\delta^{-e})$. Each radius- r ball covering $X + TX$ enlarges to a ball of radius $r(1 + \delta^{-e})$, coverable by at most $C_m(1 + \delta^{-e})^m$ radius- r balls. Absorb this factor in the exponent. $\square$

Lemma A.3.58 (Scalar-like action propagates covering numbers). *Let $X \subset B_{\mathbb{R}^m}(0, R)$ be nonempty and bounded, where $R \geq 1$. Let $0 < r \leq 1$, $0 < \delta \leq r$, and let $T \in \text{End}(\mathbb{R}^m)$ satisfy $\|T - rI\| \leq r/2$. If $0 < \rho \leq 1$ and $r\rho \geq \delta$, then*

$$\mathcal{N}_{r\rho}(X + TX) \geq c_m R^{-m} \mathcal{N}_r(X) \mathcal{N}_\rho(X). \quad (\text{A.3.100})$$

Consequently, if $\mathcal{N}_{r\rho}(X + TX) \leq K \mathcal{N}_{r\rho}(X)$ for some $K \geq 1$, then

$$\mathcal{N}_{r\rho}(X) \geq c_m K^{-1} R^{-m} \mathcal{N}_r(X) \mathcal{N}_\rho(X). \quad (\text{A.3.101})$$

Proof. The singular values of T lie in $[r/2, 3r/2]$. Choose a maximal $10rR$ -separated subset $\{x_1, \dots, x_M\} \subset X$. Since $\text{diam}(TX) \leq 3rR$ and $r\rho \leq rR$, the translates $x_i + TX$ are pairwise separated by more than $2r\rho$. Therefore

$$\mathcal{N}_{r\rho}(X + TX) \geq \sum_{i=1}^M \mathcal{N}_{r\rho}(x_i + TX) = M \mathcal{N}_{r\rho}(TX).$$

The bi-Lipschitz bounds on T and a fixed-radius covering comparison give $\mathcal{N}_{r\rho}(TX) \geq c_m \mathcal{N}_\rho(X)$. Maximality of the separated set covers X by M balls of radius $10rR$. Each such ball is coverable by at most $C_m R^m$ balls of radius r , so $M \geq c_m R^{-m} \mathcal{N}_r(X)$. Combining these inequalities proves [\(A.3.100\)](#); the upper bound assumed in the second assertion then gives [\(A.3.101\)](#). $\square$

Proposition A.3.59 (The matrix action theorem from ball generation). *Assume the following assertion for each of the three standard simple real matrix algebras E acting on $\mathbb{R}^m$: for every $\kappa, \zeta > 0$ there are $b > 0$ and an integer s such that, for all sufficiently small h , every $F \subset E \cap B(0, h^{-b})$ satisfying*

$$\mathcal{N}_r(F) \geq h^b r^{-\kappa} \quad (r \geq h), \quad F \text{ is } h^b\text{-away from proper unital subalgebras of } E$$

obeys

$$E \cap B(0, h^\zeta) \subset \mathcal{W}_s(F) + B(0, h). \quad (\text{A.3.102})$$

Here only the models with integral matrix size are included. Under this assertion, [Theorem A.3.21](#) holds. The implication is proved below, while its premise is established unconditionally in [Theorem A.3.54](#).

Proof. Fix m, κ, ς as in [Theorem A.3.21](#). Apply [Theorem A.3.23](#) to the finite list of model algebras and choose common positive exponents e_*, c_* . First choose $0 < \zeta < 1/2$ so small that all losses of the form $h^{-C_m \zeta}$ in the inverse and product bounds of [Theorem A.3.56](#) are at most $h^{-c_*/8}$. The asserted ball-generation input then fixes b and s . Choose $0 < \eta < \min(1, b/2)$, and let $c = c(m, \eta) > 0$ be the constant of [Theorem A.3.55](#) if $m \geq 2$; for $m = 1$ put $c = 1$. All these choices precede the choice of the small theorem exponent e . We take e sufficiently small that every fixed-dimensional covering, conjugation, bounded-word and perturbation loss below, when expressed as a power of h^{-1} , is at most $h^{-\min(b, c_* e_*)/100}$, this is possible because $h = \delta^{1/k}$ with $k \leq 1/c$. We also require $C_{m, 2s} e < 1/(4k)$ and, at the end, e smaller than the gain in [\(A.3.108\)](#). These are finitely many strict inequalities, so one positive e works.

Suppose the conclusion of [Theorem A.3.21](#) fails with $\varepsilon = e$. Then, writing $N_r = \mathcal{N}_r(X)$,

$$\mathcal{N}_\delta(X + X), \quad \sup_{a \in \mathcal{A}} \mathcal{N}_\delta(X + aX) < \delta^{-e} N_\delta. \quad (\text{A.3.103})$$

For $m \geq 2$, apply [Theorem A.3.55](#) to $\mathcal{A}$. It supplies $h = \delta_1 = \delta^{1/k}$, a change of basis g , and a model algebra E with $k \leq 1/c$ and

$$\|g\| + \|g^{-1}\| \leq \delta^{-5me}, \quad g\mathcal{A}g^{-1} \subset E + B(0, h).$$

For $m = 1$ use instead $h = \delta, k = 1, g = I$ and $E = \mathbb{R}$; the proper-subalgebra condition is vacuous and all subsequent estimates are the same. In this case [\(A.3.108\)](#) already contradicts the assumed upper bound on $\mathcal{N}_\delta(X)$, so the final propagation is not needed. Let $Y = gX$ and let F be the orthogonal projection of $g\mathcal{A}g^{-1}$ onto E , using the Frobenius inner product. Fixed-dimensional norm comparison gives $\text{dist}(F, g\mathcal{A}g^{-1}) \leq C_m h$. The last conclusion of [Theorem A.3.55](#), after this projection and a harmless decrease of η , makes F h^b -away from proper subalgebras.

For a linear isomorphism u of $\mathbb{R}^d$, covering each preimage of an r -ball by $O_d((1 + \|u^{-1}\|)^d)$ r -balls gives

$$\mathcal{N}_r(uZ) \geq c_d(1 + \|u^{-1}\|)^{-d} \mathcal{N}_r(Z). \quad (\text{A.3.104})$$

The analogous upper comparison uses $\|u\|$. Apply these inequalities to g and to conjugation by g ; use the $C_m h$ -closeness of F and fixed-radius covering comparison for $r \geq h$. The hypotheses on $\mathcal{A}, X$ then yield, for our choice of e ,

$$F \subset B(0, h^{-b}), \quad \mathcal{N}_r(F) \geq h^b r^{-\kappa}, \quad Y \subset B(0, h^{-e_*}), \quad \mathcal{N}_r(Y) \geq h^{e_*} r^{-\kappa} \quad (r \geq h). \quad (\text{A.3.105})$$

Notice that the losses are powers of $\delta^{-O_m(e)}$, which are $h^{-O_m(ke)}$; the bounded integer k makes the displayed parameter choice legitimate.

By [Theorem A.3.57](#), [\(A.3.103\)](#) propagates to all scales $r \geq \delta$ for bounded words in $\mathcal{A}$. Conjugation and [\(A.3.104\)](#) therefore give, uniformly for $a \in \mathcal{A}$,

$$\mathcal{N}_h(Y + Y), \quad \mathcal{N}_h(Y + (ga g^{-1})Y) \leq \delta^{-C_m e} \mathcal{N}_h(Y).$$

Since F is within $C_m h$ of $g\mathcal{A}g^{-1}$ and $Y \subset B(0, \delta^{-C_m e})$, the perturbation estimate [\(A.3.94\)](#) gives the same bound with $f \in F$ in place of $ga g^{-1}$, after enlarging C_m . Apply [Theorem A.3.57](#) at scale h to Y, F , with its small exponent $O_m(ke)$ and the fixed word length s . By [\(A.3.102\)](#) and one more use of [\(A.3.94\)](#), every $T \in E \cap B(0, h^\varsigma)$ satisfies

$$\mathcal{N}_h(Y + TY) \leq h^{-C_{m,s} ke} \mathcal{N}_h(Y). \quad (\text{A.3.106})$$

In particular $h^\zeta I$ satisfies (A.3.106). The inverse estimate (A.3.93), with an enlarged K that also bounds $\|h^{-\zeta} I\|$, shows that $h^{-\zeta} I$ is a good action at cost $h^{-C_m(\zeta+C_{m,s}ke)}$. For any $U \in E \cap B(0, 1)$, the matrix $h^\zeta U$ is covered by (A.3.106); the product estimate (A.3.90) applied to $(h^{-\zeta} I)(h^\zeta U) = U$ consequently gives

$$\sup_{U \in E \cap B(0,1)} \mathcal{N}_h(Y + UY) \leq h^{-c_*/2} \mathcal{N}_h(Y), \quad (\text{A.3.107})$$

by the choices of ζ and e . The same bound holds for $\mathcal{N}_h(Y + Y)$.

If $\mathcal{N}_h(Y) \leq h^{-\zeta-e_*}$, then (A.3.105) and Theorem A.3.23 contradict (A.3.107). Hence $\mathcal{N}_h(Y) > h^{-\zeta-e_*}$. Using (A.3.104) once more and decreasing e gives

$$\mathcal{N}_h(X) \geq h^{-\zeta-e_*/2}. \quad (\text{A.3.108})$$

Because $\zeta < 1/2$, the element $h^{1/2} I$ lies in the ball on the left of (A.3.102). Choose $q \in \mathcal{W}_s(F)$ with $\|q - h^{1/2} I\| \leq h$. Replace each leaf of q by a matrix in $g\mathcal{A}g^{-1}$ within $C_m h$, retaining the same expression tree. The finite telescoping identity for products and the norm bounds in (A.3.105) give an element $a \in \mathcal{W}_s(\mathcal{A})$ such that

$$\|a - h^{1/2} I\| \leq h\delta^{-C_{m,s}e}.$$

Squaring and using $C_{m,2s}e < 1/(4k)$ gives

$$a^2 \in \mathcal{W}_{2s}(\mathcal{A}), \quad \|a^2 - hI\| \leq h/2. \quad (\text{A.3.109})$$

The original-scale calculus Theorem A.3.57 also gives

$$\mathcal{N}_r(X + a^2 X) \leq \delta^{-C_{m,2s}e} \mathcal{N}_r(X) \quad (r \geq \delta).$$

For $j = 1, \dots, k-1$, apply Theorem A.3.58 with $r = h$, $\rho = h^j$, $T = a^2$ and $R = \delta^{-e}$. Its hypotheses hold because $h^{j+1} \geq h^k = \delta$. We obtain, with a constant independent of δ ,

$$\mathcal{N}_{hj+1}(X) \geq c_m \delta^{(C_{m,2s}+m)e} \mathcal{N}_h(X) \mathcal{N}_{hj}(X).$$

Iterating at most $1/c$ times and using (A.3.108) yields

$$\mathcal{N}_\delta(X) = \mathcal{N}_{h^k}(X) \geq \delta^{C_{m,s,c}e} \mathcal{N}_h(X)^k \geq \delta^{-\zeta-e_*/2+C_{m,s,c}e}.$$

For $e < e_*/(4(C_{m,s,c} + 1))$ this contradicts the upper bound $\mathcal{N}_\delta(X) \leq \delta^{-\zeta-e}$. The contradiction proves the stated implication. $\square$

Proof of Theorem A.3.21. For each standard simple real model algebra, the required ball-generation assertion is Theorem A.3.54. Apply Theorem A.3.59. Its other inputs are the locally proved model reduction, approximate-ring calculus, and action estimates cited in that proposition's proof. $\square$

The next implication closes the half-dimensional geometric reduction. It invokes the matrix theorem established above. We use the finite-group Graph BSG and Ruzsa–Petridis inequalities proved in Volume II.11, `lem:graph-bsg` and `lem:petridis`. Their proofs are dimension-free; rounding a bounded subset of $\mathbb{R}^m$ to a δ -lattice changes each covering count by at most a dimensional constant. A linear map whose norm and inverse norm are at most δ^{-C_e} changes these estimates by at most another fixed power of δ^{-e} .

A.3.10 The half-dimensional projection base case

Proposition A.3.60 (Half-dimensional base conditional on matrix sum-product). *Assume Theorem A.3.21 in dimension m . Then Theorem A.3.61 holds for $n = 2m$.*

Proof. The case $m = 1$ is the rank-one theorem proved in Volume III.6, so let $m \geq 2$. Fix α, κ , set $b = \alpha/2$, and choose

$$b < \varsigma < m, \quad 0 < \kappa_0 < \min(\kappa/8, \kappa).$$

Let $e_{\text{sp}} > 0$ be the matrix-theorem exponent for (m, κ_0, ς) . We choose the present exponent $e > 0$ small compared with e_{sp} after bounding the finitely many power losses below. Write $K = \delta^{-e}$.

Suppose no nonempty subset $T \subset A$ obeys $\sigma(\mathcal{E}_\delta(T; e)) \leq \delta^e$. The regular-projection lemma gives V_1 and $A_1 \subset A$ such that, with $X_1 = \pi_{V_1} A_1$,

$$\begin{aligned} \mathcal{N}_\delta(A_1) &\geq \delta^{-\alpha+3e}, & \mathcal{N}_\delta(X_1) &< \delta^{-b-e}, \\ \mathcal{N}_\delta(X_1 \cap B(x, r)) &\leq \delta^{-b-10e} r^{\kappa/(8)} & (x \in V_1, r \geq \delta). \end{aligned} \quad (\text{A.3.110})$$

Put $t = \delta^{3e/\kappa}$. Since $\sigma(\mathcal{S}(V_1, t)) \leq \delta^{2e}$, choose $V_2 \in \mathcal{E}_\delta(A_1; e) \setminus \mathcal{S}(V_1, t)$ and a witnessing $A_2 \subset A_1$. Thus

$$\Delta(V_2, V_1) > t, \quad \mathcal{N}_\delta(A_2) \geq \delta^{-\alpha+4e}, \quad \mathcal{N}_\delta(\pi_{V_j} A_2) < \delta^{-b-e} \quad (j = 1, 2). \quad (\text{A.3.111})$$

Let $F : V_1 \oplus V_2 \rightarrow \mathbb{R}^{2m}$ be the inverse of $z \mapsto (\pi_{V_1} z, \pi_{V_2} z)$. The determinant lower bound in (A.3.111) and the singular-value formula imply

$$L := \max(1, \|F\|, \|F^{-1}\|) \leq C_m t^{-1} \leq K^{C_1}. \quad (\text{A.3.112})$$

Hereafter $C_1, C_2, \dots$ are finite constants depending only on m, α, κ , enlarged when needed; fixed dimensional coefficients are absorbed by taking δ small. Put $X = \pi_{V_1} A_2 \subset X_1$ and $Y = \pi_{V_2} A_2$. Since $A_2 \subset F(X \times Y)$, product covering and (A.3.111) give

$$K^{-C_2} \delta^{-b} \leq \mathcal{N}_\delta(X), \mathcal{N}_\delta(Y) \leq K \delta^{-b}. \quad (\text{A.3.113})$$

The set

$$D = \mathcal{E}_\delta(A_2; e) \setminus (\mathcal{S}(V_1, t) \cup \mathcal{S}(V_2, t))$$

has $\sigma(D) \geq \delta^e - 2\delta^{2e} \geq \delta^{2e}$. For $V \in D$, the restrictions $P_V = \pi_V F|_{V_1}$ and $Q_V = \pi_V F|_{V_2}$ are invertible: $F(V_1) = V_2^\perp$, $F(V_2) = V_1^\perp$, and the two excluded tubes bound their determinants below. Define $\phi_V = P_V^{-1} Q_V : V_2 \rightarrow V_1$. The same singular-value comparison as in (A.3.112) gives

$$\|\phi_V\| + \|\phi_V^{-1}\| + \|P_V^{-1}\| \leq K^{C_3} \quad (V \in D). \quad (\text{A.3.114})$$

Here are the finite-scale details of the BSG reduction. Take maximal 2δ -separated representatives $X_0 \subset X$, $Y_0 \subset Y$. A witness $A_V \subset A_2$ for $V \in D$ has $\mathcal{N}_\delta(A_V) \geq \delta^{-\alpha+5e}$. Map a separated net of A_V by F^{-1} and round its two coordinates to cells represented in X_0, Y_0 . The image under F of one pair of coordinate cells is coverable by K^{C_4} radius- δ balls,

so at least $K^{-C_4}\delta^{-\alpha}$ distinct pairs remain. After applying P_V, Q_V , round both coordinate images to a common δ -lattice in V . By (A.3.114), at most K^{C_5} cells arise from each original projection ball. Therefore these pairs form a graph of density at least K^{-C_5} in vertex sets of cardinality $K^{\pm C_5}\delta^{-b}$, and their restricted sumset has cardinality at most $K^{C_5}\delta^{-b}$. The finite Graph BSG lemma from Volume II.11, followed by pullback under P_V , supplies $X_V \subset X_0$, $Y_V \subset Y_0$ such that

$$|X_V|, |Y_V| \geq K^{-C_6}\delta^{-b}, \quad \mathcal{N}_\delta(X_V + \phi_V Y_V) \leq K^{C_6}\delta^{-b}. \quad (\text{A.3.115})$$

The sets can be selected measurably in V , since there are only finitely many candidates and the cover conditions are Borel.

Normalize $\sigma|_D$ to a probability ν . Finite Cauchy–Schwarz gives

$$\mathbb{E}_{V, V' \sim \nu} |X_V \cap X_{V'}| |Y_V \cap Y_{V'}| = \sum_{(x,y) \in X_0 \times Y_0} \nu\{V : x \in X_V, y \in Y_V\}^2 \geq K^{-C_7} |X_0| |Y_0|.$$

Pigeonholing this bounded random variable yields a $V_* \in D$ and $D' \subset D$ with $\sigma(D') \geq K^{-C_8}$, for which

$$|X_V \cap X_*|, |Y_V \cap Y_*| \geq K^{-C_8}\delta^{-b} \quad (V \in D'), \quad X_* = X_{V_*}, \quad Y_* = Y_{V_*}. \quad (\text{A.3.116})$$

The Ruzsa triangle inequality, using the common intersections as middle sets, now bounds $X_* - X_V$ and $Y_* - Y_V$ by $K^{C_9}\delta^{-b}$. Apply it twice more with the small sums in (A.3.115), once at V and once at V_* ; the Petridis–Ruzsa inequality gives the first bound below:

$$\mathcal{N}_\delta(X_* + X_*) \leq K^{C_{10}}\delta^{-b}, \quad \mathcal{N}_\delta(X_* - \phi_V \phi_*^{-1} X_*) \leq K^{C_{10}}\delta^{-b} \quad (V \in D'). \quad (\text{A.3.117})$$

All divisions in these Ruzsa applications are by sets of size at least $K^{-C_8}\delta^{-b}$; rounding errors are controlled by (A.3.114). Thus C_{10} is independent of δ, e, V .

Set $\mathcal{A} = \{-\phi_V \phi_*^{-1} : V \in D'\}$. For any operator f , fix a unit $u \in V_1$, put $v = \phi_*^{-1}u$, and choose a hyperplane $W \subset V_1$ containing $-fu$. If $\|-\phi_V \phi_*^{-1} - f\| \leq r$, then $\text{dist}(\phi_V v, W) \leq r$. Theorem A.3.12 puts such V in a Schubert tube of radius $K^{C_{11}}r$. Cover $\mathcal{A}$ by r -balls, apply (A.3.11) to each pullback, and use $\sigma(D') \geq K^{-C_8}$. It follows that

$$\mathcal{N}_r(\mathcal{A}) \geq K^{-C_{12}}r^{-\kappa} \quad (r \geq \delta). \quad (\text{A.3.118})$$

For $K^{C_{11}}r > 1$ the displayed lower bound is at most one after enlarging C_{12} , so this covers all $r \geq \delta$. For every proper nonzero $W \subset V_1$, take a unit $u \in W$ and $v = \phi_*^{-1}u$. If all $a \in \mathcal{A}$ had $\text{dist}(au, W) < r$, the same tube lemma would put all of D' in one tube of radius $K^{C_{11}}r$. The Schubert bound forces $r \geq K^{-C_{13}}$. This is precisely the quantitative irreducibility hypothesis.

Finally, (A.3.110) and the size of X_* give, by covering X_* with r -balls and summing their δ -covering numbers,

$$\mathcal{N}_r(X_*) \geq K^{-C_{14}}r^{-\kappa/8} \quad (r \geq \delta). \quad (\text{A.3.119})$$

The sets $\mathcal{A}, X_*$ lie in a $K^{C_{14}}$ -ball, and $\mathcal{N}_\delta(X_*) \leq K^{C_{14}}\delta^{-b} \leq \delta^{-\varsigma - e_{\text{sp}}}$, since $b < \varsigma$. Choose e so that $C_j e < e_{\text{sp}}/10$ for $1 \leq j \leq 14$. Then (A.3.118), (A.3.119) and irreducibility imply every hypothesis of Theorem A.3.21 with exponent e_{sp} and nonconcentration exponent κ_0 . The theorem's lower bound is $\delta^{-e_{\text{sp}}} \mathcal{N}_\delta(X_*)$, while (A.3.115) and (A.3.117) bound the same sum by $2K^{C_{15}} \mathcal{N}_\delta(X_*)$. Decrease e once more so $C_{15}e < e_{\text{sp}}/2$. For small δ this is a contradiction. $\square$

A.3.11 Good pieces and quotient/sum-rank induction

He first proves the following apparently weaker projection statement.

Theorem A.3.61 (Finite-set form of He's good-piece theorem). *Suppose A is finite and satisfies the hypotheses (A.3.9)–(A.3.11), with the same parameters $(n, m, \alpha, \kappa, \varepsilon)$. Then there exists a nonempty $T \subset A$ such that*

$$\sigma(\mathcal{E}_\delta(T; \varepsilon)) \leq \delta^\varepsilon. \quad (\text{A.3.120})$$

Source and ownership. This is the finite-set restriction of He, Theorem 6, printed p. 4. The base case is reduced in Theorem A.3.60 to the matrix theorem proved in this chapter. The quotient, sum and intersection rank arrows (Propositions 9–11) and the all-rank large-set normalization needed by the last arrow have local proofs; Theorem A.3.73 assembles them. The next proposition proves that this restricted good-piece statement implies the full all-large-subsets theorem.

Proposition A.3.62 (Good-piece theorem implies the full projection theorem). *Fix n, m, α, κ as in Theorem A.3.8. If Theorem A.3.61 holds with some exponent $\varepsilon_* > 0$, then Theorem A.3.8 holds with every sufficiently small exponent $0 < \varepsilon < \varepsilon_*/10$.*

Proof. Decrease ε_* if necessary so that $0 < \varepsilon_* < 1$; the good-piece assertion remains valid under a smaller exponent by the same monotonicity of hypotheses and bad-event thresholds. Fix $0 < \varepsilon < \varepsilon_*/10$ and let A, σ satisfy (A.3.9)–(A.3.11) with ε in place of the displayed exponent. Choose a maximal δ -separated finite set $P \subset A$. Euclidean packing gives a constant C_n such that

$$C_n^{-1} \#P \leq \mathcal{N}_\delta(P) \leq \mathcal{N}_\delta(A) \leq \#P \leq C_n \mathcal{N}_\delta(A). \quad (\text{A.3.121})$$

The middle upper bound holds because the closed radius- δ balls centered on P cover A . The same packing estimate on $P \cap B(x, \rho)$, followed by (A.3.10), shows that P satisfies the size and local nonconcentration conditions of Theorem A.3.61 with exponent $\varepsilon_*/2$ for all small δ ; fixed constants are absorbed because $\varepsilon < \varepsilon_*/10$. The direction condition holds with exponent ε_* as well.

Set $\lambda = \delta^{\varepsilon_*/2}$. Starting with P , repeatedly apply Theorem A.3.61 to the current residual set R as long as $\#R \geq \lambda \#P$. Indeed, (A.3.121) and the bounds on P show, uniformly for such R , that

$$\mathcal{N}_\delta(R) \geq C_n^{-1} \#R \geq \delta^{-\alpha+\varepsilon_*}, \quad \frac{\mathcal{N}_\delta(R \cap B(x, \rho))}{\mathcal{N}_\delta(R)} \leq \delta^{-\varepsilon_*} \rho^\kappa \quad (\rho \geq \delta)$$

once δ is sufficiently small. The new good piece $P_i \subset R$ is nonempty, so the process terminates after finitely many steps with disjoint good pieces $P_1, \dots, P_J$ and residual R_0 satisfying $\#R_0 < \lambda \#P$. Each piece obeys

$$\sigma(\mathcal{E}_\delta(P_i; \varepsilon_*)) \leq \delta^{\varepsilon_*}. \quad (\text{A.3.122})$$

Let $G = \bigsqcup_i P_i$, $\theta = \delta^{\varepsilon_*/3}$, $\eta = C_n \delta^{\varepsilon_*}$ and $M = \frac{1}{2} \delta^{-b-\varepsilon_*}$. For small δ , $\theta - \lambda > \eta$. If $V \in \mathcal{B}_\delta(P; \theta, M)$ with witnessing set $B \subset P$, then $\#(\bar{B} \cap G) \geq (\theta - \lambda) \#P \geq (\theta - \lambda) \#G$ and

$\mathcal{N}_\delta(\pi_V(B \cap G)) \leq M$. Apply [Theorem A.3.7](#) to the partition of G into the P_i . Its piece event $\mathcal{B}_\delta(P_i; \eta, M)$ is contained in $\mathcal{E}_\delta(P_i; \varepsilon_*)$: for any η -large subset $T \subset P_i$, packing gives

$$\mathcal{N}_\delta(T) \geq C_n^{-1} \#T \geq \delta^{\varepsilon_*} \#P_i \geq \delta^{\varepsilon_*} \mathcal{N}_\delta(P_i),$$

and $M < \delta^{-b-\varepsilon_*}$. Thus [\(A.3.122\)](#) and the weights summing to one yield

$$\sigma(\mathcal{B}_\delta(P; \theta, M)) \leq \frac{1-\eta}{\theta-\lambda-\eta} \delta^{\varepsilon_*} \leq \delta^{\varepsilon_*/2} \quad (\text{A.3.123})$$

for all small δ .

Finally, let $V \in \mathcal{E}_\delta(A; \varepsilon)$ and choose a witnessing $A' \subset A$. Let $P' = \{p \in P : \text{dist}(p, A') \leq \delta\}$. The δ -balls centered on P' cover A' , so [\(A.3.121\)](#) gives

$$\#P' \geq \mathcal{N}_\delta(A') \geq \delta^\varepsilon \mathcal{N}_\delta(A) \geq C_n^{-1} \delta^\varepsilon \#P \geq \theta \#P.$$

Because $\pi_V P'$ is contained in the 2δ -neighborhood of $\pi_V A'$, a bounded-overlap covering comparison gives

$$\mathcal{N}_\delta(\pi_V P') \leq C_m \mathcal{N}_\delta(\pi_V A') < C_m \delta^{-b-\varepsilon} \leq M$$

for small δ . Hence $\mathcal{E}_\delta(A; \varepsilon) \subset \mathcal{B}_\delta(P; \theta, M)$. The latter is Borel because P is finite, and [\(A.3.123\)](#) bounds its measure by $\delta^{\varepsilon_*/2} \leq \delta^\varepsilon$. Taking its Borel complement as D proves [Theorem A.3.8](#) at exponent ε without assuming measurability of $\mathcal{E}_\delta(A; \varepsilon)$ for an arbitrary set A . $\square$

The sum-of-planes lemma also completes one of the induction arrows in He's argument. It is important to distinguish a proved implication between ranks from a proof of the base rank itself. In the next proposition, the larger-rank instance of [Theorem A.3.8](#) is an explicit hypothesis.

The regular slice also closes the quotient arrow when m divides n . There is a small point worth making explicit: the slice supplied by [Theorem A.3.11](#) lies in $A^{(\delta)}$, rather than in A . Moving its points back to A costs only fixed covering constants, but is necessary before contradicting the assumed failure of the good-piece theorem.

Proposition A.3.63 (Quotient-rank induction for the good-piece theorem). *Let $q \geq 3$, $n = qm$, $n' = (q-1)m$, $0 < \alpha < n$, and $\kappa > 0$. Put $\beta = (q-1)\alpha/q$ and $\kappa_1 = \kappa/(2q^2)$. If [Theorem A.3.8](#) holds for (n', m, β, κ_1) , then [Theorem A.3.61](#) holds for (n, m, α, κ) with a sufficiently small positive exponent depending on the exponent of that lower-dimensional instance.*

Proof. Let $e_* > 0$ be an exponent for the assumed (n', m, β, κ_1) instance of [Theorem A.3.8](#). Choose $\varepsilon > 0$ so small that

$$\varepsilon < \frac{\kappa}{100q^2}, \quad (2C+2)\varepsilon < e_*, \quad \left(4 + \frac{3m}{\kappa}\right)\varepsilon < e_*, \quad (\text{A.3.124})$$

where C is the constant in [Theorem A.3.11](#). Decrease ε further to absorb fixed dimensional constants in the inequalities below. Suppose, toward contradiction, that a finite $A \subset B(0, 1)$ and σ satisfy [\(A.3.9\)–\(A.3.11\)](#), but no nonempty subset of A has the asserted good-piece

property. Thus (A.3.20) holds. Apply Theorem A.3.11 and obtain $y + W$ and B . Here $W = V_q^\perp$ and $y \in V_q = W^\perp$ in that lemma's construction, so $B - y \subset B_W(0, 1)$.

Write $\tilde{B} = B - y \subset W$. Its size bound (A.3.21) and the first inequality in (A.3.124) imply the lower-rank size hypothesis at exponent e_* . Moreover, by (A.3.22) and that size bound,

$$\mathcal{N}_\delta(\tilde{B} \cap B_W(x, \rho)) \leq \delta^{-2C_\varepsilon} \rho^{\kappa_1} \mathcal{N}_\delta(\tilde{B}) \leq \delta^{-e_*} \rho^{\kappa_1} \mathcal{N}_\delta(\tilde{B}) \quad (\rho \geq \delta).$$

Let $G = \{V : J_W(V) > 0\}$ and let σ_W be the conditional pushforward of $\sigma|_G$ to $\text{Gr}(W, m)$. By Theorem A.3.5, $\sigma_W(\mathcal{S}_W(U, \rho)) \leq \delta^{-2\varepsilon} \rho^\kappa$ for $\delta \leq \rho \leq 1$. Since $\kappa > \kappa_1$ and $2\varepsilon < e_*$, this is at most $\delta^{-e_*} \rho^{\kappa_1}$; for $\rho > 1$ the required bound is automatic. All the lower-rank hypotheses therefore hold, and Theorem A.3.8 yields

$$\sigma_W(\mathcal{E}_\delta^W(\tilde{B}; e_*)) \leq \delta^{e_*}. \quad (\text{A.3.125})$$

The threshold in the internal bad event is $(m/n')\beta = (m/n)\alpha = b$.

For each $b_0 \in B$, choose one $a(b_0) \in A$ with $|a(b_0) - b_0| < \delta/2$ from the construction of B and put $T = \{a(b_0) : b_0 \in B\} \subset A$. The chosen map is injective: two distinct points b_0 are projections of distinct points of the finite set B_0 in that construction, and we choose one of those preimages for each. Fixed-radius covering comparisons give

$$C_n^{-1} \mathcal{N}_\delta(B') \leq \mathcal{N}_\delta(a(B')) \leq C_n \mathcal{N}_\delta(B') \quad (B' \subset B), \quad (\text{A.3.126})$$

and the same comparison holds for the projections in each m -plane, with C_m in place of C_n . Consequently, if $V \in \mathcal{E}_\delta(T; \varepsilon)$, the inverse image under a of a witnessing subset of T shows, for all small δ , that

$$\mathcal{E}_\delta(T; \varepsilon) \subset \mathcal{E}_\delta(B; 2\varepsilon). \quad (\text{A.3.127})$$

Indeed its covering number is at least $C_n^{-2} \delta^\varepsilon \mathcal{N}_\delta(B) \geq \delta^{2\varepsilon} \mathcal{N}_\delta(B)$, and its projected covering number is less than $C_m \delta^{-b-\varepsilon} < \delta^{-b-2\varepsilon}$. Translation by $-y$ leaves these covering numbers unchanged.

Set $h = 3\varepsilon/\kappa < 1$ and $H = \{V : J_W(V) < \delta^h\}$. The first assertion of Theorem A.3.5 gives $\sigma(H) \leq \delta^{-\varepsilon} \delta^{h\kappa} = \delta^{2\varepsilon}$. For $V \in \mathcal{E}_\delta(B; 2\varepsilon) \setminus H$, Theorem A.3.9 sends V to $V' = \pi_W V \in \mathcal{E}_\delta^W(\tilde{B}; 4\varepsilon + mh)$. The last inequality in (A.3.124) implies $\mathcal{E}_\delta^W(\tilde{B}; 4\varepsilon + mh) \subset \mathcal{E}_\delta^W(\tilde{B}; e_*)$. Using (A.3.125) and (A.3.127), we obtain

$$\sigma(\mathcal{E}_\delta(T; \varepsilon)) \leq \delta^{2\varepsilon} + \delta^{e_*} \leq \delta^\varepsilon$$

for sufficiently small δ . This contradicts (A.3.20) for the nonempty set T . $\square$

The next result is the divisible-rank starting case of He's projection theorem without spatial nonconcentration. Its extra power of δ^{-1} in the size of A replaces the missing ball bound. The later relative-size induction will extend it to every rank.

Proposition A.3.64 (Large-set projection at divisible rank). *Let $n = qm$ with $q \geq 2$, $0 < \alpha < n$, and $\kappa > 0$. There is $M = M(n, q, \kappa) > 0$ such that, for every sufficiently small $\varepsilon > 0$ and all sufficiently small $\delta > 0$, the following holds. If a finite $A \subset B(0, 1)$ and a probability σ on $\text{Gr}(n, m)$ satisfy*

$$\mathcal{N}_\delta(A) \geq \delta^{-\alpha-M\varepsilon}, \quad \sigma(\mathcal{S}(W, \rho)) \leq \delta^{-\varepsilon} \rho^\kappa \quad (W \in \text{Gr}(n, n-m), \rho \geq \delta),$$

then

$$\sigma(\mathcal{E}_\delta(A; \varepsilon)) \leq \delta^\varepsilon.$$

No ball-nonconcentration hypothesis on A is assumed.

Proof. First prove a good-piece assertion. Choose a fixed $M_0 > 2q + 3n(q-1)/\kappa + 10$. Suppose a finite set F has $\mathcal{N}_\delta(F) \geq \delta^{-\alpha-M_0u}$ and satisfies the displayed direction bound with u in place of ε , where $0 < u < \kappa/10$. We claim that some nonempty $T \subset F$ has $\sigma(\mathcal{E}_\delta(T; u)) \leq \delta^u$. If not, then for every nonempty $T \subset F$ the bad event has measure greater than δ^u . Put $h = 3u/\kappa < 1$. Choose successively $V_1, \dots, V_q$ and nonempty nested sets $F = F_0 \supset F_1 \supset \dots \supset F_q$ as follows. Having chosen the first $j-1$ planes, extend their sum to a fixed $(n-m)$ -plane W_j and choose $V_j \in \mathcal{E}_\delta(F_{j-1}; u)$ outside $\mathcal{S}(W_j, \delta^h)$. This is possible because the latter set has σ -mass at most $\delta^{2u} < \delta^u$. Take F_j to witness badness. For $j = 1$, no exclusion is needed. Then

$$\mathcal{N}_\delta(F_q) \geq \delta^{qu} \mathcal{N}_\delta(F), \quad \mathcal{N}_\delta(\pi_{V_j} F_q) < \delta^{-\alpha/q-u}, \quad J := \text{Vol}(V_1, \dots, V_q) \geq \delta^{(q-1)h}. \quad (\text{A.3.128})$$

The coordinate map $z \mapsto (\pi_{V_1} z, \dots, \pi_{V_q} z)$ has inverse norm at most $C_n J^{-1}$. Covering each projected coordinate set and subdividing the inverse images of the product balls therefore gives

$$\mathcal{N}_\delta(F_q) \leq C_n J^{-n} \prod_{j=1}^q \mathcal{N}_\delta(\pi_{V_j} F_q) < C_n \delta^{-\alpha-[q+3n(q-1)/\kappa]u}.$$

On the other hand, the first inequality in (A.3.128) gives $\mathcal{N}_\delta(F_q) \geq \delta^{-\alpha-(M_0-q)u}$. The choice of M_0 makes these incompatible for sufficiently small δ . This proves the good-piece claim.

We upgrade it to the asserted all-subsets conclusion, retaining enough size surplus for every residual set. Choose an integer $L \geq 10$ and put $u = L\varepsilon$; require $u < \kappa/10$ and take $M > (M_0 + 1)L$. Let $P \subset A$ be a maximal δ -separated net. By packing, $C_n^{-1} \#P \leq \mathcal{N}_\delta(P) \leq \mathcal{N}_\delta(A) \leq \#P \leq C_n \mathcal{N}_\delta(A)$. Set $\lambda = \delta^{u/2}$. As long as the current residual $R \subset P$ has $\#R \geq \lambda \#P$,

$$\mathcal{N}_\delta(R) \geq C_n^{-1} \lambda \#P \geq C_n^{-1} \delta^{-\alpha-M\varepsilon+u/2} \geq \delta^{-\alpha-M_0u}$$

for small δ , because $M > (M_0 + 1)L$. The direction condition at exponent u follows from the one at exponent ε . Repeatedly apply the good-piece claim, removing a nonempty piece each time. This finite procedure gives disjoint pieces $P_1, \dots, P_J$ with $\sigma(\mathcal{E}_\delta(P_i; u)) \leq \delta^u$ and a residual of size below $\lambda \#P$.

Apply the weighted aggregation [Theorem A.3.7](#) to the union of the pieces with $\theta = \delta^{u/3}$, $\eta = C_n \delta^u$, and $M_{\text{proj}} = \frac{1}{2} \delta^{-\alpha/q-u}$. Exactly as in (A.3.123), it gives

$$\sigma(\mathcal{B}_\delta(P; \theta, M_{\text{proj}})) \leq \frac{1-\eta}{\theta-\lambda-\eta} \delta^u \leq \delta^{u/2}$$

for small δ . The verification uses only the packing comparison $\mathcal{N}_\delta(X) \geq C_n^{-1} \#X$ for subsets of P , not spatial nonconcentration. Finally, if $V \in \mathcal{E}_\delta(A; \varepsilon)$, the points of P within δ of a witnessing subset of A form a set of cardinality at least $C_n^{-1} \delta^\varepsilon \#P \geq \theta \#P$, and its projection has covering number at most $C_m \delta^{-\alpha/q-\varepsilon} \leq M_{\text{proj}}$. Thus $V \in \mathcal{B}_\delta(P; \theta, M_{\text{proj}})$. Since $u/2 > \varepsilon$, the displayed measure bound proves the proposition. $\square$

For the remaining ranks, it is useful to keep the large-set normalization statement separate from the good-piece theorem. Write $L(n, m; \alpha, \kappa)$ for the following assertion: there are $M, \varepsilon_0 > 0$ such that, for every $0 < \varepsilon < \varepsilon_0$ and sufficiently small δ , every finite $A \subset B(0, 1)$ with $\mathcal{N}_\delta(A) \geq \delta^{-\alpha-M\varepsilon}$ and every probability σ satisfying (A.3.11) obey $\sigma(\mathcal{E}_\delta(A; \varepsilon)) \leq \delta^\varepsilon$. The bad event uses $b = (m/n)\alpha$. Thus Theorem A.3.64 proves this assertion when $m \mid n$.

Proposition A.3.65 (Sum-rank step for large-set normalization). *Let $q \geq 2$, $qm < n$, $0 < \alpha < n$, and $\kappa > 0$. If $L(n, qm; \alpha, \kappa/(2q))$ holds, then $L(n, m; \alpha, \kappa)$ holds.*

Proof. Let $M_*, \varepsilon_* > 0$ be the constants in the assumed rank- qm assertion. We first prove its rank- m good-piece analogue. Take a small $u > 0$ and a finite $F \subset B(0, 1)$ satisfying the direction hypothesis at exponent u and

$$\mathcal{N}_\delta(F) \geq \delta^{-\alpha-M_0u}, \quad (\text{A.3.129})$$

where M_0 will be fixed below. Suppose that every nonempty $T \subset F$ has $\sigma(\mathcal{E}_\delta(T; u)) > \delta^u$. Sample $V_1, \dots, V_q$ independently with law σ . Exactly as in the construction in Theorem A.3.66, choose nested witnesses $F = F_0 \supset F_1 \supset \dots \supset F_q$ whenever $V_j \in \mathcal{E}_\delta(F_{j-1}; u)$ and $\Delta(V_j, T_j) > \delta^{3u/\kappa}$, where T_j extends the preceding plane sum to an $(n-m)$ -plane. The conditional probability of each successful step is at least $\delta^u - \delta^{2u} \geq \delta^{2u}$. Thus with probability at least δ^{2qu} ,

$$\mathcal{N}_\delta(F_q) \geq \delta^{qu} \mathcal{N}_\delta(F), \quad \mathcal{N}_\delta(\pi_{V_j} F_q) < \delta^{-(m/n)\alpha-u}, \quad \text{Vol}(V_1, \dots, V_q) \geq \delta^{3(q-1)u/\kappa}. \quad (\text{A.3.130})$$

All selections are measurable because F is finite; choose the first witness from a fixed ordering of its subsets.

On this event $V = V_1 + \dots + V_q$ has dimension qm . The adjoint product-coordinate map described in the proof of Theorem A.3.66 has inverse norm at most $C_{n,q} \delta^{-3(q-1)u/\kappa}$. Product covering followed by subdivision of its inverse images therefore gives a fixed $C_p = C_p(n, q, \kappa)$ with

$$\mathcal{N}_\delta(\pi_V F_q) \leq \delta^{-(qm/n)\alpha-C_pu}. \quad (\text{A.3.131})$$

By Theorem A.3.6, the conditional distribution Σ of V on the direct-sum event satisfies

$$\Sigma(\mathcal{S}(W, r)) \leq \delta^{-2u} r^{\kappa/q} \leq \delta^{-e'} r^{\kappa/(2q)} \quad (r \geq \delta)$$

whenever $e' = Cu$ with $C \geq 2$. Choose $C > \max(2q, C_p + 1)$ and then $M_0 > M_*C$. For sufficiently small $u < \varepsilon_*/C$, (A.3.129) verifies the rank- qm size hypothesis at exponent e' . The subset F_q in (A.3.130) has $\mathcal{N}_\delta(F_q) \geq \delta^{e'} \mathcal{N}_\delta(F)$, and (A.3.131) places its plane V in $\mathcal{E}_\delta(F; e')$. The successful tuples have probability at least δ^{2qu} , hence their image has Σ -mass at least δ^{2qu} . Because $C > 2q$, this exceeds $\delta^{e'}$ for small δ , contradicting $L(n, qm; \alpha, \kappa/(2q))$. The good-piece claim follows.

Finally we upgrade the good-piece assertion. Fix $L \geq 10$, put $u = L\varepsilon < \min(\kappa/10, \varepsilon_*/C)$, choose $M > (M_0 + 1)L$, and let P be a maximal δ -separated net of the original set A . Set $\lambda = \delta^{u/2}$. As long as a residual $R \subset P$ has $\#R \geq \lambda \#P$, the packing comparison from Theorem A.3.64 yields

$$\mathcal{N}_\delta(R) \geq C_n^{-1} \delta^{-\alpha-M\varepsilon+u/2} \geq \delta^{-\alpha-M_0u}$$

for small δ . The direction condition at exponent u is inherited from that at exponent ε . Repeatedly remove good pieces P_i until the residual has fewer than $\lambda\#P$ points. Apply [Theorem A.3.7](#) to their union with $\theta = \delta^{u/3}$, $\eta = C_n\delta^u$, and $M_{\text{proj}} = \frac{1}{2}\delta^{-(m/n)\alpha-u}$. As in [\(A.3.123\)](#), the resulting bad-direction mass is at most $\delta^{u/2}$. A witness to $V \in \mathcal{E}_\delta(A; \varepsilon)$ produces at least $C_n^{-1}\delta^\varepsilon\#P$ net points with projection covering number at most $C_m\delta^{-(m/n)\alpha-\varepsilon}$. For small δ , these bounds imply $V \in \mathcal{B}_\delta(P; \theta, M_{\text{proj}})$. Since $u/2 > \varepsilon$, the required mass is at most δ^ε . This proves $\mathcal{L}(n, m; \alpha, \kappa)$. $\square$

Proposition A.3.66 (Sum-rank induction for the good-piece theorem). *Let $q \geq 2$ and $qm < n$, and fix $0 < \alpha < n$ and $\kappa > 0$. If [Theorem A.3.8](#) is valid for $(n, qm, \alpha, \kappa/q)$, then [Theorem A.3.61](#) is valid for (n, m, α, κ) with a sufficiently small exponent depending on the former theorem's exponent.*

Proof. Write $b = m\alpha/n$, and let $e_0 > 0$ be an exponent for the asserted rank- qm instance. Its conclusion also holds with any $0 < e_* \leq e_0$: the hypotheses with e_* are stronger, while its conclusion asks about fewer large subsets, a weaker projection lower bound, and a weaker exceptional-probability bound. Fix $e_* < e_0/2$, and take $0 < \varepsilon \ll_{n,m,q,\kappa} e_*$ with $\varepsilon < \kappa/(4q)$. We show that a finite counterexample to the rank- m good-piece assertion is impossible for sufficiently small δ .

Let A, σ satisfy [\(A.3.9\)–\(A.3.11\)](#) at rank m and exponent ε . Suppose that every nonempty $T \subset A$ obeys $\sigma(\mathcal{E}_\delta(T; \varepsilon)) > \delta^\varepsilon$. Take independent $V_1, \dots, V_q$ with law σ . We construct nested nonempty sets $A = A_0 \supset A_1 \supset \dots \supset A_q$ as the planes are sampled. At the first step, choose $V_1 \in \mathcal{E}_\delta(A_0; \varepsilon)$ and a witnessing A_1 . Since A is finite, all choices may be made from a fixed ordering of its subsets, and the resulting events and selections are measurable.

At step $j \geq 2$, condition on the previous choices, and put $S_{j-1} = V_1 + \dots + V_{j-1}$. Extend S_{j-1} to an $(n-m)$ -plane T_j ; on previous successful steps this is possible because $(j-1)m \leq n-m$. We choose the extension by a fixed Borel Gram–Schmidt rule using the standard basis. Set $h = 3\varepsilon/\kappa$. The direction hypothesis gives

$$\sigma\{V : \Delta(V, T_j) \leq \delta^h\} \leq \delta^{-\varepsilon+h\kappa} = \delta^{2\varepsilon}.$$

The counterexample assumption, applied to A_{j-1} , thus leaves conditional probability at least $\delta^\varepsilon - \delta^{2\varepsilon} \geq \delta^{2\varepsilon}$ for a $V_j \in \mathcal{E}_\delta(A_{j-1}; \varepsilon)$ satisfying $\Delta(V_j, T_j) > \delta^h$. Choose a witnessing A_j on this event. Therefore the whole successful construction has probability at least $\delta^{(2q-1)\varepsilon} \geq \delta^{2q\varepsilon}$. On it,

$$\mathcal{N}_\delta(A_q) \geq \delta^{q\varepsilon} \mathcal{N}_\delta(A), \quad \mathcal{N}_\delta(\pi_{V_j} A_q) < \delta^{-b-\varepsilon} \quad (1 \leq j \leq q). \quad (\text{A.3.132})$$

Let $V = V_1 + \dots + V_q$, and let $F : \bigoplus_{j=1}^q V_j \rightarrow V$ send $(v_1, \dots, v_q)$ to $\sum_j v_j$. By Gram–Schmidt, the absolute determinant J of F is the product, over $j \geq 2$, of the m -Jacobian of the projection of V_j onto $S_{j-1}^\perp$. Since $S_{j-1} \subset T_j$, each such Jacobian is at least $\Delta(V_j, T_j) > \delta^h$. Consequently $J > \delta^{(q-1)h}$, $\dim V = qm$, and, because $\|F\| \leq \sqrt{q}$, the least singular value of F is at least $c_{q,m}J$. The adjoint $F^* : V \rightarrow \bigoplus_j V_j$ is the map $x \mapsto (\pi_{V_1}x, \dots, \pi_{V_q}x)$. Product covers of the sets in [\(A.3.132\)](#), pulled back through $(F^*)^{-1}$ and subdivided into radius- δ

balls, give the explicit comparison

$$\mathcal{N}_\delta(\pi_V A_q) \leq C_{n,q} J^{-qm} \prod_{j=1}^q \mathcal{N}_\delta(\pi_{V_j} A_q) \leq C_{n,q} \delta^{-qb - [q+3qm(q-1)/\kappa]\varepsilon}. \quad (\text{A.3.133})$$

Choose ε so that $[q + 3qm(q-1)/\kappa]\varepsilon < e_*/2$ and $2q\varepsilon < e_*/2$. After decreasing δ to absorb the fixed constant in (A.3.133), the successful event implies $V \in \mathcal{E}_\delta^{(qm)}(A; e_*)$, where the latter is the rank- qm version of the bad event with threshold $qb = qm\alpha/n$.

By Theorem A.3.6, conditional on $\dim V = qm$ the law σ' of V satisfies a Schubert estimate with exponent κ/q and prefactor $\delta^{-2\varepsilon}$. Since $2\varepsilon < e_*$, the rank- qm theorem applies to (A, σ') at exponent e_* : its size and ball hypotheses also follow from those assumed at exponent ε . Hence $\sigma'(\mathcal{E}_\delta^{(qm)}(A; e_*)) \leq \delta^{e_*}$. The probability of $\dim V < qm$ is at most $(q-1)\delta^{\kappa-\varepsilon}$. We obtain

$$\delta^{2q\varepsilon} \leq \mathbb{P}\{V \in \mathcal{E}_\delta^{(qm)}(A; e_*), \dim V = qm\} \leq \delta^{e_*},$$

which is impossible for small δ . This proves the asserted good-piece instance. $\square$

The opposite operation in the induction is to intersect several target planes. Small projections to each plane need not yield a small projection to their intersection: a large fiber perpendicular to that intersection is the obstruction. The following finite-set trichotomy makes this alternative exact. It reconstructs the combinatorial core of He, Proposition 35, without importing the uniform-cover theorem as an unproved research input.

A.3.12 Intersection geometry and the all-rank relative theorem

Proposition A.3.67 (Coordinate intersection trichotomy). *Let $q \geq 2$, $n = q(n-m) + r$ with $1 \leq r < m < n$, and partition $\{1, \dots, n\}$ into disjoint blocks $B_1, \dots, B_q, I_0$ with $\#B_j = n-m$ and $\#I_0 = r$. Put $I_j = \{1, \dots, n\} \setminus B_j$, and let $\omega_I : \mathbb{Z}^n \rightarrow \mathbb{Z}^I$ be coordinate projection. For every nonempty finite $Z \subset \mathbb{Z}^n$ and $K \geq 1$, at least one of the following holds:*

1. $\#\omega_{I_j} Z \geq K(\#Z)^{m/n}$ for some $1 \leq j \leq q$;
2. some $y \in \mathbb{Z}^{I_0}$ has $\#(Z \cap \omega_{I_0}^{-1}(y)) \geq K(\#Z)^{(n-r)/n}$;
3. there is $Z' \subset Z$ with

$$\#Z' \geq \frac{\#Z}{2K^{q+1}}, \quad \#\omega_{I_0} Z' \leq 2K^q(\#Z)^{r/n}.$$

Proof. Write $U = \{1, \dots, n\} \setminus I_0$ and $J_j = U \setminus B_j$, so that $I_j = I_0 \cup J_j$. Each coordinate of U belongs to exactly $q-1$ of the J_j . We first establish, for any nonempty finite $F \subset \mathbb{Z}^U$, the uniform-cover bound

$$(\#F)^{q-1} \leq \prod_{j=1}^q \#\omega_{J_j} F. \quad (\text{A.3.134})$$

Indeed, let X be uniform on F , order the coordinates of U , and write H for finite Shannon entropy. The chain rule and the fact that conditioning cannot increase entropy give

$$H(X_{J_j}) = \sum_{i \in J_j} H(X_i \mid X_{J_j \cap \{1, \dots, i-1\}}) \geq \sum_{i \in J_j} H(X_i \mid X_{\{1, \dots, i-1\}}).$$

Sum in j . Every summand on the right occurs $q-1$ times, so $\sum_j H(X_{J_j}) \geq (q-1)H(X) = (q-1)\log \#F$. Since $H(X_{J_j}) \leq \log \#\omega_{J_j}F$, exponentiation proves (A.3.134).

For $y \in \omega_{I_0}Z$, set $Z_y = Z \cap \omega_{I_0}^{-1}(y)$ and $a_y = \#Z_y$. Apply (A.3.134) to each Z_y , identified with its U -coordinates, then use Hölder's inequality in y :

$$\sum_y a_y^{(q-1)/q} \leq \sum_y \prod_{j=1}^q (\#\omega_{J_j}Z_y)^{1/q} \leq \prod_{j=1}^q (\#\omega_{I_j}Z)^{1/q}. \quad (\text{A.3.135})$$

The last equality implicit in the sum is valid because distinct I_0 -fibers have distinct I_j -projections.

Put $S = \#Z = \sum_y a_y$ and $E = \sum_y a_y^2$. A second application of Hölder, with exponents $(q+1)/q$ and $q+1$, yields

$$S^{q+1} \leq \left(\sum_y a_y^{(q-1)/q} \right)^q E \leq E \prod_{j=1}^q \#\omega_{I_j}Z.$$

If alternative 1 fails, then, using $q(n-m) = n-r$ and hence $qm = (q-1)n+r$, we obtain

$$E > K^{-q} S^{q+1-qm/n} = K^{-q} S^{2-r/n}. \quad (\text{A.3.136})$$

Suppose alternative 2 fails as well, so $a_y < M := KS^{1-r/n}$ for every y . Set $t = S^{1-r/n}/(2K^q)$ and let $Y = \{y : a_y \geq t\}$. The contribution of $y \notin Y$ to E is at most $t \sum_{y \notin Y} a_y \leq tS = E_0/2$, where $E_0 = K^{-q} S^{2-r/n}$. By (A.3.136), $\sum_{y \in Y} a_y^2 > E_0/2$; since $a_y < M$,

$$\sum_{y \in Y} a_y > \frac{E_0}{2M} = \frac{S}{2K^{q+1}}, \quad \#Y \leq \frac{S}{t} = 2K^q S^{r/n}.$$

Take $Z' = \bigcup_{y \in Y} Z_y$. These are precisely the two bounds in alternative 3. $\square$

The coordinate statement becomes geometric after a quantitative change of variables. A loss of a fixed multiple of ε is inevitable because the straightening map can have norm as large as a power of δ^{-1} ; the next proposition records that loss explicitly.

Proposition A.3.68 (Transverse-plane intersection reduction). *Let $q \geq 2$, $n = q(n-m)+r$ with $1 \leq r < m < n$, and $0 < \alpha < n$. There is $C = C(n, q) \geq 1$ such that the following holds for every $\varepsilon > 0$ and all sufficiently small $\delta > 0$. Let $A \subset \mathbb{R}^n$ be bounded and $V_1, \dots, V_q \in \text{Gr}(n, m)$. Put $U_j = V_j^\perp$ and $V = \bigcap_j V_j$. Here $B_V(y, \delta)$ denotes the closed radius- δ ball in V . Assume*

$$\begin{aligned} \text{Vol}(U_1, \dots, U_q) &\geq \delta^\varepsilon, \\ \delta^{-\alpha+\varepsilon} &\leq \mathcal{N}_\delta(A) \leq \delta^{-\alpha-\varepsilon}, \\ \mathcal{N}_\delta(\pi_{V_j}A) &\leq \delta^{-(m/n)\alpha-\varepsilon} \quad (1 \leq j \leq q), \\ \mathcal{N}_\delta(A \cap \pi_V^{-1}(B_V(y, \delta))) &\leq \delta^{-((n-r)/n)\alpha-\varepsilon} \quad (y \in V). \end{aligned}$$

Then some $A' \subset A$ satisfies

$$\mathcal{N}_\delta(A') \geq \delta^{C_\varepsilon} \mathcal{N}_\delta(A), \quad \mathcal{N}_\delta(\pi_V A') \leq \delta^{-(r/n)\alpha - C_\varepsilon}.$$

Proof. The positive volume hypothesis makes $U_1 + \cdots + U_q$ a direct sum of dimension $n - r$, and hence $\dim V = r$. Choose orthonormal bases in each U_j and in V , and define an invertible map $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by sending the coordinate blocks $B_1, \dots, B_q, I_0$ of [Theorem A.3.67](#) to these bases. Then $\|F\| \leq \sqrt{q+1}$ and

$$|\det F| = \text{Vol}(U_1, \dots, U_q) \geq \delta^\varepsilon.$$

Its least singular value is therefore at least $c_n \delta^\varepsilon$. Writing $T = F^{-1}$, we have $\|T\| \leq C_n \delta^{-\varepsilon}$. The I_0 -coordinate of Tx is exactly the orthonormal coordinate vector of $\pi_V x$. Moreover, the map $\omega_{I_j} T$ has kernel U_j and factors through π_{V_j} by a linear map of norm at most $\|T\|$.

We use two elementary covering comparisons. In a Euclidean space of dimension at most n , an L -Lipschitz image of a set satisfies $\mathcal{N}_\delta(fX) \leq C_n(1+L)^n \mathcal{N}_\delta(X)$: map each ball of a radius- δ cover and subdivide its image ball. Also, the number of half-open side- δ coordinate cubes meeting a bounded set is between $c_n \mathcal{N}_\delta(X)$ and $C_n \mathcal{N}_\delta(X)$, since such a cube lies in a bounded number of radius- δ balls and such a ball meets a bounded number of cubes. Fixed constants below are absorbed by an arbitrarily small extra multiple of ε after decreasing δ .

Let $Z \subset \mathbb{Z}^n$ be the indices of the side- δ coordinate cubes meeting TA , and put $S = \#Z$. The covering comparisons, $\|F\| \leq \sqrt{q+1}$ and $\|T\| \leq C_n \delta^{-\varepsilon}$ give

$$\delta^{-\alpha+2\varepsilon} \leq S \leq \delta^{-\alpha-(n+3)\varepsilon} \quad (\text{A.3.137})$$

for sufficiently small δ . Since $\omega_{I_j} T$ factors through π_{V_j} , the m -dimensional image covering comparison gives

$$\#\omega_{I_j} Z \leq \delta^{-(m/n)\alpha - (m+3)\varepsilon} \quad (1 \leq j \leq q). \quad (\text{A.3.138})$$

For a fixed I_0 -coordinate cube, its preimage in A has π_V -image inside a ball of radius $\sqrt{r} \delta$. Cover that ball by C_n radius- δ balls, apply the assumed fiber bound to each, and then use the n -dimensional image comparison under T . Thus every I_0 -fiber of Z has size at most

$$\delta^{-((n-r)/n)\alpha - (n+3)\varepsilon}. \quad (\text{A.3.139})$$

Set $K = \delta^{-(n+10)\varepsilon}$. The lower bound in [\(A.3.137\)](#) shows that, for small δ , $KS^{m/n}$ is larger than the right side of [\(A.3.138\)](#), and $KS^{(n-r)/n}$ is larger than the right side of [\(A.3.139\)](#). Therefore the first two alternatives of [Theorem A.3.67](#) fail, and its third alternative supplies $Z' \subset Z$ with

$$\#Z' \geq \frac{1}{2} K^{-(q+1)} S, \quad \#\omega_{I_0} Z' \leq 2K^q S^{r/n}. \quad (\text{A.3.140})$$

Choose one point of A from each cube indexed by Z' , and call their set A' . A radius- δ ball in the original coordinates maps under T to a ball of radius at most $C_n \delta^{1-\varepsilon}$ and thus meets at most $C_n \delta^{-n\varepsilon}$ of the selected coordinate cubes. Consequently

$$\mathcal{N}_\delta(A') \geq c_n \delta^{n\varepsilon} \#Z' \geq \delta^{[(q+1)(n+10)+n+4]\varepsilon} \mathcal{N}_\delta(A).$$

Since the I_0 -coordinate of Tx is the coordinate of $\pi_V x$, one projected coordinate cube requires at most C_n radius- δ balls in V . The second inequality in (A.3.140) and the upper bound in (A.3.137) give

$$\mathcal{N}_\delta(\pi_V A') \leq C_n \# \omega_{I_0} Z' \leq \delta^{-(r/n)\alpha - [q(n+10)+n+5]\varepsilon}.$$

Taking $C = (q+1)(n+10) + n+5$ proves both assertions. $\square$

The large-set normalization can be proved without fixing a putative dimension α during the induction. The relevant benchmark is $(\#P)^{m/n}$ for a separated finite set P . The next inequality is the quantitative intersection counting step used below. Its fibre hypothesis concerns *every* affine $(n-r)$ -tube, not just fibres of one chosen projection.

Lemma A.3.69 (Transverse fibre-product inequality). *Let $q \geq 2$, $n = q(n-m) + r$, and $0 < r \leq n-m < m < n$. Suppose $V_1, \dots, V_q \in \text{Gr}(n, m)$, $U_j = V_j^\perp$, and*

$$J = \text{Vol}(U_1, \dots, U_q) > 0, \quad V = \bigcap_{j=1}^q V_j.$$

For every finite δ -separated $F \subset \mathbb{R}^n$, put

$$M_F = \sup_{\substack{x \in \mathbb{R}^n \\ W \in \text{Gr}(n, n-r)}} \#(F \cap (x + W^{(C_n \delta)})),$$

where C_n is a fixed dimensional constant. Then

$$(\#F)^q \leq C_{n,q} J^{-qm} M_F \prod_{j=1}^q \mathcal{N}_\delta(\pi_{V_j} F). \quad (\text{A.3.141})$$

Proof. The direct sum $U_1 + \dots + U_q$ has dimension $n-r$ and its orthogonal complement is V . Choose orthonormal bases of the U_j and V , and form the block-coordinate map $G : \mathbb{R}^n \rightarrow \mathbb{R}^n$ as in the proof of Theorem A.3.68. Its norm is bounded dimensionally and its determinant has absolute value J ; thus $\|G^{-1}\| \leq C_{n,q} J^{-1}$. Let $Z \subset \mathbb{Z}^n$ be the indices of half-open side- δ coordinate cubes meeting $G^{-1}F$. Since F is δ -separated and $\|G\|$ is bounded, each cube contains at most $C_{n,q}$ points of $G^{-1}F$. Consequently $\#F \leq C_{n,q} \#Z$.

Use the coordinate blocks $B_1, \dots, B_q, I_0$ of Theorem A.3.67, so $\#B_j = n-m$, $\#I_0 = r$, and $I_j = \{1, \dots, n\} \setminus B_j$. The entropy and Hölder calculation (A.3.134)–(A.3.135) gives, with $z_y = \#(Z \cap \omega_{I_0}^{-1}(y))$,

$$(\#Z)^{q+1} \leq \left(\sum_y z_y^2 \right) \prod_{j=1}^q \# \omega_{I_j} Z \leq (\max_y z_y) (\#Z) \prod_{j=1}^q \# \omega_{I_j} Z.$$

The I_0 -coordinate of $G^{-1}x$ is the orthonormal coordinate of $\pi_V x$, since every U_j is orthogonal to V . Thus one I_0 -fibre of Z represents points of F in an affine $V^\perp$ -tube of radius $C_n \delta$, and $\max_y z_y \leq M_F$. Furthermore $\omega_{I_j} G^{-1}$ has kernel U_j and factors through π_{V_j} by a map of

norm at most $C_{n,q}J^{-1}$. Subdividing the image of each radius- δ ball into coordinate cubes gives

$$\#\omega_{I_j}Z \leq C_{n,q}J^{-m}\mathcal{N}_\delta(\pi_{V_j}F).$$

Combining these three estimates proves (A.3.141). $\square$

Theorem A.3.70 (Relative-size projection bound, all ranks). *For $1 \leq m \leq n$ and $\kappa > 0$ there are $K = K(n, m, \kappa) > 1$ and $\varepsilon_0 > 0$ such that the following holds for $0 < \varepsilon < \varepsilon_0$ and sufficiently small $\delta > 0$. Let $P \subset B(0, 1)$ be a nonempty finite δ -separated set, put $S = \#P$, and let σ be a probability on $\text{Gr}(n, m)$. If $m < n$, assume*

$$\sigma(\mathcal{S}(W, \rho)) \leq \delta^{-\varepsilon} \rho^\kappa \quad (W \in \text{Gr}(n, n-m), \rho \geq \delta).$$

Then

$$\sigma(\mathcal{B}_\delta(P; \delta^\varepsilon, \delta^{K\varepsilon} S^{m/n})) \leq \delta^\varepsilon. \quad (\text{A.3.142})$$

Proof. We induct first on n , and within a fixed ambient dimension in descending order of m . The case $m = n$ is immediate: a δ -separated subset $B \subset P$ has $\mathcal{N}_\delta(B) \geq c_n \#B$, which exceeds $\delta^{K\varepsilon} S$ if $\#B \geq \delta^\varepsilon S$ and $K > 1$. All fixed multiplicative constants below are absorbed by strict inequalities between exponents after decreasing δ .

We first record a sampling observation used twice. Suppose λ is the conditional law of σ on an event of mass greater than δ^a , and for each sampled plane V a chosen witness $B_V \subset P$ has $\#B_V \geq \delta^a S$. Finiteness of P permits a Borel first-witness choice. For q independent λ -samples, Jensen's inequality gives

$$\mathbb{E}_\lambda \# \bigcap_{j=1}^q B_{V_j} = \sum_{x \in P} \lambda\{x \in B_{V_j}\}^q \geq \delta^{qa} S. \quad (\text{A.3.143})$$

For any fixed Schubert tube at radius δ^h , its λ -mass is at most $\delta^{\kappa h - \varepsilon - a}$. By extending each preceding plane sum to a fixed complementary-dimensional plane, as in Theorem A.3.6, one therefore excludes failure of transversality for q samples with probability at most $q\delta^{\kappa h - \varepsilon - a}$. If this is at most $\frac{1}{4}\delta^{qa}$, then (A.3.143) shows that the λ^q -probability of a transverse tuple with

$$\# \bigcap_j B_{V_j} \geq \frac{1}{4}\delta^{qa} S$$

is at least $\frac{1}{4}\delta^{qa}$. The same conclusion holds for the orthogonal complements of sampled planes, using $\Delta(V^\perp, T) = \Delta(V, T^\perp)$.

Suppose first that $m \mid n$, and write $n = qm$. For $q = 1$ the base case above applies. For $q \geq 2$ suppose the event in (A.3.142) has σ -mass greater than δ^ε . Condition on it and choose witnesses B_V with $\#B_V \geq \delta^\varepsilon S$ and $\mathcal{N}_\delta(\pi_V B_V) \leq M := \delta^{K\varepsilon} S^{1/q}$. Apply the sampling observation with $a = \varepsilon$ and $h = (q+3)\varepsilon/\kappa$. It supplies a transverse tuple with common subset F of size at least $\frac{1}{4}\delta^{q\varepsilon} S$ and volume $J \geq \delta^{(q-1)h}$. The product-coordinate estimate used in Theorem A.3.64 gives

$$\#F \leq C_{n,q} J^{-n} \prod_j \mathcal{N}_\delta(\pi_{V_j} F) \leq C_{n,q} \delta^{[qK - n(q-1)(q+3)/\kappa]\varepsilon} S.$$

Choose K so that the bracket exceeds q . This is a contradiction for small δ .

Next suppose $m < n/2$ and $m \nmid n$. Choose $q = \lfloor n/m \rfloor$, so $2 \leq q$, $qm < n$, and $qm > m$. The rank- qm case is already known in the same ambient dimension, with constant $K_* = K(n, qm, \kappa/q)$. If the rank- m bad event has mass greater than δ^ε , the preceding sampling argument gives, with conditional probability at least $\frac{1}{4}\delta^{q\varepsilon}$, a transverse tuple and a common subset F of size at least $\frac{1}{4}\delta^{q\varepsilon}S$. For $V = V_1 + \dots + V_q$, product covering as in (A.3.133) yields

$$\mathcal{N}_\delta(\pi_V F) \leq C_{n,q} \delta^{[qK - qm(q-1)(q+3)/\kappa]\varepsilon} S^{qm/n}.$$

The unconditional probability of these tuples is at least $\frac{1}{4}\delta^{2q\varepsilon}$. Set $e = (2q + 2)\varepsilon$. By Theorem A.3.6, the law of V , conditional on direct sum, has the Schubert bound required by the rank- qm theorem at exponent e and parameter κ/q . Choose K so large that

$$qK - qm(q-1)(q+3)/\kappa > (2q + 2)K_*.$$

For small δ , the common subset witnesses the rank- qm bad event at exponent e , whose conditional probability is at least $\frac{1}{4}\delta^{2q\varepsilon} > \delta^e$. This contradicts the induction hypothesis.

It remains to consider $n/2 < m < n$, so the smaller ambient dimension, rather than a larger rank, supplies the induction input. Write

$$n = q(n-m) + r, \quad q \geq 2, \quad 0 < r \leq n-m, \quad n' = n-r.$$

The rank- m theorem is known in $W \cong \mathbb{R}^{n'}$, including the identity case $m = n'$; let K_s be its constant. Fix $a = 4$, $b = 8$, $c = 2$, and choose K sufficiently large below. Put

$$H = qK/2, \quad L = \delta^{-H\varepsilon} S^{n'/n}, \quad \lambda = \delta^{c\varepsilon}, \quad M = \delta^{K\varepsilon} S^{m/n}.$$

Successively remove from P a piece $T_i = R_i \cap (x_i + W_i^{(\delta)})$ of cardinality at least L , whenever one exists in the current residual R_i . This finite procedure gives disjoint pieces T_i and a residual R with no such large affine $(n-r)$ -slice.

We claim that each removed piece has

$$\sigma(\mathcal{B}_\delta(T_i; \delta^{a\varepsilon}, M)) \leq \delta^{a\varepsilon}. \quad (\text{A.3.144})$$

Represent the affine plane with $x_i \in W_i^\perp$, and choose a maximal δ -separated $Q_i \subset \pi_{W_i} T_i \subset B_{W_i}(0, 1)$. Because T_i is separated and lies in a δ -tube, $\#Q_i \asymp_n \#T_i \geq L$. Indeed each radius- δ ball in W_i lifts to a radius- 3δ ball containing at most C_n points of T_i . Assigning each point of $\pi_{W_i} T_i$ to a nearby point of Q_i therefore has multiplicity at most C_n ; the same bound applies to every subset of T_i . Let $h_s = (a+3)\varepsilon/\kappa$. The event $J_{W_i}(V) < \delta^{h_s}$ has σ -mass at most $\delta^{(a+2)\varepsilon}$. For all other V , a witness to the event on the left of (A.3.144) transfers, by the proof of Theorem A.3.9, to a subset of Q_i of cardinality at least $\delta^{b\varepsilon} \#Q_i$ and projection covering number at most $C_n \delta^{-mh_s} M$. Since $\#Q_i \geq c_n L$, the latter is below $\delta^{bK_s\varepsilon} (\#Q_i)^{m/n'}$ provided

$$K - m(a+3)/\kappa > bK_s - Hm/n'. \quad (\text{A.3.145})$$

The conditional direction law on W_i satisfies the rank- m hypothesis at exponent $b\varepsilon$ by Theorem A.3.5. The lower-dimensional theorem bounds the transferred event by $\delta^{b\varepsilon}$. Together with the small-Jacobian event this proves (A.3.144).

If $\#R \geq \lambda S$, we claim the same bound for R . Otherwise suppose $\sigma(\mathcal{B}_\delta(R; \delta^{a\varepsilon}, M)) > \delta^{a\varepsilon}$. Condition on this bad event. Apply (A.3.143) to its witnesses, but sample the orthogonal complements $U_j = V_j^\perp$. Take

$$h_r = ((q+1)a+3)\varepsilon/\kappa.$$

The conditional Schubert tail and the sampling observation supply a transverse tuple with common subset $F \subset R$ satisfying

$$\#F \geq \frac{1}{4} \delta^{qa\varepsilon} \#R, \quad J := \text{Vol}(U_1, \dots, U_q) \geq \delta^{(q-1)h_r}, \quad \mathcal{N}_\delta(\pi_{V_j} F) \leq M.$$

Every affine $(n-r)$ -tube of radius $C_n \delta$ is covered by $C_{n,q}$ tubes of radius δ ; therefore the residual construction gives $M_F \leq C_{n,q} L$ in Theorem A.3.69. Since $\#R \geq \delta^{c\varepsilon} S$,

$$L \leq \delta^{-(H+cn'/n)\varepsilon} (\#R)^{n'/n}, \quad M \leq \delta^{(K-cm/n)\varepsilon} (\#R)^{m/n}.$$

The fibre-product inequality and $n' + qm = qn$ now imply

$$\delta^{q^2 a\varepsilon} (\#R)^q \leq C_{n,q} \delta^{[qK-H-cq-qm(q-1)((q+1)a+3)/\kappa]\varepsilon} (\#R)^q.$$

Choose K so that the bracket exceeds $q^2 a$. This contradicts small δ , proving the claim for R .

The two strict choices for K are compatible: (A.3.145) becomes easier as K grows with $H = qK/2$, while the last bracket grows like $qK/2$. Finally partition P into the removed pieces and, if $\#R \geq \lambda S$, the residual. Each part obeys a bad-direction bound $\delta^{a\varepsilon}$ at threshold M . If $\#R < \lambda S$, discard it; a global witness of cardinality $\delta^\varepsilon S$ still has at least $(\delta^\varepsilon - \lambda)S$ points in the union of the removed pieces. In either case Theorem A.3.7, with piece threshold $\eta = \delta^{a\varepsilon}$, gives

$$\sigma(\mathcal{B}_\delta(P; \delta^\varepsilon, M)) \leq C \delta^{(a-1)\varepsilon} \leq \delta^\varepsilon$$

for sufficiently small δ . This completes both inductions. $\square$

Corollary A.3.71 (Large-set normalization in all ranks). *The assertion $\mathsf{L}(n, m; \alpha, \kappa)$ holds for every $1 \leq m \leq n$, $0 < \alpha < n$, and $\kappa > 0$. Thus He's Proposition 29 needs no spatial nonconcentration assumption on the set.*

Proof. The case $m = n$ was noted above. For $m < n$, let K be given by Theorem A.3.70, put $e = 2\varepsilon$, and choose M_0 so that $M_0 m/n > 2K + 2$. Suppose $\mathcal{N}_\delta(A) \geq \delta^{-\alpha-M_0\varepsilon}$ and the direction condition holds at exponent ε . Let $P \subset A$ be a maximal δ -separated net and $S = \#P \asymp_n \mathcal{N}_\delta(A)$. For $V \in \mathcal{E}_\delta(A; \varepsilon)$, a witnessing subset of A yields, by assignment to nearby points of P , a set $P' \subset P$ with $\#P' \geq c_n \delta^\varepsilon S$ and

$$\mathcal{N}_\delta(\pi_V P') \leq C_n \delta^{-(m/n)\alpha-\varepsilon}.$$

For small δ , the first bound exceeds $\delta^\varepsilon S$, while the second is below $\delta^{Ke} S^{m/n}$, because $S \geq \delta^{-\alpha-M_0\varepsilon}$ and $M_0 m/n > 2K + 2$. The direction condition at exponent e follows from the one at exponent ε . Consequently $\mathcal{E}_\delta(A; \varepsilon) \subset \mathcal{B}_\delta(P; \delta^e, \delta^{Ke} S^{m/n})$. Theorem A.3.70 bounds its σ -mass by $\delta^e \leq \delta^\varepsilon$. $\square$

We can now complete the *conditional* intersection arrow in He’s rank induction. The large-set assertion L is essential here: it gives an upper bound on the size of a putative counterexample without assuming that a smaller subset inherits spatial nonconcentration. For $m = n$, we use the convention that $L(n, n; \alpha, \kappa)$ is the identity-projection assertion, requiring no direction hypothesis. It holds with any size-surplus constant greater than 2: every δ^ε -large subset then has covering number at least $\delta^{-\alpha-\varepsilon}$, so its bad event is empty.

Proposition A.3.72 (Intersection-rank induction conditional on large-set normalization). *Let $q \geq 2$ and $n = q(n - m) + r$, where $0 < r \leq n - m < m < n$. Put $\beta = (n - r)\alpha/n$, with $0 < \alpha < n$, and fix $\kappa > 0$. Assume $L(n, m; \alpha, \kappa)$ and $L(n - r, m; \beta, \kappa)$, and assume [Theorem A.3.8](#) in dimensions (n, r) at parameters $(\alpha, \kappa/q)$. Then [Theorem A.3.61](#) holds in dimensions (n, m) at parameters (α, κ) .*

Proof. Let M_n be a size-surplus constant for $L(n, m; \alpha, \kappa)$, and M_s one for $L(n - r, m; \beta, \kappa)$; if $m = n - r$, use the trivial identity case. Let $e_r > 0$ be an admissible exponent of the rank- r theorem. We will choose the good-piece exponent $u > 0$ sufficiently small in terms of these fixed data. Suppose that a finite $A \subset B(0, 1)$ satisfies (A.3.9)–(A.3.11) with u in place of ε , but every nonempty $T \subset A$ has

$$\sigma(\mathcal{E}_\delta(T; u)) > \delta^u. \quad (\text{A.3.146})$$

In particular this holds for A ; therefore $L(n, m; \alpha, \kappa)$ implies

$$\mathcal{N}_\delta(A) < \delta^{-\alpha - M_n u}. \quad (\text{A.3.147})$$

We first exclude a large $(n - r)$ -dimensional slice. Choose $h = 3u/\kappa$, $C_s > 2 + 3m/\kappa$, and then a constant $H > M_s C_s + 2$. If an affine $(n - r)$ -plane $x + W$ satisfies

$$\mathcal{N}_\delta(S) \geq \delta^{-\beta - Hu}, \quad S = A \cap (x + W^{(\delta)}),$$

take the representative $x \in W^\perp$, which leaves the affine plane unchanged, and put $B = \pi_W(S - x) \subset B_W(0, 1)$. Every point of $S - x$ is within δ of B , so $\mathcal{N}_\delta(B) \geq c_n \mathcal{N}_\delta(S) \geq \delta^{-\beta - M_s C_s u}$ for small δ . By [Theorem A.3.5](#), the pushforward to $\text{Gr}(W, m)$ of σ , conditioned on $J_W(V) > 0$, satisfies the Schubert hypothesis at exponent $C_s u$; the exceptional set $J_W(V) < \delta^h$ has mass at most δ^{2u} . Apply $L(n - r, m; \beta, \kappa)$ to B at exponent $C_s u$. It gives

$$\sigma_W(\mathcal{E}_\delta^W(B; C_s u)) \leq \delta^{C_s u}. \quad (\text{A.3.148})$$

Indeed, if $V \in \mathcal{E}_\delta(S; u)$ has $J_W(V) \geq \delta^h$, a witnessing $S' \subset S$ maps to $B' = \pi_W(S' - x) \subset B$ with $\mathcal{N}_\delta(B') \geq c_n \delta^u \mathcal{N}_\delta(B)$. The projections $\pi_V(S' - x)$ and $\pi_V B'$ are within δ , so the covering comparison and [Theorem A.3.9](#) yield

$$\mathcal{N}_\delta(\pi_{\pi_W V} B') \leq C_n \delta^{-mh} \mathcal{N}_\delta(\pi_V S') < \delta^{-(m/n)\alpha - C_s u}.$$

For small δ , B' witnesses the event in (A.3.148). But (A.3.146) for S gives this event conditional probability at least $\delta^u - \delta^{2u} > \delta^{2u}$, contradicting (A.3.148), since $C_s > 2$. Consequently every affine $(n - r)$ -slice obeys

$$\mathcal{N}_\delta(A \cap (x + W^{(\delta)})) < \delta^{-\beta - Hu}. \quad (\text{A.3.149})$$

Take independent $V_1, \dots, V_q$ with law σ , and write $U_j = V_j^\perp$. We construct nested nonempty sets $A = A_0 \supset A_1 \supset \dots \supset A_q$ by requiring $V_j \in \mathcal{E}_\delta(A_{j-1}; u)$ and choosing a witnessing A_j . At each step after the first, extend $U_1 + \dots + U_{j-1}$ to an m -plane T_j by a fixed Borel Gram–Schmidt rule and require $\Delta(U_j, T_j) > \delta^h$. The complementary-plane identity $\Delta(U_j, T_j) = \Delta(V_j, T_j^\perp)$ and (A.3.11) show that the excluded set has conditional probability at most δ^{2u} . Thus (A.3.146) makes the whole construction succeed with probability at least δ^{2qu} . On success,

$$\mathcal{N}_\delta(A_q) \geq \delta^{qu} \mathcal{N}_\delta(A), \quad \mathcal{N}_\delta(\pi_{V_j} A_q) < \delta^{-(m/n)\alpha - u}, \quad \text{Vol}(U_1, \dots, U_q) \geq \delta^{3(q-1)u/\kappa}. \quad (\text{A.3.150})$$

In particular $V = \bigcap_j V_j$ has dimension r .

Let C_g be the constant in Theorem A.3.68. Choose once and for all

$$D > \max\{M_n, H, q + 1, 3(q - 1)/\kappa, 1\}.$$

The lower size hypothesis for A_q at exponent Du follows from (A.3.9) and (A.3.150); its upper size hypothesis follows from (A.3.147). The projection and transversality hypotheses follow from (A.3.150); the fiber hypothesis follows from (A.3.149). Hence Theorem A.3.68, applied with $\varepsilon = Du$, supplies $A' \subset A_q$ with

$$\mathcal{N}_\delta(A') \geq \delta^{(C_g D + q)u} \mathcal{N}_\delta(A), \quad \mathcal{N}_\delta(\pi_V A') \leq \delta^{-(r/n)\alpha - C_g Du}. \quad (\text{A.3.151})$$

The complementary-plane identity also transfers the Schubert hypothesis from V_j to U_j . Apply Theorem A.3.6 to $U_1, \dots, U_q$, then take orthogonal complements. Conditional on their direct-sum event, the law of V satisfies the rank- r Schubert bound $\delta^{-2u} \rho^{\kappa/q}$. Take u small enough that

$$2u < e_r, \quad (C_g D + q)u < e_r, \quad 2qu < e_r.$$

The original A satisfies all three hypotheses of the rank- r theorem at exponent e_r : the hypotheses at u are stronger, and the just-derived direction law has the required prefactor. That theorem bounds the conditional mass of $\mathcal{E}_\delta^{(r)}(A; e_r)$ by δ^{e_r} . Yet (A.3.151) puts every successful tuple in that event, whose unconditional probability is at least $\delta^{2qu} > \delta^{e_r}$; conditioning on an event of probability at most one only increases this lower bound. This contradiction proves the good-piece theorem. $\square$

Proposition A.3.73 (Closure of He’s rank induction). *Assume Theorem A.3.21 in every matrix dimension required by the half-dimensional case. Then Theorem A.3.61 and Theorem A.3.8 hold for every $1 \leq m < n$, $0 < \alpha < n$, and $\kappa > 0$.*

Proof. Order pairs (n, m) , $0 < m < n$, lexicographically by $(n, \min(m, n - m), m)$. The order is well founded. For $n = 2m$, Theorem A.3.60 gives the good-piece theorem from the assumed matrix theorem. If $m \mid n$ and $n = qm$ with $q \geq 3$, Theorem A.3.63 uses the already established full theorem in ambient dimension $(q - 1)m < n$. If $m < n/2$ and $m \nmid n$, take $q = \lfloor n/m \rfloor$; then $qm < n$ and $n - qm < m$, so (n, qm) is earlier in the order. Theorem A.3.66 gives the good-piece theorem. In the remaining case $m > n/2$ and $m \nmid n$, write $n = q(n - m) + r$ with $q \geq 2$ and $0 < r \leq n - m$. The pair (n, r) is earlier: either $r < n - m$, or $r = n - m$ and then $r < m$. The all-rank normalization Theorem A.3.71 supplies both

L-hypotheses of [Theorem A.3.72](#); that proposition therefore gives the good-piece theorem from the earlier rank- r full theorem. In every case [Theorem A.3.62](#) converts the good-piece conclusion to the full theorem before the next induction step. All dimension and exponent parameters are fixed at each finite step, so choosing a smaller positive exponent when an implication requires it preserves the hypotheses. $\square$

Proof of Theorems A.3.8 and A.3.61. The matrix theorem has just been proved. Its half-dimensional application and the remaining rank arrows therefore satisfy the hypotheses of [Theorem A.3.73](#), which gives both stated conclusions for every admissible rank. $\square$

To use a discretized theorem for a fractal set, one first turns a Frostman measure into point sets whose cell masses are comparable. The construction below is a finite-scale fact; it does not rely on He’s theorem.

A.3.13 From finite sets to analytic sets

Lemma A.3.74 (Frostman mass bins give regular discrete sets). *Let $0 < s < n$, $0 < \eta < 1$, and let ν be a compactly supported probability on $B(0, 1) \subset \mathbb{R}^n$ such that $\nu(B(x, r)) \leq C_\nu r^s$ for every x and $0 < r \leq 2$. For every sufficiently small dyadic $\delta = 2^{-k}$ there are a number $a = a(\delta) \in [0, n + 2\eta]$, a finite $A_\delta \subset \text{spt } \nu$ containing one point in each of a collection $\mathcal{Q}_\delta$ of dyadic δ -cubes, and a constant $c = c(n, \eta) > 0$ such that*

$$\delta^{a+\eta} < \nu(Q) \leq \delta^a \quad (Q \in \mathcal{Q}_\delta), \quad (\text{A.3.152})$$

$$\nu\left(\bigcup_{Q \in \mathcal{Q}_\delta} Q\right) \geq c, \quad \mathcal{N}_\delta(A_\delta) \geq \delta^{-s+2\eta}, \quad (\text{A.3.153})$$

$$\mathcal{N}_\delta(A_\delta \cap B(x, \rho)) \leq \delta^{-2\eta} \rho^s \mathcal{N}_\delta(A_\delta) \quad (x \in \mathbb{R}^n, \delta \leq \rho \leq 1). \quad (\text{A.3.154})$$

The lower bound in (A.3.153) and the upper bound in (A.3.154) hold after decreasing the admissible δ threshold in terms of n, s, η, C_ν .

Proof. At most $C_n \delta^{-n}$ dyadic δ -cubes meet $B(0, 1)$. Those with mass at most $\delta^{n+\eta}$ have total mass at most $C_n \delta^\eta$, hence less than $1/2$ for small δ . Divide the remaining masses into at most $J = \lceil (n + 2\eta)/\eta \rceil$ classes of the form $(\delta^{a+\eta}, \delta^a]$ with a a nonnegative multiple of η . One class has total mass $M \geq 1/(2J)$; let $\mathcal{Q}_\delta$ be its cubes. Choose one point of $\text{spt } \nu \cap Q$ from each $Q \in \mathcal{Q}_\delta$, and call the resulting set A_δ . If a mass-bearing point lies on a dyadic boundary, the half-open convention assigns it to one cube, so the choice is unambiguous.

Write $N = \#\mathcal{Q}_\delta = \#A_\delta$. The mass range gives

$$M\delta^{-a} \leq N < M\delta^{-a-\eta}. \quad (\text{A.3.155})$$

The Frostman bound gives $\nu(Q) \leq C'_{n,s} C_\nu \delta^s$. Since every selected cube has $\nu(Q) > \delta^{a+\eta}$, we obtain $a \geq s - \eta - O_{n,s,C_\nu}(1/\log(1/\delta))$. A radius- δ ball meets at most C_n dyadic δ -cubes, so $\mathcal{N}_\delta(A_\delta) \geq N/C_n$. Combine this with (A.3.155) and $M \geq 1/(2J)$; for sufficiently small δ , all fixed constants are at most $\delta^{-\eta}$, proving the cardinality assertion in (A.3.153).

For $\rho \geq \delta$, every selected cube containing a point of $A_\delta \cap B(x, \rho)$ lies inside $B(x, \rho + \sqrt{n}\delta)$. The Frostman bound and the lower mass in (A.3.152) show that the number of these cubes is at most $C''_{n,s} C_\nu \rho^s \delta^{-a-\eta}$. Since $N \geq M\delta^{-a}$ and $\mathcal{N}_\delta(A_\delta) \geq N/C_n$, we get

$$\frac{\mathcal{N}_\delta(A_\delta \cap B(x, \rho))}{\mathcal{N}_\delta(A_\delta)} \leq C'''_{n,s,\eta,C_\nu} \delta^{-\eta} \rho^s.$$

Decrease the δ threshold once more to absorb the fixed coefficient into $\delta^{-\eta}$. This is (A.3.154). $\square$

We now make the analytic finite-to-continuum passage, including the measurable choice of covers, explicit. The following implication uses the discretized theorem proved above. A measure is *carried by* an arbitrary set X if some Borel subset $B \subset X$ has full measure; this convention does not assert that X itself is Borel. We use the analytic compact-selection theorem registered as a Category 2 interface in Volume III.1: if E is analytic and $s < \dim_{\text{H}} E$, there is compact $K \subset E$ with $\dim_{\text{H}} K > s$.

Theorem A.3.75 (Analytic finite-to-continuum transfer). *Fix $1 \leq m < n$, $0 < \alpha < n$, and $\kappa > 0$, and put $\kappa_0 = \min\{\kappa, \alpha/2\}$. There exist $\tau, \gamma > 0$, depending only on these parameters, with the following property. If $E \subset B(0, 1)$ is analytic, $\dim_{\text{H}} E > \alpha - \tau$, and σ is a nonzero finite Borel measure on $\text{Gr}(n, m)$ satisfying*

$$\sigma(\mathcal{S}(W, \rho)) \leq C_\sigma \rho^\kappa \quad (W \in \text{Gr}(n, n-m), 0 < \rho \leq 1), \quad (\text{A.3.156})$$

then σ is not carried by

$$X_\gamma(E) = \{V \in \text{Gr}(n, m) : \dim_{\text{H}} \pi_V E \leq (m/n)\alpha + \gamma\}. \quad (\text{A.3.157})$$

Proof. Let $\varepsilon_* > 0$ be a parameter for which Theorem A.3.8 holds. Its assertion remains true after decreasing ε_* : every hypothesis becomes stronger and every conclusion weaker. Hence we may require $\varepsilon_* < \min\{\alpha/4, (n-\alpha)/4, 1/4\}$. Put $\tau = \varepsilon_*/20$, $\gamma = \tau$, $s = \alpha - \tau$, and $\eta = (m/n)\alpha + \varepsilon_*/2$. By analytic compact selection choose compact $K \subset E$ with $\dim_{\text{H}} K > s$. Frostman's theorem supplies a probability ν supported on K with

$$\nu(B(x, r)) \leq C_\nu r^s \quad (x \in \mathbb{R}^n, 0 < r \leq 2). \quad (\text{A.3.158})$$

Normalize σ to a probability μ ; its bound (A.3.156) then has a possibly larger fixed coefficient, harmless for small δ .

Suppose for contradiction that μ is carried by $X_\gamma(E)$, and fix a Borel full- μ set $B \subset X_\gamma(E)$. We need no measurability assertion about the whole exceptional set. Fix $k_0 \geq 1$. For every $V \in B$, the strict inequality $\dim_{\text{H}} \pi_V E < \eta$ gives a countable cover of $\pi_V K$ by balls of radii $r_j < 2^{-k_0}$ with $\sum_j r_j^\eta < 1$. Group balls with $2^{-(k+1)} < r_j \leq 2^{-k}$ and replace their centers by rational points of $\mathbb{R}^n$, enlarging each radius to $4 \cdot 2^{-k}$. The k th group, denoted U_k , is a union of at most $C_\eta 2^{k\eta}$ open balls of that radius. Since ν is a probability and $\pi_V K \subset \bigcup_{k \geq k_0} U_k$, we have $\sum_{k \geq k_0} \nu\{x : \pi_V x \in U_k\} \geq 1$. Choose a fixed $c_0 > 0$ with $c_0 \sum_{k \geq 1} k^{-2} < 1$; then at least one $k \geq k_0$ has

$$\nu\{x : \pi_V x \in U_k\} > c_0 k^{-2}. \quad (\text{A.3.159})$$

For each k , let F_k be the set of directions for which some union U of at most $C_\eta 2^{k\eta}$ rational-centered radius- $4 \cdot 2^{-k}$ balls satisfies (A.3.159). This is Borel: for each fixed open U , the map $V \mapsto \nu\{x : \pi_V x \in U\}$ is lower semicontinuous by Fatou's lemma, and the family of finite rational-ball unions is countable. We have $B \subset \bigcup_{k \geq k_0} F_k$. Consequently, since $\mu(B) = 1$, some arbitrarily large k satisfies

$$\mu(F_k) \geq c_0 k^{-2}. \quad (\text{A.3.160})$$

Indeed otherwise countable subadditivity would bound $\mu(\bigcup_{k \geq k_0} F_k)$ by $c_0 \sum_{k \geq k_0} k^{-2} < 1$.

Fix such a large k , put $\delta = 2^{-k}$, and partition the dyadic δ -cubes meeting K by their ν -masses. Cubes with mass at most $\delta^{n+\tau}$ have total mass $O_n(\delta^\tau)$. The others belong to at most $J = J(n, \tau)$ bins

$$\mathcal{Q}_l = \{Q : \delta^{a_l+\tau} < \nu(Q) \leq \delta^{a_l}\}, \quad a_l \in \{0, \tau, 2\tau, \dots\}.$$

Write $B_l = \bigcup_{Q \in \mathcal{Q}_l} Q$ and $M_l = \nu(B_l)$. Call a bin *good* if $M_l \geq \delta^\tau$, and, for each good bin, choose one $x_Q \in K \cap Q$ and let $A_l = \{x_Q : Q \in \mathcal{Q}_l\}$. All these sets and bins are fixed before choosing a direction.

Every good A_l satisfies the hypotheses of Theorem A.3.8 for sufficiently small δ . To check the exponents, the Frostman bound on one cube and its lower bin mass imply

$$a_l \geq s - \tau - O_{n,s,C_\nu}(1/\log(1/\delta)) = \alpha - 2\tau - o(1).$$

Since a radius- δ ball meets at most C_n grid cubes, while $\#A_l \geq M_l \delta^{-a_l}$, it follows that

$$\mathcal{N}_\delta(A_l) \geq C_n^{-1} M_l \delta^{-a_l} \geq \delta^{-\alpha+4\tau} \geq \delta^{-\alpha+\varepsilon_*}. \quad (\text{A.3.161})$$

The last inequality uses $4\tau < \varepsilon_*$. A selected cube whose representative lies in $B(x, \rho)$ is contained in $B(x, \rho + \sqrt{n}\delta)$. Comparing its mass with (A.3.158), and then using $M_l \geq \delta^\tau$, gives, for $\delta \leq \rho \leq 1$,

$$\frac{\mathcal{N}_\delta(A_l \cap B(x, \rho))}{\mathcal{N}_\delta(A_l)} \leq C \delta^{-2\tau} \rho^s \leq \delta^{-3\tau} \rho^{\kappa_0} \leq \delta^{-\varepsilon_*} \rho^{\kappa_0}. \quad (\text{A.3.162})$$

For $\rho > 1$ the target bound is trivial. Finally (A.3.156), after normalization and absorption of its fixed coefficient, gives the direction hypothesis (A.3.11) with κ_0 .

For $V \in F_k$, choose a witnessing union U . Removing the low-mass cubes and bins not good from (A.3.159) loses only $O(J\delta^\tau) < c_0/(2k^2)$. Hence some good l satisfies

$$\nu(B_l \cap \{x : \pi_V x \in U\}) > c_0/(2Jk^2). \quad (\text{A.3.163})$$

The set $F_{k,l}$ of directions satisfying (A.3.163) for at least one allowed rational-ball union U is Borel by the same lower-semicontinuity argument, now integrating over the fixed measurable set B_l . As F_k is covered by the $F_{k,l}$, one good bin has

$$\mu(F_{k,l}) \geq c_0/(Jk^2) > \delta^{\varepsilon_*} \quad (\text{A.3.164})$$

when k is sufficiently large.

For $V \in F_{k,l}$, let $A'_l \subset A_l$ consist of the representatives of the cubes $Q \in \mathcal{Q}_l$ that meet $\{x : \pi_V x \in U\}$. The bin upper mass in (A.3.163) shows that $\#A'_l \geq c_0 \delta^{-a_l} / (2Jk^2)$. On the other hand $\#A_l \leq M_l \delta^{-a_l - \tau} \leq \delta^{-a_l - \tau}$. Covering a grid-separated set by δ -balls loses at most a factor C_n , so

$$\frac{\mathcal{N}_\delta(A'_l)}{\mathcal{N}_\delta(A_l)} \geq \frac{c_0}{2C_n Jk^2} \delta^\tau \geq \delta^{\varepsilon_*}.$$

Each representative is within $\sqrt{n}\delta$ of a point projecting into U , and U is a union of at most $C_\eta \delta^{-\eta}$ radius- 4δ balls. Thus

$$\mathcal{N}_\delta(\pi_V A'_l) \leq C_{n,\eta} \delta^{-\eta} < \delta^{-(m/n)\alpha - \varepsilon_*}$$

for sufficiently small δ , because $\eta = (m/n)\alpha + \varepsilon_*/2$. Every $V \in F_{k,l}$ is therefore a bad direction for A_l in Theorem A.3.8, contradicting (A.3.164). This completes the proof. $\square$

The preceding theorem has the analytic-set scope of He's Theorem 2 and is slightly stronger because its lower dimension hypothesis has slack τ . Its only external leaf beyond the locally proved discrete theorem is the Category 2 analytic compact-selection/Frostman interface already declared in Volume III.1. The quantifier “carried by” is important: the proof needs only a Borel full-mass subset of the exceptional-direction set, not an unproved Borelness assertion for that whole set. To pass to a Hausdorff-dimension bound, we compare the exceptional set for analytic E with the Borel exceptional set for one compact subset of E .

Corollary A.3.76 (Analytic-set exceptional bound). *Fix $1 \leq m < n$, $0 < \alpha < n$ and $\xi > 0$. Choose the gain $\gamma > 0$ from Theorem A.3.75 with $\kappa = \min\{\alpha/2, \xi/(2m)\}$ small enough that $(m/n)\alpha + \gamma < m$. If $E \subset \mathbb{R}^n$ is analytic with $\dim_H E = \alpha$, then*

$$\dim_H \{V \in \text{Gr}(n, m) : \dim_H \pi_V E \leq (m/n)\alpha + \gamma\} \leq m(n - m) - 1 + \xi. \quad (\text{A.3.165})$$

Proof. Set $D = m(n - m)$ and let τ be the slack from Theorem A.3.75. By analytic compact selection, choose compact $K \subset E$ with $\dim_H K > \alpha - \tau$. If E is unbounded, first restrict it to a sufficiently large ball; a similarity then places K inside $B(0, 1)$ without changing any projected Hausdorff dimensions. Put

$$X_E = \{V : \dim_H \pi_V E \leq (m/n)\alpha + \gamma\}, \quad X_K = \{V : \dim_H \pi_V K \leq (m/n)\alpha + \gamma\}.$$

Because $K \subset E$, we have $X_E \subset X_K$. By Volume III.2, L-III.2.04, X_K is Borel; the non-strict sublevel is the countable intersection of strict sublevels with thresholds decreasing to $(m/n)\alpha + \gamma$. If $\dim_H X_E > D - 1 + \xi$, choose t with $D - 1 + \xi < t < \dim_H X_E \leq \dim_H X_K$. The Frostman theorem already used in Volumes I and III gives a nonzero finite Borel measure σ carried by a compact subset of X_K such that $\sigma(B(V, r)) \leq Mr^t$.

By Theorem A.3.4, for every complementary plane W and $0 < \rho < 1/2$,

$$\sigma(\mathcal{S}(W, \rho)) \leq CM\rho^{(t-D+1)/m}.$$

Since $(t - D + 1)/m > \xi/m > \xi/(2m)$, multiplying σ by one sufficiently small positive constant makes $\sigma(\mathcal{S}(W, \rho)) \leq \rho^{\xi/(2m)}$ for all W and all $0 < \rho \leq 1$. This contradicts Theorem A.3.75 applied to K , because the rescaled nonzero measure is carried by X_K . Hence (A.3.165) holds. $\square$

The compact comparison avoids asserting any Borelness or capacitability for X_E itself. For $m = 1$ the reader should use Volume III.6, not this higher-rank interface.

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